INCH-POUND

MIL-P-24764
29 November 1991

MILITARY SPECIFICATION

POWER SUPPLIES, SHIPBOARD, ELECTRONIC

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for a standard family of shipboard electronic power supplies for use in Naval ship and shore systems including combat and other mission critical applications. These power supplies are coordinated under the Standard Hardware Acquisition and Reliability Program (SHARP).

1.2 Classification. Electronic power supplies whose initial application is for Navy shipboard, subsurface ship, or shore use are classified as one of the following families.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Family S1. These are Shipboard Power Supply Products (SPSP) units of the following styles in accordance with the associated detail specification (see supplement 1 for list of associated specifications). All styles have a length of 10.80 inches (274.3 mm) and a height of 2.48 inches (63.0 mm).

Style 1 - The width is 5.00 inches (127.0 mm).

Style 2 - The width is 5.80 inches (147.3 mm).

Style 3 - The width is 6.80 inches (172.7 mm).

Style 4 - The width is 8.80 inches (223.5 mm).

Style 5 - The width is 11.80 inches (299.7 mm).

1.2.2 Family S2. These are Standard Electronic Module (SEM) compatible units of the following classes and formats in accordance with the associated detail specification (see supplement 1 for list of associated specifications).

Classes

Class III - For primary utilization in shipboard and subsurface ship applications (and where suitable, in shore applications) which may be exposed to nuclear radiation (this class assignment is in agreement with the class description specified in MIL-M-28787).

Formats

Format A - The basic size has a span of 2.62 inches (66.6 mm), a thickness of 0.290 inches (7.37 mm), and a total height of 1.95 inches (49.5 mm) including keying pins. The size may increase in span, thickness, or simultaneously in both span and thickness within the constraints of MIL-STD-1389. See appendix A of MIL-STD-1389.

Format B - The basic size has a span of 2.74 inches (69.6 mm), a thickness of 0.290 inches (7.37 mm), and a total height of 1.95 inches (49.5 mm) including keying pins. The size may increase in span, thickness,

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or simultaneously in both span and thickness within the constraints of MIL-STD-1389. See appendix B of MIL-STD-1389.

Format C - The basic size has a span of 5.88 inches (149.4 mm), a thickness of 0.280 inches (7.11 mm), and a total height of 4.06 inches (103.1 mm) including keying pins. The size may increase in thickness within the constraints of MIL-STD-1389. See appendix C of MIL-STD-1389.

Format D - The basic size has a span of 5.88 inches (149.4 mm), a thickness of 0.280 inches (7.11 mm), and a total height of 4.83 inches (122.7 mm) including keying pins. The size may increase in thickness within the constraints of MIL-STD-1389. See appendix D of MIL-STD-1389.

Format E - The basic size has a span of 5.88 inches (149.4 mm), a thickness of 0.380 inches (9.65 mm), and a total height of 6.68 inches (169.7 mm) including keying pins. The size may increase in thickness within the constraints of MIL-STD-1389. See appendix E of MIL-STD-1389.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-N-290	- Nickel Plating (Electrodeposited).
PPP-B-00636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bags, Sleeves and Tubing.
MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-W-8939	- Welding, Resistance, Electronic Circuit Modules.
MIL-Q-9858	- Quality Program Requirements

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MIL-S-19500	- Semiconductor Devices, General Specification for.
MIL-C-28754	- Connectors, Electrical, Modular, and Component Parts, General Specification for.
MIL-M-28787	- Modules, Standard Electronic, General Specification for.
MIL-C-28859	- Connector Component Parts, Electrical Backplane, Printed Wiring, General specification for.
MIL-A-28870	- Assemblies, Electrical Backplane, Printed-Wiring, General Specification for.
MIL-M-38510	- Microcircuits, General Specification for.
MIL-H-38534	- Hybrid Microcircuits, General Specification for.
MIL-C-39003	- Capacitors, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability, General Specification for.
MIL-R-39008	- Resistors, Fixed, Composition (Insulated), Established Reliability, General Specification for.
MIL-G-45204	- Gold Plating, Electrodeposited.

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MIL-P-50884 - Printed Wiring, Flexible and Rigid-Flex.

MIL-P-55110 - Printed Wiring Board, General Specification for.

STANDARDS

FEDERAL

FED-STD-595 - Colors used in Government Procurement.

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings, and in Specifications, Standards and Technical Documents.

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-130 - Identification Marking of U.S. Military Property.

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-275 - Printed Wiring for Electronic Equipment.

MIL-STD-454 - Standard General Requirements for Electronic Equipment.

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electro-magnetic Interference.

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MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-481	-	Configuration Control-Engineering Changes (Short Form), Deviations and Waivers
MIL-STD-701	-	Lists of Standard Semiconductor Devices.
MIL-STD-740-1	-	Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-756	-	Reliability Modeling and Prediction.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-883	-	Test Methods and Procedures for Microelectronics.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-961	-	Military Specifications and Associated Documents, Preparation of.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.

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MIL-STD-1399 Section 300	-	Interface Standard for Shipboard Systems.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1562	-	Lists of Standard Microcircuits.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric).
MIL-STD-1772	-	Certification Requirements for Hybrid Microcircuit Facilities and Lines.
MIL-STD-2000	-	Standard Requirements for Soldered Electrical and Electronic Assemblies.
MIL-STD-2036	-	General Requirements for Electronic Equipment Specifications
MIL-STD-2038	-	Requirements for Employing Standard Power Supplies.
MIL-STD-2073-1	-	DoD Materiel Procedures for Development and Application of Packaging Requirements.
MIL-STD-45662	-	Calibration Systems Requirements.

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HANDBOOKS

FEDERAL

Handbook H4-2	-	Federal Supply Code for Manufacturers (United States and Canada) Code to Name.
Handbook H6	-	Federal Item Name Directory for Supply Cataloging.

MILITARY

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment.
DOD-HDBK-263	-	Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric).

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General

3.1.1 Associated detail specifications. The individual power supply requirements shall be as specified herein and in accordance with the associated detail specification. In the event of any conflict between the requirements of this specification and the associated detail specification, the latter shall govern.

3.1.1.1 Specification format. Associated detail specifications for both family S1 and family S2 shall be slash sheets to this specification and shall be written in accordance with MIL-STD-2038.

3.1.2 Definitions. The definitions of terms used in the document shall be in accordance with 6.5 of this document and definitions in MIL-STD-2038.

3.2 Qualification. Power supplies furnished under this specification shall be products which are authorized by the SPSP Quality Assurance Activity (SPSP-QAA) for listing on the applicable qualified products list at the time of delivery (see 4.4 and 6.4).

3.3 Electrical performance requirements. Power supplies shall be in accordance with the electrical requirements herein and in the associated detail specification.

3.3.1 Input power.

3.3.1.1 Alternating current (AC) input power. Unless otherwise specified in the associated detail specification, power supplies shall operate and maintain specified performance when the input power has the characteristics in accordance with MIL-STD-1399, section 300, for type I, 115 volts alternating current (VAC) root mean square (RMS) line-to-line, 60 hertz (Hz), 3-phase, ungrounded power except in accordance with 3.3.1.1.1 through 3.3.1.1.4 for voltage and frequency. Harmonic current limitations are specified in 3.3.1.10 and hold-up time is specified in 3.3.2.11.

3.3.1.1.1 Worst case voltage excursion. The worst case voltage excursion from nominal user voltage resulting from user voltage tolerance (average of the three line-to-line voltages), voltage cyclic variation, and voltage transient limits combined, except under fault conditions, shall be plus or minus 25

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er material or part except that which possibly may be inflicted to
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percent. Power supplies shall be capable of continuous operation at nominal AC input voltage plus or minus 25 percent. The turn-on voltage threshold is specified in 3.3.1.6.

3.3.1.1.2 Spike voltage. Unless otherwise specified in the associated detail specification, power supplies shall withstand, without damage, the voltage spike in accordance with MIL-STD-1399, section 300 (see 4.7.2.1). The transient response of the output voltage to this spike voltage shall be nonoscillatory or highly damped. The response shall be limited to initial transients (during the first 20 microseconds (us)) with an amplitude less than 5 percent of the nominal output voltage. If present, the secondary transients (after 20 us until the end of the input spike pulse) shall have an amplitude less than 2.5 percent of the nominal output voltage. Unless otherwise specified in the associated detail specification, any additional overshoot or undershoot excursions shall be within the ripple and noise envelope specified in 3.3.2.3.

3.3.1.1.3 Frequency tolerance. Power supplies shall operate over a frequency range of 47 to 63 Hz including transients and worst case excursions.

3.3.1.1.4 Emergency condition voltage excursion. The power supplies shall not be damaged by an emergency condition input power line voltage of plus 35 percent from nominal user voltage (up to 12 hours duration). The power supply outputs shall remain within plus or minus 5 percent of the nominal output voltage. Upon restoration of normal input power, the power supplies shall recover automatically and provide the performance specified herein.

3.3.1.2 Direct current (DC) input power, 155 volts direct current (VDC). Unless otherwise specified in the associated detail specification, power supplies shall operate and maintain specified performance when operating from DC power derived from step down transformers and multi-pulse rectification of ship service 440-volt, 60-Hz, 3-phase, ungrounded type I power with characteristics in accordance with MIL-STD-1399, section 300. The AC power frequency shall be within the range specified in 3.3.1.1.3. The nominal DC input voltage shall be 155 VDC with a worst case voltage excursion from 155 VDC of plus or minus 30 percent. The positive and negative peaks of any ripple voltage shall remain within the 30 percent DC voltage tolerance levels. Ripple voltage shall be either 23 percent peak-to-peak maximum with a fundamental frequency up to six times the AC source power frequency, or it shall be 5 percent peak-to-peak maximum with a

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fundamental frequency of 12, 18, or 24 times the AC source power frequency. This DC input power shall be applied to any two of the three AC input terminals. Power supplies shall be capable of continuous operation while using DC input power as specified herein. The turn-on voltage threshold is specified in 3.3.1.6.

3.3.1.3 Auxiliary DC input. Unless otherwise specified in the associated detail specification, power supplies shall operate and maintain specified performance when operating from DC power applied to the auxiliary input terminals separate from the AC input terminals. The DC voltage shall be in accordance with 3.3.1.2. Intended uses of the auxiliary DC input include back-up power, uninterruptible power, and extension of the power supply hold-up capability.

3.3.1.4 Alternate DC input power. When specified in the associated detail specification, power supplies shall operate from DC input voltages other than the voltage specified in 3.3.1.2. Alternate input sources may be any single voltage (such as 28 VDC or 270 VDC with associated tolerances) providing the potential for multiple system application of the power supply. The alternate DC input voltage and its characteristics shall be as specified in the associated detail specification.

3.3.1.5 Input protection. Power supplies shall not be damaged by polarity reversal, momentary interruptions (up to 30 seconds), loss of a single phase (with 3-phase input), low input line voltage, or a short circuit on the input power lines. The maximum input current drawn under any steady state input voltage conditions less than the maximum specified input voltage shall be no greater than 2.0 times the input steady state true RMS or DC current at maximum rated load and nominal input voltage. The applicable turn-on current paragraph (see 3.3.1.9.1 or 3.3.1.9.2) shall apply for the maximum input current drawn during any transient input condition. The power supply outputs shall not exceed the overvoltage requirements specified in 3.3.2.14 during the abnormal input power conditions. Upon restoration of normal input power, the power supplies shall recover automatically and perform as specified herein. A power supply failure shall not cause damage to the input power lines.

3.3.1.6 Turn-on threshold. Power supply turn-on is defined as the input voltage where the output voltage rises to a level greater than or equal to 95 percent of the nominal level, at maximum load. Unless otherwise specified in the associated detail specification, the input voltage threshold for power supply turn-on shall be between 120 and 135 VDC, or for an AC

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input, between 83 and 96 VAC. The threshold for an alternate DC input (see 3.3.1.4) shall be specified in the associated detail specification.

3.3.1.7 Turn-off threshold and hysteresis. Power supply turn-off is defined as the input voltage where the output voltage drops to a level less than 95 percent of the nominal level, at maximum load. Unless otherwise specified in the associated detail specification, the input voltage threshold for power supply turn-off shall be between 90 and 102 VDC, or for an AC input between 72 and 84 VAC. In addition, the power supply input voltage hysteresis (input voltage turn-on threshold of 3.3.1.6 minus input voltage turn-off threshold) shall be a minimum of 18 VDC (14.4 VAC). The power supply outputs are not required to meet the static output regulation requirements (see 3.3.2.5) for input voltages less than nominal DC voltage minus 30 percent or nominal AC voltage minus 25 percent regardless of the turn-off threshold. The threshold and hysteresis for an alternate DC input (see 3.3.1.4) shall be specified in the associated detail specification.

3.3.1.8 Input current balance. The RMS values of the input currents on the 3-phase AC input lines shall be within 5 percent of each other when the source voltage is balanced within 1 percent between input lines.

3.3.1.9 Turn-on current. The requirements specified in 3.3.1.9.1 and 3.3.1.9.2 apply to all AC, DC, and auxiliary power inputs to the power supply.

3.3.1.9.1 Step input. During the entire turn-on cycle, the peak inrush current (see 4.7.2.2) resulting from a step input voltage and with the input capacitors discharged shall be limited to 10 times the input steady state true RMS or DC current at maximum rated load and nominal input voltage (this shall apply for power supplies with rated input power of 3 kilowatts or less). For power supplies with rated input power above 3 kilowatts, the inrush current shall be in accordance with MIL-STD-1399, section 300, for type I power. Unless otherwise specified in the associated detail specification, the rate of rise of the inrush current is not controlled and, therefore, can be much greater than 10 amperes /(millisecond-kilowatt) of nominal input power. The rate of change of a step input voltage shall be greater than or equal to 1 volt per us.

3.3.1.9.2 Slowly rising input. For turn-on with slowly rising input voltage with a rate of change less than 1 volt per millisecond or for remote turn-on (see 3.3.3.1), the rate of rise of input current shall be no greater than 10 amperes /(milli-second-kilowatt) of nominal input power. This specified rate of

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rise of input current is related to the maximum value of external input inductance specified in the associated detail specification. The peak input current during the entire turn-on cycle shall be limited to 2.0 times the input steady state true RMS or DC current at maximum rated load and nominal input voltage.

3.3.1.10 Harmonic current limitation. Each power supply shall minimize harmonic distortion effects on the electrical system. Unless otherwise specified in the associated detail specification, power supplies shall limit the harmonic content of the input current so that no individual harmonic line current from the second harmonic through 20 kilohertz (kHz) exceeds a magnitude of 100/n percent of the unit's full load fundamental current, where n is the harmonic multiple number. Harmonic line current requirements apply when power supplies are operating (see 4.7.2.3) in the mode that generates the highest input current harmonic current amplitude.

3.3.1.10.1 Harmonic current (3 percent) limitation. Unless otherwise specified in the associated detail specification, individual power supplies are not required to meet the 3 percent harmonic current limit in accordance with MIL-STD-2036, section 300 of MIL-STD-1399, or MIL-STD-461. Systems required to meet the 3 percent harmonic current limit may reject the harmonic current with an AC to DC converter and provide DC power with characteristics in accordance with 3.3.1.2 or 3.3.1.3 to the individual power supplies.

3.3.1.11 Power factor. The power factor (see 4.7.2.4) shall be between 0.8 and unity.

3.3.1.12 Electrical bonding. Power supplies shall have a minimum of one zero-volt reference connector pin connected directly to external parts of the supply (except other connector pins or parts requiring electrical isolation). The zero-volt reference pin(s) shall have a current rating equal to or greater than the current rating of the input power return pins. Direct current resistance between the connector pin and any part of the chassis shall be 0.1 ohm maximum. Non-conductive protective finishes shall be omitted (or removed) from bonding surfaces. Design shall assure permanence of low impedance bonds in the presence of humid and saline atmospheres over the power supply life.

3.3.1.13 External input filter. When specified in the associated detail specification, an external input filter is required to enable the power supplies to meet conducted emissions in accordance with MIL-STD-461, requirement CE03. The input impedance limit (see 3.10.9) of the power supply shall be as

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specified in the associated detail specification so the input filter can be designed by the user to operate together with the power supply.

3.3.1.14 Input isolation resistance. Input power connections and chassis ground shall be isolated by greater than 10 megohms resistance at 800 VDC. Input power connections and DC output voltage and return shall be isolated by greater than 10 megohms resistance at 800 VDC. The test voltage shall be 500 VDC if the maximum input power voltage is 100 VDC or less and no user system power line tests are ran which would cause this voltage value to be exceeded.

3.3.1.15 Input capacitive loading. The capacitive loading on any AC input line to chassis ground shall be not greater than 0.01 microfarads. The capacitive load on any DC only input line to chassis shall be not greater than 0.05 microfarad per kilowatt.

3.3.2 Output power.

3.3.2.1 Output voltage and current. The power supplies shall deliver the DC voltages and rated currents in accordance with the associated detail specification. Where more than one output voltage may be selected, the output voltage shall be determined by the wiring interconnection of the system. Unless otherwise specified in the associated detail specification, any output adjustment devices (other than pin programming) shall be internal to the power supply (see 3.10.6), and shall be designated as factory, depot, or shop adjust only. If the adjustment entails a moving part (for example, potentiometer), the part shall be secured after adjustment to prevent movement during vibration and shock.

3.3.2.2 Output load characteristics. The power supplies shall provide the output voltage and currents in accordance with the associated detail specification for load conditions which include resistive, capacitive (30 microfarads per rated output ampere), and constant current loads. The requirements shall be met for input line variations within the specified limits, through all environmental extremes specified, and for any combination of these conditions. Unless otherwise specified in the associated detail specification, the threshold voltage at which a constant current load shall be assumed to operate at full load capacity shall be less than or equal to 2.0 VDC. Additional capacitive loading for parallel outputs shall be in accordance with 3.3.2.20.

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heat dissipators. This preferred arrangement shall be used except where it

3.3.2.3 Total ripple and noise. Unless otherwise specified in the associated detail specification, total peak-to-peak ripple and noise in a 10 Hz to 5 megahertz (MHz) bandwidth (see 4.7.2.5), including spikes, hash, periodic, and aperiodic components, shall not exceed 1.3 percent of the nominal output voltages of each power supply or 130 millivolts (mV), whichever is greater, while operating over the range of 10 to 100 percent of full rated load current and over the specified input voltage range. In addition, the ripple and noise in a 10 Hz to 5 MHz bandwidth shall not exceed 2 percent of the nominal output voltages of each power supply or 200 mV, whichever is greater, when operated at loads in excess of 100 percent which do not result in more than a 5 percent decrease in output voltage or at loads below the specified minimum. The peak-to-peak ripple and noise in a 10 Hz to 20 MHz bandwidth shall not exceed 6 percent of the nominal output voltages of each power supply.

3.3.2.3.1 Low frequency ripple and noise. The peak-to-peak ripple and noise in a 10 Hz to 10 kHz bandwidth impressed upon the output voltage shall not exceed 0.5 percent of the nominal output voltage or 50 mV, whichever is greater, (see 4.7.2.5.1). This requirement shall be met at full rated load current and over the specified input voltage range.

3.3.2.3.2 High frequency ripple and noise. Unless otherwise specified in the associated detail specification, the peak-to-peak ripple and noise in a bandwidth from 10 kHz to 1 MHz impressed upon the output voltage shall not exceed 0.5 percent of the nominal output voltage or 50 mV, whichever is greater, (see 4.7.2.5.2). This requirement shall be met over the range of rated resistive load current and over the specified input voltage range.

3.3.2.3.3 Single spectral line amplitude. When specified in the associated detail specification, the amplitude of any single spectral line on the outputs of the power supply shall meet the requirements under any allowable combination of input voltage and output load current.

3.3.2.4 Common mode output current. The common mode output current from any power output terminal and return terminal to the chassis shall not exceed 200 milliamperes peak-to-peak (mAp-p) in a 10 Hz to 20 MHz bandwidth when measured in accordance with 4.7.2.6 for any load current from 10 percent to full rated load.

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3.3.2.5 Static line and load regulation. Unless otherwise specified in the associated detail specification, the steady state output voltage shall remain within plus or minus 1 percent of the specified nominal output voltage at 25 degrees celsius (°C) for all allowable combinations of input voltage and output current. Static line and load regulation shall be measured in accordance with 4.7.2.7.

3.3.2.6 Static regulation temperature effects. The line and load regulation shall not exceed plus or minus 2 percent of the nominal output voltages of each power supply type when operated over the temperature range specified in 3.5.2.1 (a) and 3.5.2.1 (b) and for all allowed simultaneous variations of line voltage and load current.

3.3.2.7 Dynamic load regulation. The transient response of the output voltage to a step change (see 4.7.2.8) in load current shall be nonoscillatory or highly damped. The response shall be limited to an initial single half cycle overshoot (undershoot) with an amplitude less than 3 percent of the nominal output voltage. If present, the second half cycle undershoot (overshoot) shall have an amplitude less than 1.5 percent of the nominal output voltage. Unless otherwise specified in the associated detail specification, any additional overshoot or undershoot excursions shall be within the ripple and noise envelope specified in 3.3.2.3.

3.3.2.8 Dynamic line regulation. The transient response of the output voltage to a step change (see 4.7.2.9) in input voltage shall be nonoscillatory or highly damped. The response shall be limited to an initial single half cycle overshoot (undershoot) with an amplitude less than 3 percent of the nominal output voltage. If present, the second half cycle undershoot (overshoot) shall have an amplitude less than 1.5 percent of the nominal output voltage. Unless otherwise specified in the associated detail specification, any additional overshoot or undershoot excursions shall be within the ripple and noise envelope specified in 3.3.2.3.

3.3.2.9 Interaction of multiple output supplies. Unless otherwise specified in the associated detail specification, the requirements of this specification apply to each output with all possible combinations of output loading.

3.3.2.10 Hold-up time. Unless otherwise specified in the associated detail specification, each power supply output voltage shall remain within 3 percent line regulation limits for a minimum of 50 us at full rated load current following the loss of input power. This hold-up time requirement shall be

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following the loss of input power when operating at any allowable input voltage down to and including nominal AC voltage minus 25 percent or nominal DC voltage minus 30 percent. When a power interrupt signal (see 3.3.4.3) is available from the power supply, the hold-up time requirement shall be met following the loss of input power as indicated by the power interrupt signal.

3.3.2.11 Power turn-on/turn-off response. Each power supply output voltage shall rise to and remain within the regulation limits within 250 milliseconds (ms) after application of the input power (with remote on/off control turned on as specified in 3.3.3.1) and under any allowable load. Unless otherwise specified in the associated detail specification, the minimum rise time under full load conditions shall be 15 ms. The maximum transient component of excursion of output voltage (overshoot or undershoot) resulting from turn-on or turn-off shall not exceed plus or minus 5 percent of the nominal output voltage. The output voltage shall rise as specified even with an externally applied positive or negative bias on the output pins. The positive bias shall be any voltage up to and including the upper regulation limit of the full output voltage. The negative bias shall be a current source of 1 ampere or 10 percent of full load (whichever is less) applied into the negative output pins with the current return path through the positive output pins.

3.3.2.12 Control turn-on/turn-off response. Each power supply output voltage shall rise to and remain within the regulation limits within 250 ms after turn-on of the remote on/off control (see 3.3.3.1) and under any allowable load. Unless otherwise specified in the associated detail specification, the minimum rise time under full load conditions shall be 15 ms. The maximum transient component of excursion of output voltage (overshoot or undershoot) resulting from turn-on or turn-off shall not exceed plus or minus 5 percent of the nominal output voltage. The output voltage shall rise as specified even with an externally applied positive or negative bias on the output pins in accordance with 3.3.2.11.

3.3.2.13 Overcurrent protection. Each output of a power supply shall be provided with overcurrent and short circuit protection with automatic recovery. Overcurrent shall not exceed 125 percent of full rated load current. Recovery from overcurrent shall occur within 250 ms (recovery time after removal of the overcurrent). The maximum transient component of excursion of output voltage (overshoot or undershoot) resulting from overcurrent recovery shall not exceed plus or minus 5 percent of the nominal output voltage of each power supply type. The power supply shall operate under any overcurrent indefinitely. Unless otherwise

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specified in the associated detail specification, when one or more outputs of a multiple output power supply go into an overcurrent mode, the other outputs are not required to meet their regulation limits, but shall not exceed the overvoltage requirements specified in 3.3.2.14.

3.3.2.14 Overvoltage protection. The voltage of each output of a power supply shall not exceed 15 percent (or 1.2 VDC, whichever is greater) above its highest nominal output voltage. Except when allowed in the associated detail specification, the power supply shall not activate the output overvoltage circuitry in the event of power supply turn-on, short circuit recovery, load or line transients or noise spikes. Activations of these types shall be regarded as a failure of the power supply. Unless otherwise specified in the associated detail specification, over-voltage crowbar protection shall not be used.

3.3.2.14.1 Overvoltage latch. Unless otherwise specified in the associated detail specification, the overvoltage protection circuit shall latch all outputs off when an overvoltage condition occurs on one or more outputs. The power supply output voltage shall reset when the input voltage is taken to zero and then reapplied. If the overvoltage condition has not been cleared, the protection circuit shall trip again.

3.3.2.15 Remote voltage sensing. Unless otherwise specified in the associated detail specification, power supplies shall be capable of remote voltage sensing up to 10 inches (length of load leads and sense leads) from the power supply output terminals and compensating for at least a 3 percent voltage drop of the highest nominal output voltage in the load leads between the power supply output voltage terminals and the corresponding load terminals. When the associated detail specification requires remote voltage sensing, there shall be an application warning in the associated detail specification that the allowed load lead inductance to the sense point together with the load capacitance can affect stability.

3.3.2.15.1 Open sensing lead protection. Each power supply type requiring remote sensing shall provide protection against excessive output voltage overshoot in the event of opening one or both remote sensing leads. The output voltages (at the supply output terminals) of each power supply shall completely recover to and remain within plus or minus 5 percent of the nominal supply output voltage within 100 ms (recovery time) after the occurrence of an open sensing lead. The maximum excursion of output voltage (measured across the power supply output voltage terminals), during this 100 ms interval shall not exceed plus 10 percent or minus 5 percent of the nominal output

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voltages of each power supply type. Turning on the power supply with reversed sense leads shall not cause damage to the power supply nor cause the power supply output to exceed the overvoltage limits. Opening of a load lead (with sense leads connected) shall not cause damage to the power supply.

3.3.2.16 Output impedance. The output impedance of each output shall be minimized consistent with the magnitude of the output voltage and other design requirements over the frequency range of 10 Hz to 25 kHz. Unless otherwise specified in the associated detail specification, the output impedance shall not exceed 30 milliohms.

3.3.2.17 Output isolation resistance. The chassis ground shall be isolated from the output voltage and return connections by greater than 10 megohms resistance at 50 VDC. The outputs of multiple output power supplies shall be isolated in accordance with the associated detail specification.

3.3.2.18 Nonpolarized operation. Unless otherwise specified in the associated detail specification, all power supplies shall meet all requirements when operated as either positive or negative power supplies.

3.3.2.19 Reverse voltage protection. Unless otherwise specified in the associated detail specification, the power supply shall be protected against reverse voltage on its output terminals by clamping the reversed voltage to 1.0 VDC or less while sinking up to 1 ampere or 10 percent of its full load current (whichever is less) continuously. This requirement is designed to provide protection when an unenergized power supply has a low level leakage current applied to its output, through load circuitry, from another energized, opposite polarity power supply.

3.3.2.20 Parallel operation. When applicable, the requirement for parallel operation shall be specified in the associated detail specification. The following minimum requirements shall be specified:

(a) The number of power supplies of equal power rating to be paralleled.

(b) The method of current sharing.

(c) The current sharing differential.

(d) The maximum output capacitance of each power supply.

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(e) Allowed impedance imbalance between each paralleled output and the load.

Failure modes that may cause a system of parallel power supplies to fail shall have a failure rate not greater than 5 percent of the failure rate determined by the mean time between failures (MTBF) of a single power supply (see 3.7.1). Failure of any power supply (other than a short in the power supply output circuitry) shall not cause failure of the output power bus. Power supplies operating in parallel shall drive the output capacitance of the paralleled supplies in addition to the loading required in 3.3.2.2.

3.3.3 Control signals.

3.3.3.1 Output voltage remote on/off. Unless otherwise specified in the associated detail specification, all power supply outputs shall be capable of being controlled on and off using the remote switch interface. This interface shall accept transistor-transistor logic (TTL) level (0 VDC, minimum, 5.5 VDC, maximum) signals. A logic high (plus 2.0 VDC, minimum, 100 microamperes, maximum) or open circuit shall turn the power supply off. A logic low (plus 0.8 VDC, maximum, 3.2 milliamperes (mA), maximum) or connecting the remote switch pin to the remote switch return shall turn the power supply on. When specified in the associated detail specification, remote on/off input terminals shall be electrically isolated from the power supply circuitry and chassis. Unless otherwise specified, this isolation shall be 10 megohms minimum at 100 VDC.

3.3.3.2 Output voltage programming. When specified in the associated detail specification, each power supply output voltage shall be selectable to either value specified (typically only two different values) using the assigned voltage programming contacts. An open circuit shall select the lower value output voltage and a closed circuit connecting the voltage programming contact to the voltage programming return contact shall select the higher output voltage. Spurious noise signals on the voltage programming contacts shall not be directly transmitted to the output being controlled. Unless otherwise specified in the associated detail specification, there shall be a minimum of 15 decibels (dB) rejection of signals (in the frequency range of 100 Hz to 50 kHz) appearing on the voltage programming contacts.

3.3.3.3 Synchronization. When specified in the associated detail specification, the power supply shall be designed such that the switching frequency can be synchronized to the switching

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frequency of another power supply or power converter. The synchronization signal will be as defined in the associated detail specification.

3.3.4 Status signals.

3.3.4.1 Isolation. Unless otherwise specified in the associated detail specification, status signal outputs shall be isolated from the other power supply circuitry and chassis by a minimum of 10 megohms resistance at 100 VDC.

3.3.4.2 Over-temperature indication. The power supplies shall contain two internal, thermally activated, normally open switching devices to indicate excessive temperature and unsafe temperature. A nominal unsafe temperature is 10°C above the worst case transient critical component temperature (see 3.12.1.2). An excessive temperature shall be nominally 11°C to 15°C below the nominal unsafe temperature set point. The tolerance of the temperature sensing devices shall be plus or minus 5°C, maximum. The unsafe indicator activation shall occur before any component suffers damage. Both indicators shall reset to normally open prior to the resumption of power supply operation at or below an average thermal interface (see 6.5.7) temperature of 60°C. The current rating of the contacts shall be at least 0.5 amperes resistive at 28 VDC. The voltage drop of the closed contacts shall not exceed 0.4 VDC for any current less than 0.5 amperes. When specified in the associated detail specification, the normally open contact switching device may be replaced by a normally high level signal or a high impedance "off" state device. An excessive temperature would be indicated by a low level (0.4 VDC, maximum, 5.0 mA, minimum) signal. The maximum applicable external voltage shall be 30 VDC and the "off" state leakage current at 30 VDC shall not exceed 200 microamperes.

3.3.4.3 Power interrupt signal. When specified in the associated detail specification, the power supplies shall monitor internal circuitry and shall provide a low level (0.4 VDC, maximum, 5.0 mA, minimum) signal to indicate normal prime power input in accordance with 3.3.1 and a high level signal, or a high impedance "off" state, to indicate that prime power is low or has been interrupted. The power interrupt signal shall be a positive output with respect to the power interrupt signal return. The maximum applicable external voltage shall be 30 VDC and the "off" state leakage current at 30 VDC shall not exceed 200 microamperes. As the input voltage is applied to the power supply, the power interrupt signal shall go to a low level when the input voltage is greater than or equal to minus 25 percent of the nominal DC voltage (minus 20 percent of the nominal AC voltage) but less than or equal

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to the turn-on threshold voltage specified in 3.3.1.6. The power interrupt signal shall remain a low level until the input voltage drops to within the range of less than minus 30 percent of the nominal DC voltage (minus 25 percent of the nominal AC voltage) but greater than the maximum turn-off voltage threshold specified in 3.3.1.7. The power supply interrupt signal response time (including time to settle to 10 percent of its steady state value) shall be 50 us maximum.

3.3.4.4 Output status signal. The power supplies shall monitor each output voltage and shall provide a single low level signal (0.4 VDC, maximum, 5.0 mA, minimum) to indicate that each power supply output voltage is equal to or greater than 91 plus or minus 4 percent of its lowest nominal output but less than 106 percent of its highest nominal output voltage. When the output voltage drops below 91 plus or minus 4 percent, the output status signal shall go to a high level or a high impedance "off" state. The output status signal shall go to a high level or high impedance "off" state when the output voltage is within the range or equal to or greater than 106 percent of the highest nominal, but less than or equal to the overvoltage limit specified in 3.3.2.14. The output status signal shall be a positive output with respect to the output status signal return. The maximum applicable external voltage shall be 30 VDC and the "off" state leakage current at 30 VDC shall not exceed 200 microamperes. The output status signal shall go to a high level or high impedance "off" state when the output voltage remote on/off control is off (see 3.3.3.1). The output status signal shall settle to within 10 percent of its steady state value within 2 ms of an initial high or low transition. When two or more power supplies are operated in parallel, the output status signal is not required to indicate the correct status of its own output, since it could have voltage applied to its output from a paralleled power supply.

3.3.4.5 Malfunction signal. When specified in the associated detail specification, the power supplies shall provide a signal to indicate when an internal fault condition exists. The malfunction signal shall be a positive output with respect to the malfunction signal return. A low level signal (0.4 VDC, maximum, 5.0 mA, minimum) shall indicate normal power supply operation and a high level signal or high impedance "off" state shall indicate that an internal fault exists. The maximum applicable external voltage shall be 30 VDC and the "off" state leakage current at 30 VDC shall not exceed 200 microamperes. The combinations of conditions which result in a malfunction indication shall be as shown in table I. The malfunction signal will indicate a malfunction when conditions A AND C AND D are "yes" AND B is "no" OR when C AND E are "yes" OR when F is "yes."

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No other combination of conditions shall cause a malfunction indication. The malfunction signal shall settle to within 10 percent of its steady state value within 2 ms of an initial high or low transition. When power supplies are paralleled, it may be necessary to add a malfunction signal to be capable of detecting a faulty power supply (see 3.3.4.4).

TABLE I. Malfunction combinations. 1/

Output malfunction (MF)	Output under-voltage (A)	Output over-current (B)	Allowed input voltage (C)	On/off remote switch ON (D)	Other internal faults (E)	Output over-voltage (F)
1	Yes	No	Yes	Yes	-	-
2	-	-	Yes	-	Yes	-
3	-	-	-	-	-	Yes

1/ Dash denotes no requirement. $MF = (A\ \overline{B}\ C\ D) + (C\ E) + F$

3.3.4.6 Status signal noise. Noise on the power interrupt signal, output status signal and malfunction signal outputs shall be between the limits of minus 500 mV peak to 500 mV peak (referenced to the signal return) in a 100 MHz bandwidth. This requirement shall apply when the output is at a low level (0.1 to 5.0 mA current).

3.3.4.7 Visual indicators. The power supplies shall light a green light emitting diode (LED) to indicate the status of the output voltage. The LED shall be on when the output status signal is low and shall be off when the output status signal is high or off. When specified in the associated detail specification, power supplies shall light a green LED to indicate prime power is applied. The LED shall be on when the power interrupt signal is low and shall be off when the power interrupt signal is high or off. When specified in the associated detail specification, the power supplies shall light a green LED to indicate the status of internal malfunctions. The LED shall be on when the malfunction signal is low and shall be off when the malfunction signal is high or off. The LED shall have a minimum luminous intensity of 0.4 millicandelas when viewed from a line centered at the indicator and perpendicular to the surface of the power supply. The minimum luminous intensity at 30 degrees (see 3.4.4) shall be 0.2 millicandelas.

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3.3.5 Efficiency. Unless otherwise specified in the associated detail specification, the efficiency of the power supply (see 4.7.2.10) shall be 70 percent minimum for family S1 and 80 percent minimum for family S2.

3.3.6 Internal frequencies. Power supply internal oscillator frequencies shall be 18 kHz or greater.

3.4. Physical requirements.

3.4.1 Size and weight.

3.4.1.1 Family S1. The power supply size and weight shall be within the limits specified in table II. The power supplies shall meet the detailed dimensioning requirements specified herein (see appendix A) and in the associated detail specifications. The general power supply outline dimensions are shown on figure 1.

TABLE II. Size and weight.

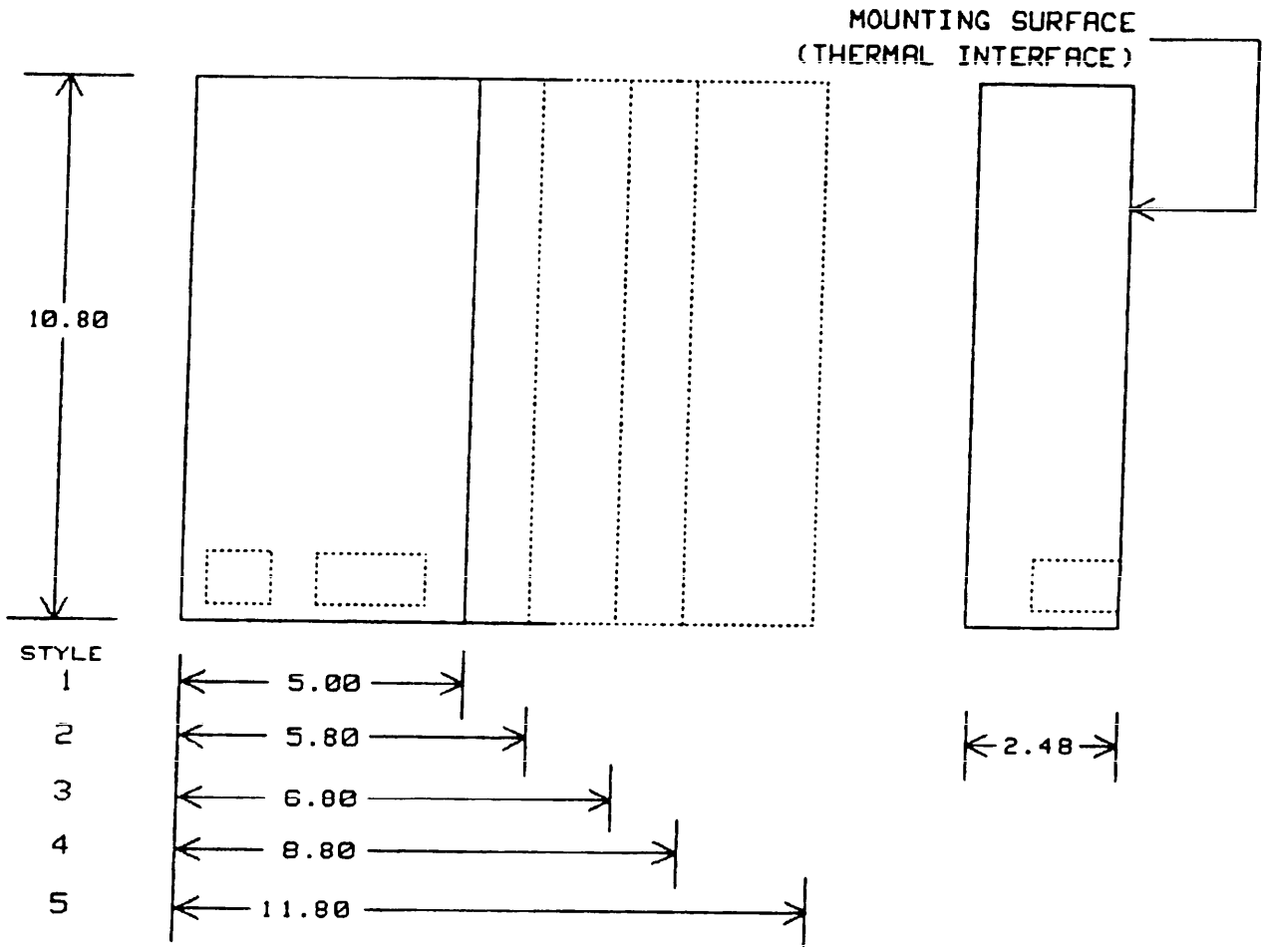
Style	Maximum pounds	Maximum length (inches (mm))	Width 1/ (inches (mm))	Height 1/ (inches (mm))
1	12	10.80(274.3)	5.00(127.0)	2.48 (63.0)
2	12	10.80(274.3)	5.80(147.3)	2.48 (63.0)
3	14	10.80(274.3)	6.80(172.7)	2.48 (63.0)
4	18	10.80(274.3)	8.80(223.5)	2.48 (63.0)
5	24	10.80(274.3)	11.80(299.7)	2.48 (63.0)

1/ The dimension tolerance is plus or minus 0.03 inches (0.76) millimeters (mm).

3.4.1.2 Family S2. The power supply size for formats A through E shall be in accordance with MIL-STD-1389, appendices A through E and in accordance with the associated detail specifications. The power supplies shall be designed for minimum weight consistent with the requirements stated herein and in the associated detail specification.

3.4.2 Connector location and contact assignments.

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NOTE:

1. Dimensions are in inches.

FIGURE 1. Family S1 power supply outline dimensions.

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3.4.2.1 Family S1. The input and output connectors shall both be installed on the same side of the power supply in accordance with appendix A. The contact assignments shall be in accordance with the associated detail specification. Figure 2 shows typical functions which would appear on the contacts. Equivalently marked, physically compatible connectors and contacts shall be approved by SPSP-QAA prior to use in a power supply.

3.4.2.2 Family S2. Unless otherwise specified in the associated detail specification, the input/output connector shall be in accordance with MIL-STD-1389, appendices A through E. Electrical functions and contact assignments shall be in accordance with the associated detail specification.

3.4.3 Hold-down screw locations.

3.4.3.1 Family S1. The hold-down screw locations shall be in accordance with appendix A.

3.4.3.2 Family S2. The hold-down screw locations shall be in accordance with MIL-STD-1389, appendices A through E.

3.4.4 Visual indicator location. The location of the visual indicators shall be as specified in the associated detail specification. The indicators shall be mounted to permit viewing from a minimum angle of 30 degrees in any direction from a line centered at the indicator and perpendicular to the surface of the power supply.

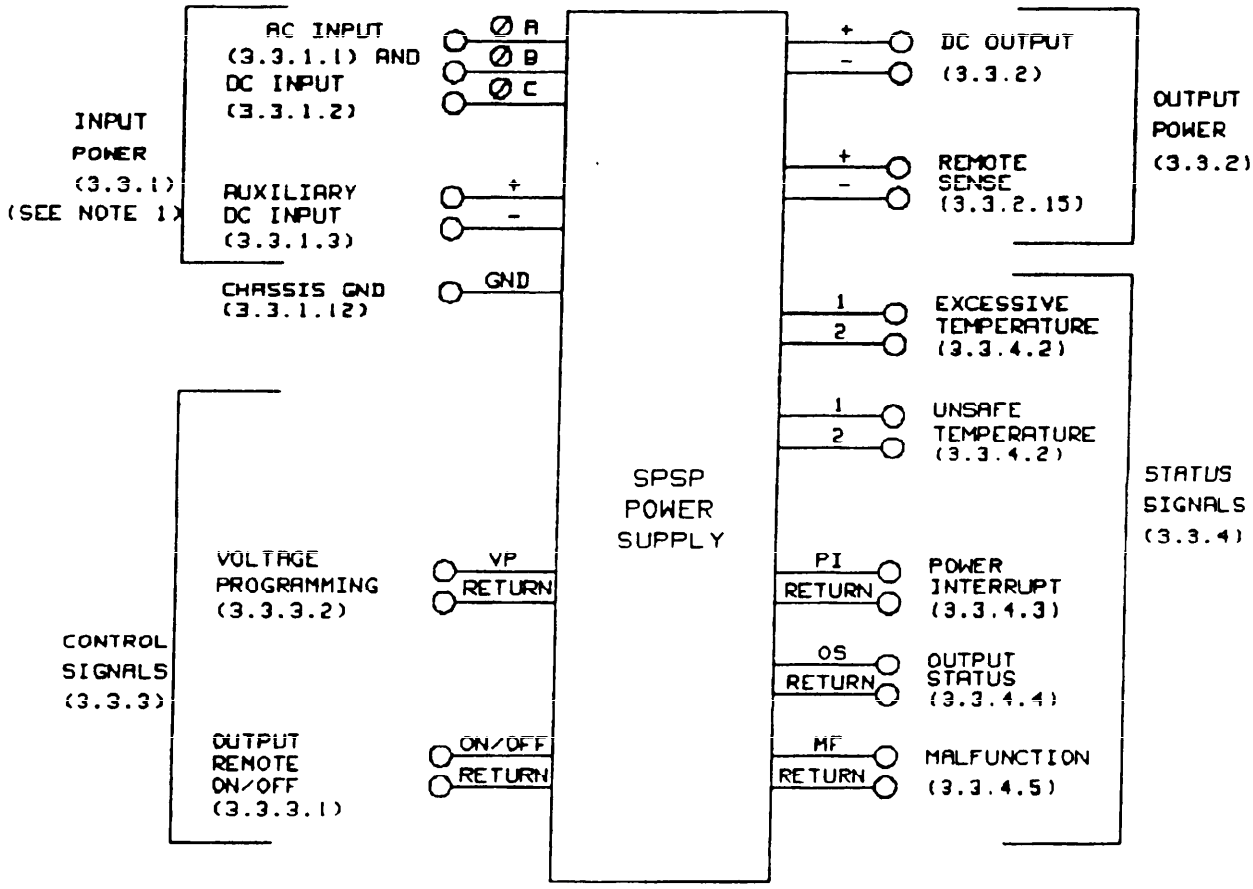
3.4.5 Modular construction.

3.4.5.1 Family S1. Power supplies shall be internally structured and functionally partitioned in a modular format (see 3.11.2). The selection of specific assembly design and packaging techniques shall reflect the reliability and maintainability requirements specified herein and in the associated detail specification.

3.4.5.1.1 Modules and modular construction. The terms "modules" and "modular construction" when applied to family S1 shall not be construed to mean Standard Electronic Modules (SEM) in accordance with MIL-STD-1389 (see 6.5.6).

3.4.5.2 Family S2. Unless otherwise specified in the associated detail specification, internal modular construction for family S2 is permitted but not required.

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NOTES:

1. Power supplies operating from DC only (in accordance with either 3.3.1.2 or 3.3.1.4) would require two inputs instead of the five inputs as shown above.

2. Unless contact limited, all functions except output power shall appear on the input connector.

FIGURE 2. Family S1 power supply electrical interfaces.

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3.4.6 Human engineering. Guidelines in applying human engineering design criteria and principles in the design of the power supplies shall be in accordance with MIL-STD-1472.

3.5 Environmental requirements.

3.5.1 Visible degradation. Power supplies shall not show evidence of visible degradation (see 4.7.1.2) as a result of being exposed to the environmental conditions specified herein. Deterioration, corrosion, or change in tolerance limits of any internal or external parts that could in any manner prevent the power supply from meeting operational service or maintenance requirements shall provide reason for the SPSP-QAA to consider the power supply as having failed to withstand the conditions of an environmental exposure. Leakage of electrolyte or impregnating compounds, discoloration of printed wiring boards, and so forth shall be considered damage and cause for rejection.

3.5.2 Operating environmental requirements. The power supplies shall withstand, without damage, the operating environmental requirements of table III (family S1) or table IV (family S2). Power supplies shall meet the electrical performance requirements specified herein and in the associated detail specification during and following exposure to the environmental tests.

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TABLE III. Operating environmental requirements (family S1).

Environment	Requirement	Test	Method and condition
Temperature: low	3.5.2.1a	4.7.3.2a	-10°C
high	3.5.2.1b	4.7.3.2b	60°C
transient	3.5.2.1c	4.7.3.2c	80°C
Vibration	3.5.2.2.1	4.7.3.3.1	MIL-STD-202, method 204, condition A
Mechanical shock	3.5.2.3	4.7.3.4	MIL-S-901, grade A, type A, class I
Long-term stability	3.5.2.4	4.7.3.5	1,000 hours
Radiation hardness ionizing dose rate	3.5.2.5	4.7.3.6.1	MIL-STD-883, method 1023
total ionizing dose	3.5.2.5	4.7.3.6.2	MIL-STD-883, method 1019
Dripproof	3.5.2.6	4.7.3.7	MIL-STD-108
Airborne sound	3.5.2.7	4.7.3.8	MIL-STD-740-1
Structureborne vibratory acceleration	3.5.2.7	4.7.3.8	MIL-STD-740-2

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TABLE IV. Operating environmental requirements (family S2).

Environment	Requirement	Test	Method and condition
Temperature: low	3.5.2.1a	4.7.3.2a	-10°C
high	3.5.2.1b	4.7.3.2b	60°C
transient	3.5.2.1c	4.7.3.2c	80°C
Vibration	3.5.2.2.2	4.7.3.3.2	MIL-STD-202, method 201, and method 214 condition I, letter D
Long-term stability	3.5.2.4	4.7.3.5	1,000 hours
Radiation hardness ionizing dose rate	3.5.2.5	4.7.3.6.1	MIL-STD-883, method 1023
total ionizing dose	3.5.2.5	4.7.3.6.2	MIL-STD-883, method 1019

3.5.2.1 Operating temperature. Power supplies shall withstand the operating temperature extremes as specified herein and under the conditions specified in 4.7.3.2.

(a) Low temperature. Power supplies shall operate continuously without damage and maintain specified performance at rated load when subjected to a temperature of minus 10°C. The power supplies shall be capable of a temperature stabilized start under rated load conditions and shall meet specified performance.

(b) High temperature. Power supplies shall operate continuously without damage and maintain specified performance at rated load when subjected to a temperature of 60°C.

(c) Transient thermal operation. Power supplies shall operate without damage and maintain high temperature end-of-life performance (see 3.5.4.2) at rated load when subjected to a temperature of 80°C for 1 hour. The parts derating (see 3.10.2) shall not apply while

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operating at this temperature; however, no part shall exceed its maximum rating. This represents conditions caused by system maintenance or loss of cooling medium.

3.5.2.2 Vibration.

3.5.2.2.1 Family S1. Power supplies shall operate without damage during vibration test (see 4.7.3.3.1) and shall not exhibit any resonances below 50 Hz.

3.5.2.2.2 Family S2. Power supplies shall operate without damage during simple harmonic motion and random vibration tests (see 4.7.3.3.2).

3.5.2.3 Mechanical shock (family S1). The power supplies shall operate without damage during the mechanical shock test (see 4.7.3.4).

3.5.2.4 Long-term stability. The output voltages of each power supply shall not change more than 0.2 percent when operated at 60°C for 1000 hours (see 4.7.3.5).

3.5.2.5 Radiation hardness. Power supplies shall be radiation hardened to the requirements herein and in 3.5.3.7. The power supplies shall meet post irradiation electrical requirements following operating exposure to an ionizing dose rate of $1X10^{9}$ rad (silicon (Si)) per second (20 to 100 nanoseconds (ns) pulse width) (see 4.7.3.6.1)and a total ionizing dose of $3X10^{3}$ rad Si (see 4.7.3.6.2). The power supply may shut down during exposure and require the input voltage to be taken to zero and then reapplied before resuming operation.

3.5.2.6 Dripproof (family S1). Power supplies shall be dripproof (see 4.7.3.7). Acceptance and approval of the power supply as meeting the dripproof requirement shall not be construed as assurance that the system or enclosure to which the power supply is mounted will be dripproof.

3.5.2.7 Airborne sound and structureborne vibratory acceleration (family S1). Unless otherwise specified in the associated detail specification, the power supplies shall meet airborne sound and structureborne vibratory acceleration acceptance limits for solidly mounted equipment (see 4.7.3.8). Airborne sound shall be in accordance with grade C of MIL-STD-740-1 except the limits shall be reduced by 28 dB for each octave band. Structureborne vibratory acceleration shall be in accordance with type III equipment of MIL-STD-740-2.

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3.5.3 Nonoperating environmental requirements. Power supplies shall withstand, without damage, the nonoperating environmental requirements of table V (family S1) or table VI (family S2). Power supplies shall meet the electrical performance requirements specified herein and in the associated detail specification following exposure to the environmental tests. Power supplies shall not be required to operate continuously during the nonoperating environmental tests; however, periodic operation of the power supply may be required by specific test procedures to measure electrical performance.

TABLE V. Nonoperating environmental requirements (family S1).

Environment	Requirement	Test	Method and condition
Humidity	3.5.3.1	4.7.3.9.1	MIL-STD-810, method 507, procedure III
Thermal shock	3.5.3.2	4.7.3.10.1	MIL-STD-202, method 107, condition A-3
Salt fog	3.5.3.4	4.7.3.12	MIL-STD-810, method 509, procedure I
Storage temperature	3.5.3.6	4.7.3.14	-55°C to 85°C
Radiation hardness neutron fluence	3.5.3.7	4.7.3.15	MIL-STD-883, method 1017
Solvents	3.5.3.8	4.7.3.16	MIL-STD-202, method 215

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TABLE VI. Nonoperating environmental requirements (family S2).

Environment	Requirement	Test	Method and condition
Humidity	3.5.3.1	4.7.3.9.2	MIL-STD-810, method 507, procedure III
Thermal shock	3.5.3.2	4.7.3.10.2	MIL-STD-202, method 107, condition A-3
Mechanical shock	3.5.3.3	4.7.3.11	MIL-STD-202, method 213, condition I
Salt fog	3.5.3.4	4.7.3.12	MIL-STD-810, method 509, procedure I
Durability	3.5.3.5	4.7.3.13	as specified
Storage temperature	3.5.3.6	4.7.3.14	-55°C to 85°C
Radiation hardness neutron fluence	3.5.3.7	4.7.3.15	MIL-STD-883, method 1017
Solvents	3.5.3.8	4.7.3.16	MIL-STD-202, method 215

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3.5.3.1 Humidity. The power supplies shall be capable of operational performance without degradation when exposed to the effects of up to 95 percent relative humidity, including conditions wherein condensation takes place in and on the power supplies (see 4.7.3.9). Energizing of the power supply within 1 hour following the exposure and subsequent electrical performance testing shall verify conformance.

3.5.3.2 Thermal shock. The power supplies shall withstand without physical damage or evidence of deterioration the thermal shock of minus 55°C to 85°C (see 4.7.3.10).

3.5.3.3 Mechanical shock (family S2). Power supplies shall withstand a shock of 100 gravities acceleration peak value, normal duration of 6 ms, sawtooth waveform, and a velocity change of 12.3 feet per second (see 4.7.3.11).

3.5.3.4 Salt fog. The power supplies shall withstand the effects of exposure to a salt-sea atmosphere (see 4.7.3.12). Failure mechanisms shall include pits, crack formations, or intergranular attacks that produce a concentrated attack that weakens a cross section. Surface corrosion shall not be evidence of failure.

3.5.3.5 Durability (family S2). Power supplies shall withstand a minimum of 500 cycles of insertion and extraction into a test fixture that represents a typical card cage rail assembly without loss of mechanical or electrical integrity (see 4.7.3.13). For formats C, D, and E, power supplies shall also withstand a minimum of 500 cycles of lateral displacement to simulate the use of thermal clamping devices. There shall be no exposure of nickel or copper underplating on the module connector contacts upon completion of the durability test.

3.5.3.6 Storage temperature. The power supplies shall not be damaged, nor shall operational performance be degraded when restored to the operating temperature range, after being exposed to the nonoperating temperature range of minus 55°C to 85°C (see 4.7.3.14).

3.5.3.7 Non-operating radiation hardness. The power supplies shall meet post irradiation electrical requirements following exposure to a neutron fluence of $2X10^{12}$ neutrons per centimeter squared (cm^2) (see 4.7.3.15).

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3.5.3.8 Solvents. Adhesive backed labels used by power supplies shall not peel off of the power supply surface during or as a result of the solvent test (see 4.7.3.16).

3.5.4 Power supply life.

3.5.4.1 Shelf life. The power supplies shall have a minimum nonoperating shelf life of 10 years without refurbishment, provided the storage temperature is 40°C or less for a minimum of 95 percent of the storage time. Storage at a temperature between 40 and 80°C for up to 5 percent of the time shall not effect the shelf life. The useful shelf life shall be extendable to at least 20 years if refurbishment is performed at the end of the initial 10 years.

3.5.4.2 End-of-life (EOL). Power supplies shall be designed in such a manner that component aging effect shall not degrade the operational limits by more than the amount specified in table VII. Parameters not specified in table VII shall not exceed the limits of the electrical performance characteristics specified herein or in the associated detail specification over the useful life of the power supplies.

TABLE VII. End-of-life.

Requirement paragraph	Parameter	Initial limit	Allowed aging	Total specified end-of-life
3.3.2.3	Total ripple and noise	1 percent 100 mV	0.5 percent 50 mV	1.5 percent 150 mV
3.3.2.3.1	Low frequency ripple/noise	0.5 percent 50 mV	0.1 percent 10 mV	0.6 percent 60 mV
3.3.2.3.2	High frequency ripple/noise	0.5 percent 50 mV	0.25 percent 25 mV	0.75 percent 75 mV
3.3.2.5	Static regulation	1 percent	0.5 percent	1.5 percent
3.3.2.13	Overcurrent protection	125 percent maximum	2 percent	127 percent maximum
3.3.2.14	Overvoltage protection	115 percent or 1.2 VDC	1 percent or 0.1 VDC	116 percent or 1.3 VDC
3.3.1.14 3.3.2.17 3.3.4.1	Isolation resistance	10 megohms minimum	10 times less	1 megohm minimum

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3.6 Electromagnetic requirements.

3.6.1 Electromagnetic interference (EMI). Each power supply type shall be designed with the output as both a positive and a negative supply to meet the requirements in MIL-STD-461 for class A4 and class A5. Each power supply type shall meet the applicable emission and susceptibility requirements of part 5, part 6, and section 5 of part 1 of MIL-STD-461. EMI testing to the above requirements (see 4.7.4) shall only be performed using the primary output polarity. Power supplies shall meet these requirements with all applicable types of input power.

3.6.1.1 EMI requirements. Unless otherwise specified in the associated detail specification, the power supply shall meet the following requirements:

(a) CE01 shall apply for families S1 and S2. This requirement is not applicable to the power supply output leads. The limits for CE01, AC leads, shall be in accordance with the associated detail specification. The limits for CE01, DC leads, shall be in accordance with MIL-STD-461.

(b) CE03 shall apply for families S1 and S2. When required, the use of an external EMI filter to meet the CE03 requirement shall be specified in the associated detail specification. CE03 is not applicable to the power supply output leads.

(c) CS01 shall apply for families S1 and S2.

(d) CS02 shall apply for families S1 and S2.

(e) CS11 (electromagnetic pulse protection (EMP)) shall apply for family S1. Power supplies shall operate without damage during the EMP test.

(f) RE01 shall apply for family S1.

(g) RE02 shall apply for family S1. This requirement is applicable to a maximum frequency of 1 gigahertz (GHz) for broadband emissions and 10 GHz for narrowband emissions. The RE02 requirement applies to both the surface with the power supply marking (installed configuration) and the mounting surface (connector side).

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(h) RS01 shall apply for families S1 and S2.

(i) RS02 shall apply for families S1 and S2.

(j) RS03 shall apply for families S1 and S2. The radiated field in part 5 of MIL-STD-461 for below deck metallic hulls shall be used.

3.7 Reliability.

3.7.1 Reliability requirement. Each power supply shall have a minimum predicted MTBF of 100,000 hours. A reliability prediction shall be performed in accordance with 4.7.5.1.

3.7.2 Environmental stress screening. Power supplies shall meet the environmental stress screening (also known as preconditioning, burn-in, manufacturing screening) in accordance with 4.7.5.2. Power supplies shall be 100 percent tested to the stress screening tests as specified in 4.7.5.2. There shall be no evidence of deterioration or physical damage after the stress screening.

3.8 Maintainability.

3.8.1 Maintenance. Unless otherwise specified, each power supply and its subassemblies shall be repairable.

3.8.2 Service and access. The power supplies shall be designed for maximum ease of maintenance (see 4.7.6). Parts requiring adjustment or replacement shall be readily accessible without major disassembly (see 3.10.6).

3.8.3 Test points. Each power supply shall have internally accessible test points to the extent necessary in field and depot level maintenance to allow isolation of the failed subassembly without removal of the power supply subassembly. Protection shall be provided in test point circuitry susceptible to damage caused by the grounding of test points.

3.8.4 Part protection. The power supplies shall be so designed that parts will not be susceptible to damage during servicing and maintenance of the power supplies.

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3.9 Parts, materials, and processes.

3.9.1 Selection. Parts, materials, and processes shall be selected as specified herein.

3.9.2 Standard parts. Maximum utilization shall be made of standard parts. Standard parts are those parts specified in:

(a) MIL-STD-454 or

(b) other standards and specifications, as specified herein, where approval of these parts prior to use is not required.

3.9.3 Nonstandard parts. Nonstandard parts shall not be used without approval in accordance with MIL-STD-2038.

3.9.4 Choice of parts and materials. Items shall be selected and used that have the broadest characteristics and of the greatest allowable tolerances that will fulfill the performance requirements of the power supply. Performance shall not be dependent on the selection of special values for individual parts.

3.9.5 Materials. Materials used in the manufacture of the power supplies shall conform to the requirements specified herein and in the materials paragraphs of MIL-STD-2036. Materials shall be certified in accordance with applicable specifications where required. When a material is not specified, a material shall be used which will enable the power supplies to satisfy the requirements specified herein. Acceptance or approval of a constituent material shall not be construed as an assurance of the acceptance of the finished product.

3.9.5.1 Encapsulation and embedment. The encapsulation and embedment of electronic parts and assemblies shall be in accordance with MIL-STD-454, requirement 47.

3.9.5.2 Moisture and fungus resistance. The power supplies shall be designed so that the materials used are moisture resistant and non-nutrient for fungus. Fungus nutrient material may be used in hermetic assemblies and other accepted and qualified uses. Other fungus nutrient materials, which are deemed necessary and are approved by the command or agency concerned, shall be treated by a method which will render the resulting exposed surface fungus resistant. For fungus-inert material, see requirement 4 of MIL-STD-454.

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3.9.5.3 Flammability. Materials used in fabrication of any power supply shall be nonburning or self-extinguishing in accordance with MIL-STD-454, requirement 3. Materials shall be self-extinguishing within 5 seconds after removal of the flame if tested in accordance with MIL-STD-202, method 111.

3.9.5.4 Use of toxic material. Materials which are capable of producing dangerous gases or having other harmful toxic effects over the temperature range of minus 62 to 85°C, including fire, shall not be used. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the SPSP-QAA to the appropriate departmental medical service who will act as an advisor to the SPSP-QAA. The power supply manufacturer shall furnish to the SPSP-QAA the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

3.9.5.5 Dissimilar metals. The use of dissimilar metals in direct contact with one another shall be avoided where possible. When dissimilar metals are used, protection against galvanic corrosion shall be applied in accordance with MIL-STD-889. The preceding does not apply where dissimilar metals are joined to provide information internal to the power supply (for example, temperature sensing).

3.9.5.6 Processes. Materials used in the power supplies shall be corrosion resistant types or shall be processed to resist corrosion. Gold, nickel, rhodium, tin, lead-tin alloys, or sufficiently thick plating of these metals are satisfactory without additional protection or treatment other than buffing or cleaning. Other materials shall have protective finishes in accordance with the processes in MIL-STD-2036 (except as modified herein). The protective treatment of surfaces shall enable the power supply to meet the requirements specified herein. Acceptance or approval of a protective treatment shall not be construed as an assurance of acceptance of the finished product.

3.9.5.7. Prohibited materials. Prohibited materials in accordance with MIL-STD-2036 shall not be used.

3.9.5.8 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the power supplies shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent

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practicable without jeopardizing the intended use. The term recovered materials means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.10 Electrical design and construction.

3.10.1 Component selection. Electronic components and hardware used in power supplies shall have a demonstrated quality level and environmental performance equivalent to or better than that of available military parts. Nonhermetically sealed packaged relays having hermetically sealed equivalents shall not be used.

3.10.1.1 Germanium semiconductors. Germanium semiconductors shall not be used.

3.10.1.2 Discrete semiconductors. Discrete semiconductors shall be in accordance with MIL-S-19500 and shall be selected according to the following priority list. Devices listed in (b), (c), and (d) shall be approved by the SPSP-QAA prior to use.

(a) MIL-S-19500 JANTX devices listed in MIL-STD-701.

(b) Other MIL-S-19500 JANTX devices.

(c) Devices being considered for a MIL-S-19500 JANTX detail specification. Devices shall be equal to or better than MIL-S-19500 JANTX devices.

(d) Other devices. Devices shall be equal to or better than MIL-S-19500 JANTX devices.

3.10.1.3 Integrated circuits. Integrated circuits shall be in accordance with the following requirements.

3.10.1.3.1 Quality requirements. Integrated circuits shall be in accordance with the requirements of MIL-M-38510, class B. The power supply manufacturer shall use MIL-M-38510 JAN QPL devices when available or acquire other devices in accordance with the "Provisions for use of MIL-STD-883 in conjunction with compliant non-JAN devices" paragraph of MIL-STD-883. Equivalent

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specifications shall be submitted to the SPSP-QAA for approval prior to initial qualification. Equivalent specifications, excluding DESC standard military drawings (SMD), shall include:

(a) The screening shall be to MIL-STD-883, method 5004, class B.

(b) Quality conformance shall be demonstrated in accordance with MIL-STD-883, method 5005, groups A, B, C, D, and E (if applicable), class B.

(c) Generic data is acceptable for demonstrating quality conformance in accordance with MIL-STD-883, method 5005, groups C and D, class B. A generic family shall be electrically and structurally similar integrated circuits. They are designed to perform the same type of basic circuit function using the same basic circuit element configuration and differ only in the number of identically specified circuits which they contain. They are designed for the same supply, bias, and signal voltages and for input-output compatibility with each other under an established set of loading rules. They are enclosed in packages of the same construction and outline, differing only in the number of active external package leads included or used and made from the same materials by use of the same processes.

3.10.1.3.2 Selection requirements. Integrated circuits shall be in accordance with the requirements of MIL-M-38510 and shall be selected according to the following priority list. Devices listed in (b), (c), (d) and (e) shall be approved by the SPSP-QAA prior to use.

(a) MIL-M-38510 JAN microcircuits listed in the table of preferred devices in MIL-STD-1562.

(b) Other MIL-M-38510 microcircuits not listed in the tables of logistics or continuous replacements, inactive or suspended military activity, or not recommended under any circumstances in MIL-STD-1562.

(c) Other microcircuits listed in the table of potential standardization candidates in MIL-STD-1562 as preferred for new designs.

(d) DESC SMD microcircuits not listed in the table of logistics or continuous replacements, inactive or suspended military activity, or not recommended under any circumstances in MIL-STD-1562.

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windings.

(e) Other microcircuits (see 3.10.1.5)

3.10.1.4 Passive components. Passive components shall be selected according to the following priority list. Devices listed in (c), (d), (f), and (g) shall be approved by the SPSP-QAA prior to use.

(a) Established reliability (ER) specification parts (minimum level R if multiple sources exist) listed in MIL-STD-454, requirements 2, 33, and 57 (if multiple sources of level R parts do not exist, level P parts may be used to achieve multiple sources). Passive components listed in MIL-STD-454, requirements 10, 50, and 58.

(b) ER parts (minimum level M if required to achieve multiple sourcing) listed in MIL-STD-454, requirements 2, 33 and 57.

(c) Other ER parts.

(d) Conventional military specification parts.

(e) ER parts that use the Weibull failure rate prediction method, as specified in MIL-C-39003, shall require a B minimum failure rate level.

(f) Other passive components.

(g) Carbon composition resistors may be used in special applications (high voltage or high current surge) when approved as nonstandard parts. Specifications for these nonstandard parts (including MIL-R-39008 carbon composition resistors) shall include appropriate parametric and quality requirements.

3.10.1.5 Other microcircuits. All other microcircuits shall include the following information in the non-standard part specification.

(a) Device nomenclature, marking, configuration, group A electrical requirements for the full military temperature range which will insure form, fit, function, and interchangeability.

(b) The required sampling and screening plans which will allow the procured devices to be in accordance with the "Provisions for the use of MIL-STD-883 in

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conjunction with compliant non-JAN devices" paragraph of MIL-STD-883. Hybrids shall meet the requirements of MIL-H-38534 for a class B product assurance level.

(c) Hybrid microcircuit suppliers shall be certified to MIL-STD-1772 or in the process of receiving MIL-STD-1772 certification and will be audited by SPSP-QAA.

3.10.1.6 Magnetic components. Transformers and inductors shall be in accordance with MIL-STD-454, requirement 14, grade 4 or 5, class S. Magnetic components shall be 100 percent subjected to a thermal shock screen in accordance with MIL-STD-202, method 107, test condition B, except the lowest temperature shall be minus 55 plus 0, minus 5°C instead of minus 65°C.

3.10.2 Parts derating. Electrical parts shall not be stressed more than the percentage of their rated value specified in table VIII, in the environments specified in 3.5, (except for 80°C transient thermal, 3.5.2.1 (c)), at nominal input voltage and at maximum rated output power. Parts shall not be stressed more than their maximum rated value at maximum input voltage and maximum rated output power. Electrical parts which are stressed more than the indicated percentages shall be approved by the SPSP-QAA.

3.10.3 Internal wiring and cabling. The use of internal wiring and cabling shall be minimized to the greatest extent possible. When internal wiring and cabling is required, it shall be in accordance with MIL-STD-454, requirements 69 and 71.

3.10.4 Printed wiring and printed wiring assemblies. Printed wiring and printed wiring assemblies shall conform to the following requirements. Equivalent materials, processes, and requirements shall be utilized only when approved by the SPSP-QAA. These equivalent materials, processes, and requirements shall be documented and forwarded (see 6.3.1) to the SPSP-QAA for review and approval.

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TABLE VIII. Parts derating.

Part type	Derating parameter	Percent of rated (or value indicated)
Resistors		
Carbon composition	Power/voltage	50/80
Film high stability	Power/voltage	50/80
Wirewound accurate	Power/voltage	50/80
Wirewound chassis mounting	Power/voltage	50/80
Variable wirewound	Power/voltage	50/80
Variable non-wirewound	Power/voltage	50/80
Thermistor	Power/voltage	50/80
Tantalum nitride chip	Power/voltage	50/80
Capacitors		
All	Ripple voltage	50
	Ripple current	70
Ceramic	Voltage	50
Glass	Voltage	50
Mica	Voltage	60 dipped 40 molded
Plastic	Voltage	50
Mylar	Voltage	50
Paper	Voltage	50
Tantalum solid	Voltage	50
(100 percent current surge test required)	Reverse voltage	2 percent of rated DC up to 85°C
	Reverse voltage	1 percent of rated DC 85°C to 125°C
Tantalum wet	Voltage	60
	Reverse voltage	0
Tantalum foil	Voltage	50
Aluminum electrolytic	Voltage	70
(Resin end-seal protection and 99.96 percent A1 foil purity)	Reverse voltage	0

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TABLE VIII. Parts derating - Continued.

Part type	Derating parameter	Percent of rated (or value indicated)
Switches	Contact current	50 resistive 25 inductive
	Voltage	50
Connectors	Contact current	50
	Working voltage	25 of dielectric withstanding voltage
Magnetic devices	Current density	2.0 mA per circular mil
	Current density (foil conductors)	4.0 mA per square mil
	Voltage (continuous)	60
	Voltage (surge)	90
	Hot spot temperature (operating)	30°C below insulation rating (see 3.12.1)
	Temperature rise	40°C maximum (see 3.12.1)
	Insulation breakdown voltage	25
RF coils	Current	50
Relays	Contact power	25
	Contact current (continuous)	50 resistive 25 inductive
	Contact current (surge)	80
	Coil energize/ dropout voltage	vendor's nominal rating
	Vibration	75 3/

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normal duration of 9 ms, sawtooth waveform, and a velocity change

TABLE VIII. Parts derating - Continued.

Part type	Derating parameter	Percent of rated (or value indicated)
Transistors		
Bipolar, power, and FET	Power 1/	50
	Forward current	50
	Reverse voltage (steady state)	75 across any junction
	Transient peak	80 across any junction
	Junction temperature	105°C maximum (see 3.12.1)
Diodes		
Switching, general purpose rectifier, and thyristor	Current (surge)	70
	Current (continuous)	50
	Peak inverse voltage	65
Zener	Power	50
All	Junction temperature	105°C maximum (see 3.12.1)
Microcircuits		
All linear	Current	70
	Voltage (signal)	75
	Voltage (surge)	80
	Voltage reverse junction (signal)	65
	Voltage reverse junction (surge)	85

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TABLE VIII. Parts derating - Continued.

Part type	Derating parameter	Percent of rated (or value indicated)
Digital 2/	Supply voltage	Hold to vendor's nominal rating
	Fanout	80
All	Junction temperature	105°C maximum (see 3.12.1)

1/ The maximum ratings as determined by the safe operating area (SOA) curves for power switching transistors shall not be exceeded.

2/ Many families of digital microcircuits exhibit additional characteristics that may require derating (for example, toggle frequency, hold times).

3/ This derating includes any contribution by the device mounting.

3.10.4.1 Rigid printed wiring boards. The design of rigid printed wiring boards shall be in accordance with MIL-STD-275. The quality of rigid printed wiring boards shall be in accordance with MIL-P-55110.

3.10.4.2 Printed wiring assemblies. The design and quality of printed wiring assemblies shall be in accordance with MIL-STD-2000.

3.10.4.3 Flexible printed wiring and assemblies. The design and quality of flexible printed wiring and assemblies shall be in accordance with MIL-P-50884, type 1, class B.

3.10.5 Thick film multilayer interconnect boards (MIBs) and assemblies. All thick film MIBs and assemblies shall conform to the following requirements. Equivalent materials, processes, and requirements shall be utilized only when approved by the SPSP-QAA. These equivalent materials, processes, and requirements shall be documented and forwarded to the SPSP-QAA for review and approval.

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3.10.5.1 Thick film MIBs. The design of thick film MIBs shall be in accordance with MIL-STD-1389, appendix F. The quality of thick film MIBs shall be in accordance with the requirements of MIL-M-28787, appendix B (see 6.5.6).

3.10.5.2 Thick film MIB assemblies. The design and quality of thick film MIB assemblies shall be in accordance with MIL-STD-2000.

3.10.6 Internal adjustments. No externally accessible adjustments shall be permitted. Any output adjustment devices (other than pin programming) shall be internal to the power supply and shall be designated as factory, depot, or shop adjust only. If adjustment entails a moving part (for example, a potentiometer), the part shall be secured after adjustment to prevent movement during vibration and shock.

3.10.6.1 Range of adjustments. The range of all adjustments internal to the power supply shall be such that no damage can be induced as a result of the adjustment being set anywhere in its range.

3.10.7 Failure limitation. The design shall limit failures to the power supply in which the fault occurs; that is, a failure in one power supply shall not induce a subsequent failure outside of the failed power supply. The design shall limit failure propagation within the power supply to the greatest extent possible.

3.10.8 Powered socket. The power supplies shall not be removed or inserted into a powered socket. Marking on the power supply shall contain this precautionary information.

3.10.9 Design of the power supply for stability. The power supply shall be designed to maximize the stability of the power supply when operating with shipboard power source impedances (instability, if it occurs, is caused by the source impedance being higher than the power supply negative input impedance and forming a negative resistance oscillator). To achieve this stability, the design shall utilize features such as damping to minimize the gain or peaking at resonance of its input filter network and shall incorporate self turn-off under low input voltage conditions (see 3.3.1.7). The manufacturer shall measure the power supply input impedance over the input voltage range from maximum down to power supply turn-off and over a frequency range of 50 Hz to 100 kHz or up to the voltage feedback loop gain cross-over frequency (whichever is greater) to evaluate the design. The associated detail specification shall set limits for the input

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impedance values. An accurate computer model of the input network will be generated by the manufacturer. The measured data, computer model, and curves obtained from this computer model shall be submitted to the SPSP-QAA for evaluation. The associated detail specification shall define the criteria for an acceptable model. The computer model will be made available to system manufacturers utilizing the SPSP power supplies.

3.11 Mechanical design and construction.

3.11.1 General design features. The power supplies shall be designed such that their physical characteristics enhance their ability to meet the electrical performance requirements (see 3.3) when exposed to the environments specified in 3.5 and 3.6. The designs shall incorporate features which will provide for operational use, repair, handling, and storage without the use of special tools, handling equipment or techniques.

3.11.2 Internal modularity.

3.11.2.1 Connectorized. For family S1 and where applicable, for family S2, the internal subassemblies of the power supply shall be configured such that all input and output terminations are made through connectors or solderless terminals. Keying shall be provided such that each subassembly type is uniquely keyed. Connectors used on internal subassemblies shall withstand 100 insertions and extractions without degradation.

3.11.2.2 Modular interchangeability. Power supply types shall be designed such that maximum commonality of internal subassemblies is achieved among power supply types. Internal subassemblies of a given type shall be mechanically and electrically interchangeable regardless of the power supply in which they are used when operated within the required power supply design limits.

3.11.3 Power supply interchangeability. Power supplies of a given key code shall be mechanically and electrically interchangeable regardless of the system in which they are used when operated within the required power supply design limits.

3.11.4 Conformal coating. The conformal coating shall be in accordance with MIL-STD-2000 and shall be a continuous, homogeneous, fully cured material that covers all components, leads (except connector contacts' leads), and circuitry, except grounding surfaces. The coating thickness may vary with the irregularity of the printed board assembly. For types UR and SR, the coating thickness is allowed to be 0.010 inch, maximum.

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3.11.5 External connections. The method of external connection to the power supplies shall be as specified herein (see 3.4.2) and in the associated detail specification. When multiple connector contacts are specified for an interface function (such as an output) for current capacity, redundancy, and so forth, all multiple contacts for each function shall be used to interface the power supply to external sources and loads.

3.11.6 Mounting.

3.11.6.1 Family S1. The power supplies shall utilize corrosion resistant steel threaded fasteners in accordance with requirement 12 of MIL-STD-454 for socket head cap screws. The fasteners shall not require treatment other than passivation to meet the requirements specified herein. Mounting fasteners shall be accessible from the side of the power supply opposite the mounting surface (see figure 1). The size, location, and torque requirements of fasteners shall be in accordance with appendix A and the associated detail specification. When required by power supply design to maintain tension in mounting fasteners during environmental stresses, locking hardware shall be integral to the power supply.

3.11.6.2 Family S2. Mounting shall be in accordance with appendices A through E of MIL-STD-1389.

3.11.7 Finishes and protective treatments.

3.11.7.1 External surfaces (family S1). External aluminum alloy surfaces of the power supply, except the mounting surface, shall be anodized in accordance with MIL-A-8625 (type optional) or chromate conversion coated in accordance with MIL-C-5541 (class optional). External chromate conversion coated surfaces and steel surfaces of the power supply, except the mounting surface, shall be painted. Painting of anodized surfaces is optional. The finished color of all external surfaces, except the mounting surface, shall be semigloss light gray in accordance with color chip 26307 of FED-STD-595. The surface finish of the power supply shall be free of any imperfections that have a detrimental effect upon the performance of the unit.

3.11.7.1.1 Mounting surface. Unless otherwise specified in the associated detail specification, the mounting surface (see figure 1) shall be aluminum alloy anodized in accordance with MIL-A-8625, type II, class 1 or 2, or type III. Type II class 1 finishes shall be coated with dichromate sealer. Other sealants shall be approved by SPSP-QAA prior to use. The color of a type

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II class 2 finish shall be optional except that it shall not be a clear finish. The finish shall not exceed 32 microinches RMS after anodization. The mounting surface shall be flat within 0.0005 inch per inch and the overall flatness shall not exceed 0.005 inch.

3.11.7.2 Frame surface (family S2). Power supply frame surface treatments and frame surface finish shall be in accordance with appendices A through E of MIL-STD-1389.

3.11.8 Sealing device.

3.11.8.1 Family S1. A sealing device (gasket, O-ring, and so forth) shall be provided on the mounting surface of the power supply in accordance with appendix A. The sealing device shall not degrade the performance of the power supply. The thermal interface between the cooling plate and the power supply mounting surface shall meet the requirements specified in 3.12 with a sealing device installed on the power supply. The sealing device shall withstand all environments and conditions specified herein. All gasketing, including mounting fastener and O-rings, shall withstand a minimum of 25 installations and removals and still maintain seal integrity during the dripproof test herein.

3.11.8.2 Family S2. A sealing device is not applicable to family S2.

3.11.9 Mechanical configuration requirements (family S1). The mechanical configuration, including keying, shall be in accordance with appendix A and the associated detail specification.

3.11.10 Mechanical configuration requirements (family S2). The mechanical configuration for power supplies is in accordance with appendices A through E of MIL-STD-1389. (When referring to MIL-STD-1389, replace "module" with "power supply"). Power supplies shall meet the requirements specified without mechanical or electrical degradation.

3.11.10.1 Depth (family S2). Unless otherwise specified in the associated detail specification, the only parts of the power supply that extend below the interface plane are the keying pins, contacts, and pin shields. When specified in the associated detail specification, the body of the power supply on multiple increment power supplies is permitted to extend below the interface plane within the limits specified in appendices A through E of MIL-STD-1389.

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3.11.10.2 Rib structure (family S2). The power supply ribs shall perform the following functions:

(a) Alignment during insertion or extraction.

(b) Retention.

(c) Cooling.

The rib configuration is shown in appendices A through E of MIL-STD-1389.

3.11.10.2.1 Rib strength. Individual power supply ribs shall withstand a torque of 10 inch-pounds (1.1 newton-meters) minimum maintained for 10 to 15 seconds (see 4.7.7.1).

3.11.10.3 Pin shields (family S2). Power supplies shall be provided with shields to protect the contacts. Details of the pin shields are specified in appendices A through E of MIL-STD-1389. Power supplies are required to have pin shields adjacent to each outside row of power supply contacts. Pin shields shall be of a nonconducting material, or if of a conducting material, the outside surface of the shield shall be treated in a manner that will prevent conduction into the base conducting material.

3.11.10.3.1 Pin shield retention. The pin shield shall withstand, without visible separation from its base, a force of 4 pounds (18 newtons) minimum (see 4.7.7.2). The requirement shall be met after exposure to all manufacturing process temperatures including preconditioning.

3.11.10.4 Fin and header structure (family S2). The power supply fin for formats A and B and the header structure for formats C, D, and E shall be as specified in appendices A through E of MIL-STD-1389.

3.11.10.4.1 Power supply torque. Format A and B power supplies shall withstand a 6 inch-pound (0.68 newton-meter) minimum torque and formats C, D, and E shall withstand a 25 inch-pound minimum torque (2.83 newton-meters) (see 4.7.7.3).

3.11.10.4.2 Header torque. Format C, D and E power supplies shall withstand a 10 inch-pound (1.1 newton-meter) header torque without detrimental effect to the mechanical or electrical properties of the power supply (see 4.7.7.4).

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3.11.10.4.3 Fin/header cantilever load. Power supplies shall withstand a force of 2 pounds (9 newtons) minimum for formats A and B, and 5 pounds (22 newtons) for formats C, D and E (see 4.7.7.5).

3.11.10.5 Connector requirements (family S2). Unless otherwise specified in the associated detail specification, the power supply connector shall be as specified in appendices A through E of MIL-STD-1389 for the applicable format. Multiple increment modules may increase contact row quantities with each row of contacts complete.

3.11.10.5.1 Connector integrity, formats A and B. Each assembled format A and B power supply connector shall withstand a minimum axial force along the contact pin length in either direction equal to the product of 12 ounces (3.3 newtons) minimum multiplied by the number of contacts without any visible separation or bending (see 4.7.7.6). (For example, 12 ounces (3.3 newtons) minimum multiplied by 40 contacts equals 480 ounces (133 newtons). The total computed force shall be applied simultaneously to all power supply connector contacts simulating module insertion and extraction.)

3.11.10.5.2 Connector integrity, formats C, D, and E. Each assembled format C, D or E power supply connector shall withstand, without damage or visible separation, a minimum axial force normal to the interface plane equal to 100 pounds (445 newtons) on insertion and 4 ounces (1.1 newtons) per contact on extraction (see 4.7.7.6). The total computed force shall be applied simultaneously to all power supply connector contacts simulating module insertion and extraction.

(a) On insertion, a 100 plus 5, minus 0 pound (445 plus 22.3, minus 0 newton) force shall be uniformly applied at two locations on the top surface of the header. These two locations are centered on the header width and are located 0.75 inch (19.1 mm) to 1.25 inches (31.8 mm) from both ends of the header. This force shall be uniformly applied by two plates (0.5 inch (12.7 mm) square) forcing the power supply connector interface plane against a plate with clearance provisions for the rows of contacts.

(b) On extraction, a 4.0 plus 0, minus 0.1 ounce (1.1 plus 0, minus 0.03 newton) force per contact shall be applied at the two extractor holes. A suitable means of clamping the body of the power supply connector shall be employed to prevent extraction of the power

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surface with the power supply marking (installed

supply from the test fixture. The force shall be obtained in 2 to 10 seconds and maintained for 10 to 15 seconds.

3.11.10.5.3 Contacts. The number of contacts on the power supply connector and their location shall be as specified in the associated detail specification or in appendices A through E of MIL-STD-1389 for the applicable format. The contact configuration is controlled, (see figure 3), only on that part of the contact protruding from the module connector surface (the interface plane).

3.11.10.5.4 Connector contact integrity. Each contact, as mounted in the connector, shall withstand an axial force of 20 ounces (5.6 newtons) minimum applied in 2 to 10 seconds along the length of the contact blade in either direction and maintained for 10 to 15 seconds (see 4.7.7.7).

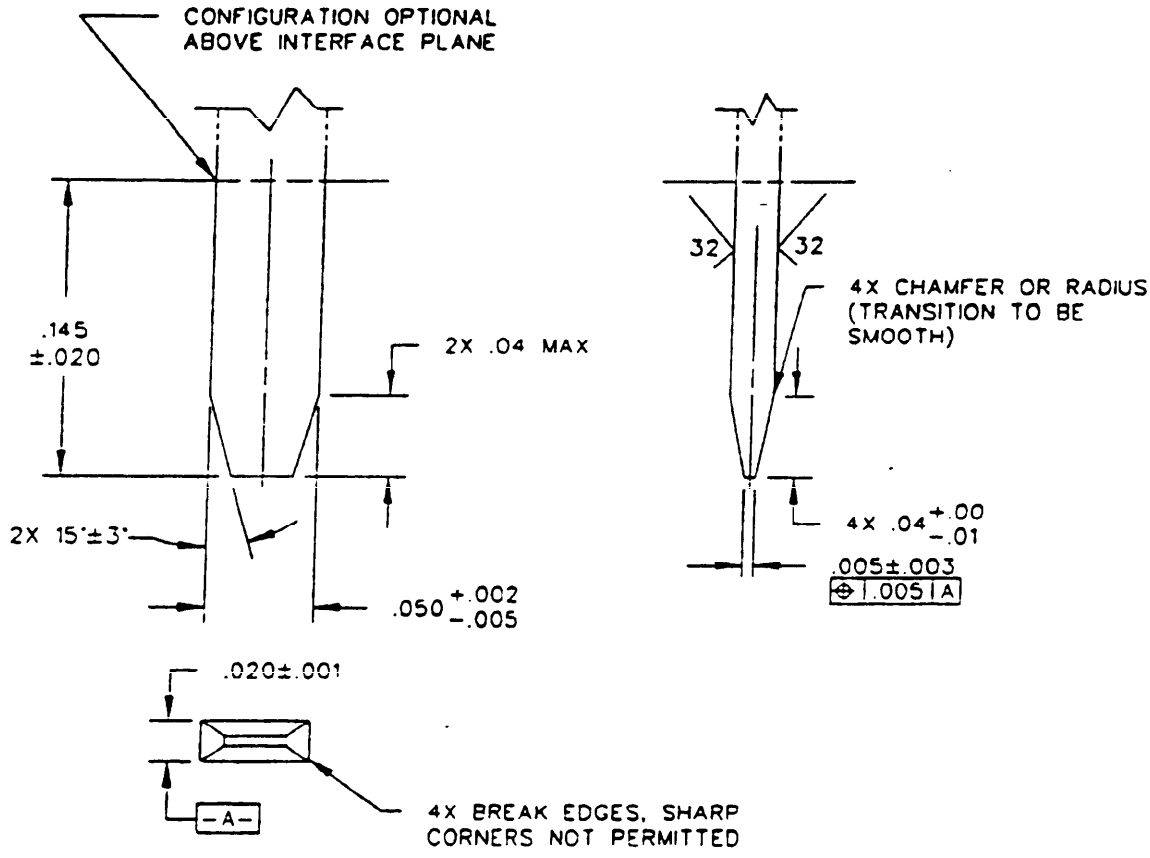
3.11.10.5.5 Contact material. The material used for the power supply contacts shall be a solid metal selected according to the manufacturing process used. Approved materials are those specified in MIL-C-28754.

3.11.10.5.6 Contact plating. The contacts shall be gold plated in accordance with MIL-G-45204, type II, grade C, class 1. Contacts shall have nickel underplating of 0.00005 inch (0.00127 mm) minimum in accordance with QQ-N-290, class 1.

3.11.10.6 Keying (family S2). Each power supply is assigned an alpha or alphanumeric key code. The first letter indicates the style and angular position of the keying pin in the alpha keying pin location and the last letter designates the style and angular position of the keying pin in the beta location.

3.11.10.6.1 Keying pin orientation. Keying pins shall be orientated to agree with the basic angle specified for the power supply by the code letters on figures 4, 5, or 6. The code letters indicate the only possible first and last letters, in that order, of the power supply key code for the combination of keying pin styles indicated in that particular figure. Code letter combinations other than those shown entirely within figures 4, 5, or 6 are not allowed. Power supply keying pin styles and orientations shall be taken entirely from either figure 4, figure 5, or figure 6 - no other combinations (such as mixing an alpha orientation from figure 4 and a beta orientation from figure 5) are allowed. Figure 7 illustrates an example of power supply keying pin configuration and specifies the tolerance for angular positioning.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.025	0.005	0.127	0.020	0.508	0.050	1.270
0.002	0.051	0.01	0.25	0.04	1.02	0.145	3.683
0.003	0.076						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Family S2 blade contact.

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BETA / LOCATION / ALPHA	A,J 0°	B,K 45°	C,L 90°	D,M 135°	E,N 180°	F,P 225°	G,Q 270°	H,R 315°
A,J 0°	A-A A-J J-A J-J	A-B A-K J-B J-K	A-C A-L J-C J-L	A-D A-M J-D J-M	A-E A-N J-E J-N	A-F A-P J-F J-P	A-G A-Q J-G J-Q	A-H A-R J-H J-R
B,K 45°	B-A B-J K-A K-J	B-B B-K K-B K-K	B-C B-L K-C K-L	B-D B-M K-D K-M	B-E B-N K-E K-N	B-F B-P K-F K-P	B-G B-Q K-G K-Q	B-H B-R K-H K-R
C,L 90°	C-A C-J L-A L-J	C-B C-K L-B L-K	C-C C-L L-C L-L	C-D C-M L-D L-M	C-E C-N L-E L-N	C-F C-P L-F L-P	C-G C-Q L-G L-Q	C-H C-R L-H L-R
D,M 135°	D-A D-J M-A M-J	D-B D-K M-B M-K	D-C D-L M-C M-L	D-D D-M M-D M-M	D-E D-N M-E M-N	D-F D-P M-F M-P	D-G D-Q M-G M-Q	D-H D-R M-H M-R
E,N 180°	E-A E-J N-A N-J	E-B E-K N-B N-K	E-C E-L N-C N-L	E-D E-M N-D N-M	E-E E-N N-E N-N	E-F E-P N-F N-P	E-G E-Q N-G N-Q	E-H E-R N-H N-R
F,P 225°	F-A F-J P-A P-J	F-B F-K P-B P-K	F-C F-L P-C P-L	F-D F-M P-D P-M	F-E F-N P-E P-N	F-F F-P P-F P-P	F-G F-Q P-G P-Q	F-H F-R P-H P-R
G,Q 270°	G-A G-J Q-A Q-J	G-B G-K Q-B Q-K	G-C G-L Q-C Q-L	G-D G-M Q-D Q-M	G-E G-N Q-E Q-N	G-F G-P Q-F Q-P	G-G G-Q Q-G Q-Q	G-H G-R Q-H Q-R
H,R 315°	H-A H-J R-A R-J	H-B H-K R-B R-K	H-C H-L R-C R-L	H-D H-M R-D R-M	H-E H-N R-E R-N	H-F H-P R-F R-P	H-G H-Q R-G R-Q	H-H H-R R-H R-R

FIGURE 4. Family S2 style 1/2 keying chart (viewing connector as shown on figure 7).

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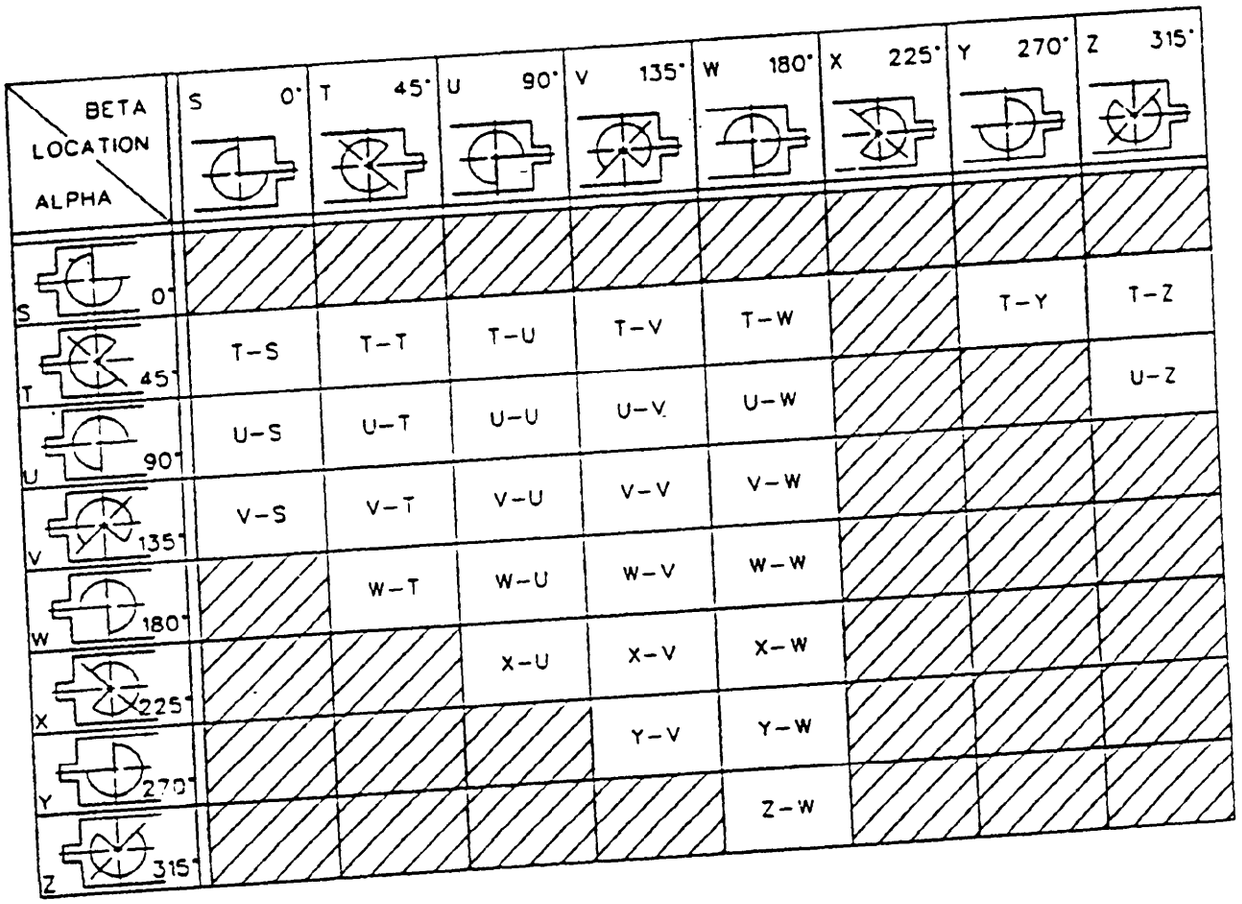


FIGURE 5. Family S2 style 1/1 keying chart (viewing connector as shown on figure 7).

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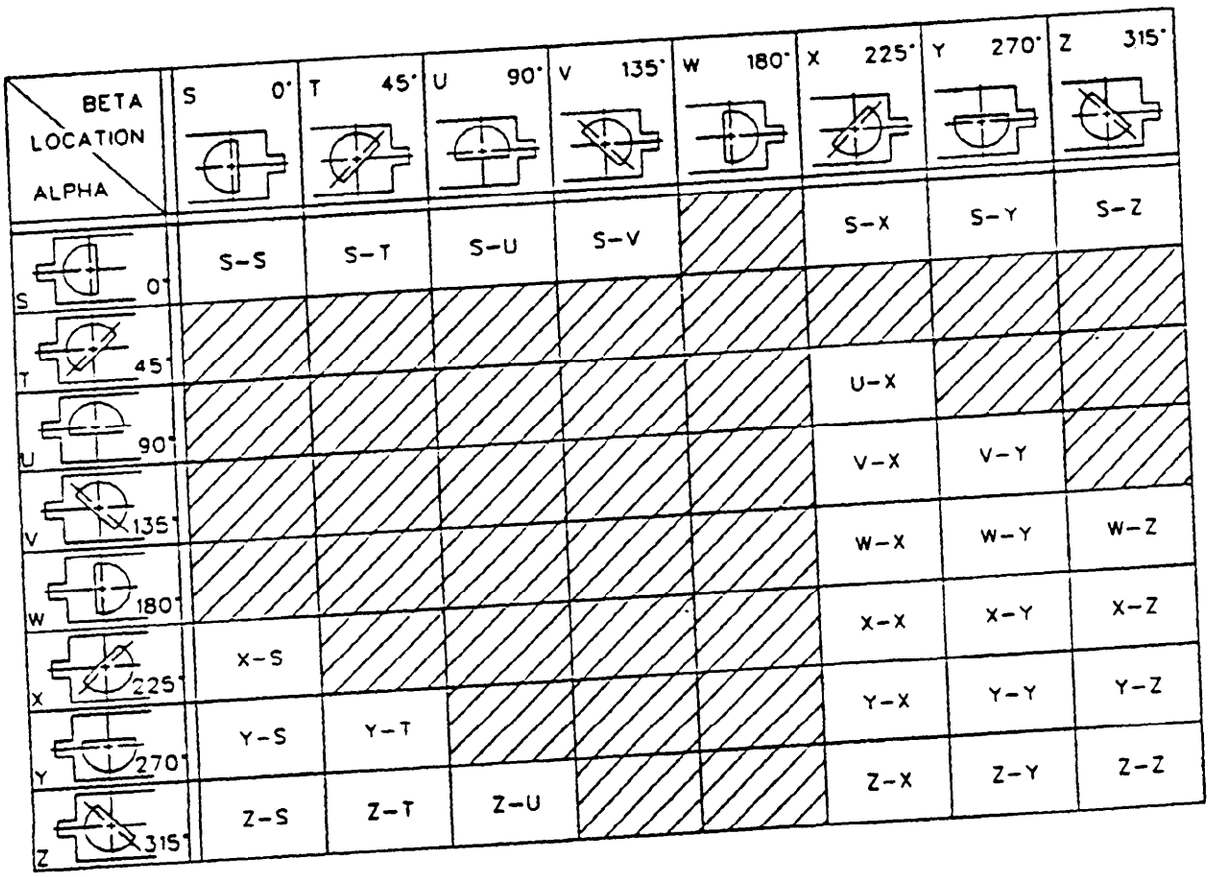
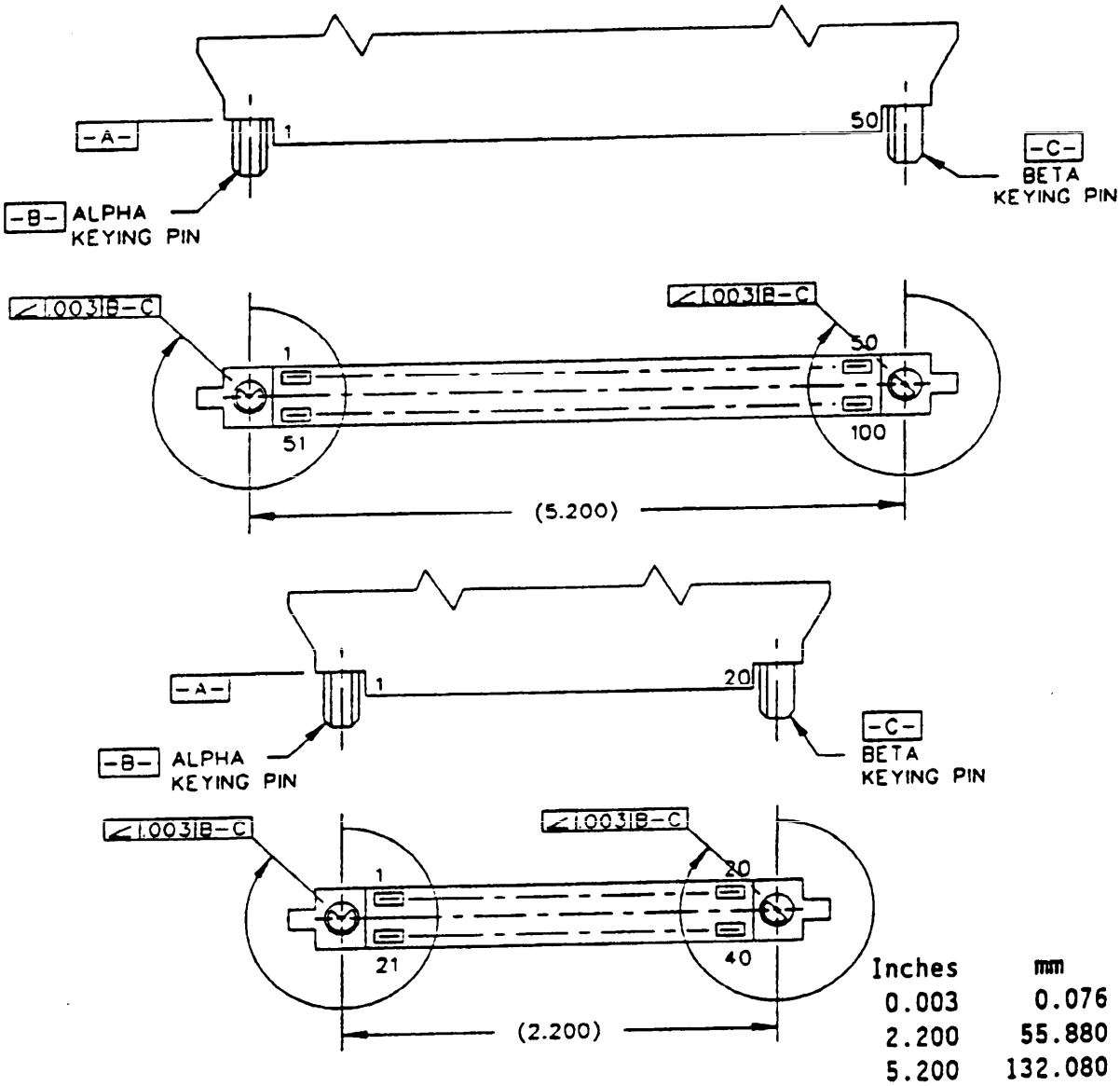


FIGURE 6. Family S2 style 2/2 keying chart (viewing connector as shown on figure 7).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 7. Family S2 keying pin orientation.

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3.11.10.6.2 Keying pin locations. There are two keying pin locations on each power supply, one designated alpha and one designated beta. The alpha and beta keying pin locations are near the lowest and highest numbered connector contacts in the first row, respectively, as illustrated on figure 7.

3.11.10.6.3 Keying pin styles. Keying pin styles shall be as specified on figure 8.

3.11.10.6.4 Keying pin sets. Only the keying pin styles and keying pin locations specified in table IX are permitted.

TABLE IX. Keying pin styles and locations.

Location		Style of Combination	Notes
Alpha (a)	Beta (b)		
Style 1	Style 2	Style 1/2	See figure 4
Style 1	Style 1	Style 1/1	See figure 5
Style 2	Style 2	Style 2/2	See figure 6

3.11.10.6.5 Keying pin integrity requirement. When installed in the module, the keying pins shall meet the following integrity requirements.

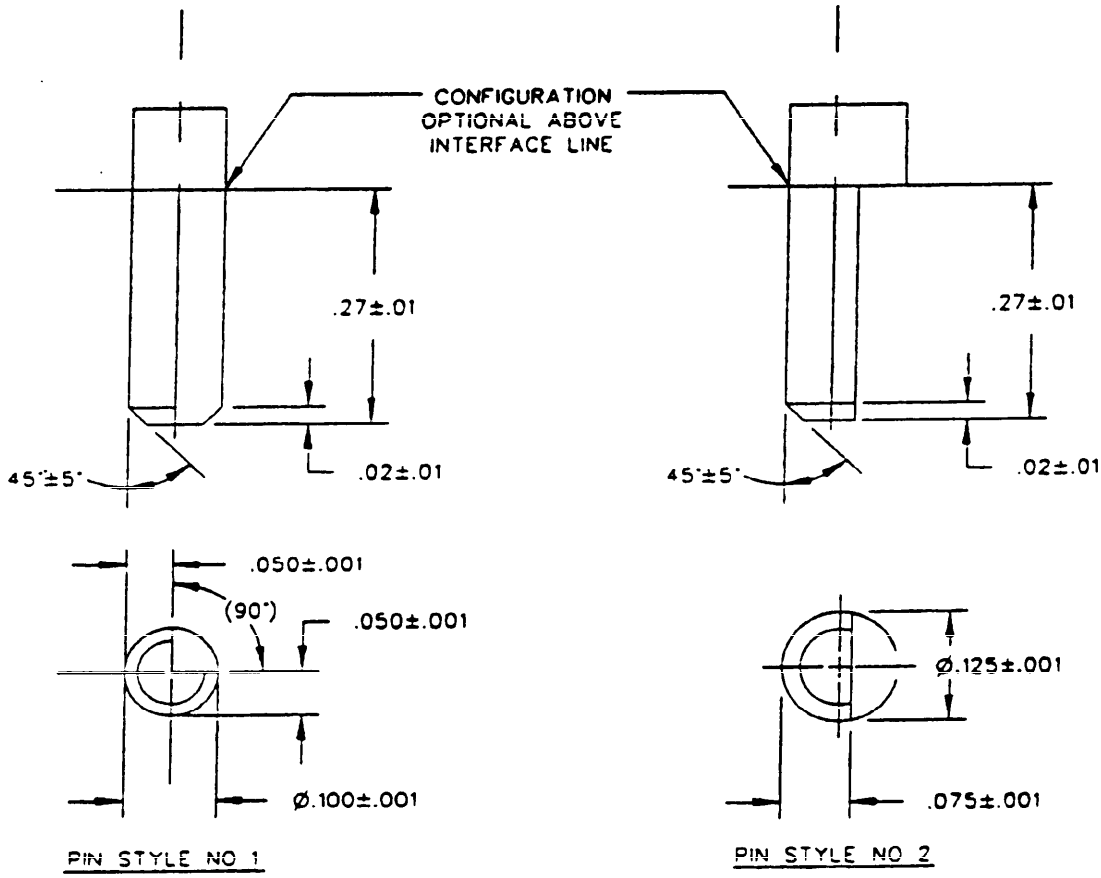
3.11.10.6.5.1 Keying pin torque. Each keying pin shall withstand a torque of 20 inch-ounces (0.14 newton-meter) minimum applied in 2 to 10 seconds and maintained for 10 to 15 seconds (see 4.7.7.8).

3.11.10.6.5.2 Keying pin pullout. Each keying pin shall withstand a pullout force of 9 pounds (40 newtons) minimum applied in 2 to 10 seconds and maintained for 10 to 15 seconds (see 4.7.7.9).

3.11.10.6.5.3 Keying pin cantilever load. Each keying pin shall withstand a cantilever load of 10 pounds (44.5 newtons) minimum applied in 2 to 10 seconds and maintained for 10 to 15 seconds (see 4.7.7.10).

3.11.10.6.5.4 Keying pin pushout. Each keying pin shall withstand a pushout force of 40 pounds (178 newtons) minimum (see 4.7.7.11). The force shall be applied in the opposite direction to the force in 3.11.10.6.5.2. This test applies only to modules with formats C, D, and E.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.001	0.025	0.02	0.51	0.075	1.905	0.125	3.175
0.01	0.25	0.050	1.270	0.100	2.540	0.27	6.86

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 8. Family S2 keying pin styles.

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3.12 Thermal design and construction.

3.12.1 Component temperatures.

3.12.1.1 Critical component temperatures (CCT). The CCT is the maximum component temperature allowed for a power supply operating at the worst case average thermal interface (see 6.5.7) temperature of 60°C. The CCT (junction for semiconductors, winding hot spot for transformers, and case hot spot for other passive components) shall be in accordance with 3.12.1.1.1 through 3.12.1.1.3. In no case shall the CCT exceed a temperature of 105°C (see 4.7.8.1).

3.12.1.1.1 Semiconductors. The CCT for semiconductor devices shall not exceed 105°C.

3.12.1.1.2 Transformers and inductors. The CCT for transformers and inductors shall not exceed 100°C.

3.12.1.1.3 Other parts. The CCT of other parts shall not exceed the individual component's maximum operating temperature minus 20°C and shall be specified on the component's hottest external area.

3.12.1.2 Transient critical component temperature (TCCT). The TCCT is the maximum component temperature allowed under worst case transient average thermal interface temperature of 80°C. The maximum TCCT shall be 20°C greater than that specified in 3.12.1.1. In no case shall the TCCT exceed the absolute maximum specified part temperature (see 4.7.8.2).

3.12.2 Thermal requirements.

3.12.2.1 Cooling system interface (family S1). The average temperature rise from the cooling plate to the power supply mounting surface shall be no greater than 7.5°C when the power supply is mounted as specified in 3.11.6.1 (for example, the cooling plate surface shall be held at 52.5°C or less to insure a power supply mounting surface temperature of no greater than 60°C). Requirements for the cooling plate finish shall be the same as for the power supply mounting surface (see 3.11.7.1.1).

3.12.2.2 Thermal requirements (family S2). Unless otherwise specified herein or in the associated detail specification, thermal requirements shall be in accordance with appendices A through E of MIL-STD-1389.

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3.12.3 Thermal grease. No thermal grease shall be required at the thermal interface of the power supply (mounting surface for family S1 or ribs for family S2) during operation. Internal construction of the power supply shall minimize the use of thermal grease to the greatest extent possible.

3.13 Proprietary parts, processes, or techniques. A power supply shall not be designed such that utilization of proprietary parts, processes, or techniques will result in restricting or eliminating additional power supply designs from meeting the same performance requirements. Proprietary parts are considered to be non-standard parts (see 3.9.3) and usage shall be minimized to the greatest extent possible.

3.14 Identification and marking.

3.14.1 Power supply marking. Power supplies shall be identified and marked with appropriate identifiers as specified herein. The following minimum information shall be marked on the power supplies:

(a) Certification mark.

(b) Power supply part number, revision letter, and amendment number.

(c) Power supply key code.

(d) Manufacturer's identification and manufacturer's power supply assembly number.

(e) Serial number.

(f) Date code.

(g) Electrostatic discharge (ESD) marking.

(h) Precautionary and safety marking.

(i) Input voltages (family S1).

(j) Output voltages (family S1).

(k) Output currents (family S1).

(l) Mounting fastener torque (family S1).

(m) Name (family S2).

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(n) Connector contact identification (family S2).

(o) Extractor fin identification (family S2).

(p) Internal subassembly marking.

Markings shall be a minimum of 0.06 inch (1.52 mm) high (excluding the key code for family S2) and applied in accordance with MIL-STD-130. Marking shall be a contrasting color to the surrounding power supply area. All marking shall be permanent and legible in accordance with MIL-STD-1285.

3.14.1.1 Marking location.

3.14.1.1.1 Family S1. The marking shall be located in accordance with the associated detail specification.

3.14.1.1.2 Family S2. The marking shall be located in accordance with appendices A through E of MIL-STD-1389 and the associated detail specification.

3.14.1.2 Certification mark. All power supplies which meet the requirements of this specification and the associated detail specifications shall have the qualification certification mark "QUAL" marked in accordance with the associated detail specification. Authorization to mark "QUAL" must be obtained from the SPSP-QAA (see 6.10). Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in the associated detail specification shall not bear "QUAL". In the event the item fails to meet the requirements of this specification and the associated detail specification, the manufacturer shall remove the "QUAL" from the sample tested and also from all items represented by the sample. The "QUAL" certification mark shall not be used on power supplies acquired to contractor drawings or specifications.

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3.14.1.3 Part or identifying number (PIN) and revision status. The power supply PIN, revision letter, and amendment number shall be marked as specified herein (see 6.6) and in the associated detail specification. The PIN for each power supply shall be assigned in accordance with MIL-STD-2038. A power supply shall be marked with the revision status of the associated detail specification to which it was tested when manufactured.

3.14.1.4 Key code.

3.14.1.4.1 Family S1. Each power supply shall be marked with a key code assigned in accordance with MIL-STD-2038. Keying and key codes shall be in accordance with appendix A.

3.14.1.4.2 Family S2. Each power supply shall be marked with a key code assigned in accordance with MIL-STD-2038. Keying and key codes shall be in accordance with 3.11.10.6.

3.14.1.5 Manufacturer's identification. Each power supply shall be marked with either the manufacturer's identification code or manufacturer's name. The manufacturer's code, if utilized, shall be a numerical code as listed in Handbook H4-2. The power supply shall be marked with the manufacturer's power supply assembly number. In addition, a revision level, dash number, or other appropriate designation shall be included with the power supply assembly number to identify the level to which the power supply was constructed.

3.14.1.6 Serial number. Each power supply shall have a serial number including the manufacturer's designation marked on the surface of the power supplies. The serial number shall consist of five digits with significant digits prefixed with zeroes, as required. The serial number shall be affixed to the power supply prior to electrical acceptance tests.

3.14.1.6.1 Serial number sequence. Each power supply manufacturer shall serialize each power supply manufactured under the requirements of the SPSP. The serial number for any given power supply type shall start with number 1 and continue in numerical sequence as many times as the power supply is manufactured, regardless of contract or customer.

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3.14.1.6.2 Manufacturer designation. A single or double alpha character will be assigned to each manufacturer contracted to produce power supplies. The designation shall be prefixed to the serial number. Request for a manufacturer designation shall be submitted to the SEM Design Review Activity (SEM-DRA, see 6.5.8).

3.14.1.7 Date code. Each power supply shall be marked with a four-digit date code designating the year and week of manufacture. The first two digits of the code shall be the last two numbers of the year and the third and fourth digits of the code shall be the calendar week. When the number of the week is single digit, it shall be preceded by a zero. The date code for a given power supply shall be the calendar week in which the last major manufacturing assembly process occurred prior to the final acceptance inspection, plus or minus 1 week.

3.14.1.8 Electrostatic discharge (ESD). Power supplies that are determined by the manufacturer or by SPSP-QAA to be sensitive to ESD by prior knowledge of device technologies or by testing to this specification (see 4.3.6) shall be marked in the areas specified by SPSP-QAA or in the associated detail specification with the ESD identifier in accordance with MIL-STD-1285.

3.14.1.9 Precautionary and safety marking.

3.14.1.9.1 Family S1. Unless otherwise specified in the associated detail specification, precautionary and safety markings shall read:

"CAUTION

*OPERATE WITH HEAT SINK
*HAZARDOUS LINE VOLTAGE
*DISCONNECT POWER BEFORE
REMOVING OR INSTALLING UNIT"

When required, additional precautionary and safety labels (installation and removal instructions, high surface temperature caution, and so forth) shall be specified in the associated detail specification or approved by the SPSP-QAA.

3.14.1.9.2 Family S2. Unless otherwise specified in the associated detail specification, precautionary and safety marking shall read:

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"CAUTION

*DISCONNECT POWER BEFORE
REMOVING OR INSTALLING UNIT"

When the power supply size is multiple thickness and space permits, the caution note shall be on the top of the power supply. When required, additional precautionary and safety labels (see 3.14.1.9.1) shall be specified in the associated detail specification or approved by the SPSP-QAA.

3.14.1.10 Input voltage (family S1). The input voltage of the power supply shall be indicated in accordance with the types of input power (see 3.3.1).

3.14.1.11 Output voltages (family S1). Each output voltage of a power supply type shall be listed with programming levels separated by slashes and outputs separated by commas. For example, the output voltages for a dual output, programmable power supply may be listed as:

Output voltage: 5.05/5.25, 12/15

3.14.1.12 Output currents (family S1). The minimum and maximum rated currents for each output listed according to 3.14.1.11 shall be marked in the same sequence as the output voltage levels. For example:

Output current: 11 to 110, 1 to 10

3.14.1.13 Mounting fastener torque (family S1). Each power supply shall be marked with the mounting fastener torque requirements as specified in the associated detail specification.

3.14.1.14 Name (family S2). Each power supply shall be marked with its name. The name and manufacturer's identification shall be orientated such that both are readable from the same point of view. The name marked on the power supply shall agree with the name in the title of the associated detail specification; however, abbreviation in accordance with MIL-STD-12 is permissible. The SEM-DRA is responsible for generation of an approved name. The names in Handbook H6 shall be used if they appropriately describe the power supply. When Handbook H6 does not list a name which appropriately describes the power supply, a name shall be developed in accordance with MIL-STD-961.

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3.14.1.15 Connector contact identification (family S2). Unless otherwise specified in the associated detail specification, each connector shall have contacts identified by numbers on the connector pin shield in accordance with appendices A through E of MIL-STD-1389.

3.14.1.16 Extractor fin identification (family S2). Fins on formats A and B multiple thickness power supplies which are used for extraction must be marked. See appendices A and B of MIL-STD-1389.

3.14.1.17 Internal subassembly marking. Internal subassemblies shall be marked with the following information:

(a) Manufacturer's part number, revision letter, and amendment number.

(b) Serial number.

(c) Manufacturer's identification.

(d) Date code.

3.15 Safety (personnel hazard). Safety (personnel hazard) shall be in accordance with requirement 1 of MIL-STD-454 and as specified herein (see 3.3.1.12).

3.16 Workmanship. Workmanship of the power supply shall be in accordance with military specifications and standards as specified herein. General workmanship (workmanship pertaining to those areas not specifically stated herein) shall be in accordance with requirement 9 of MIL-STD-454.

3.16.1 Metallic surfaces. Metallic surfaces shall be free of burrs, cracks, and sharp edges. Porosity, blowholes, fissures, or voids that are discernible by the unaided eye shall not exceed 10 percent of the total metallic surface area. Connector contacts, keying pins, ribs and fins shall not be bent or damaged.

3.16.1.1 Scratches. Scratches or modifications on the surface of aluminum parts which have been processed in accordance with MIL-A-8625 shall be treated with chemical film in accordance with MIL-C-5541 and painted to match the finish color. Racking points only will be allowed on aluminum surfaces which have been treated in accordance with MIL-A-8625, type III, class 2. Scratches or modifications on other metallic surfaces shall be repaired utilizing a protective coating that will guard against corrosion and will match the finish color. The area of repair shall not exceed 5 percent of the total surface of the power supply.

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3.16.2 Nonmetallic surfaces. Nonmetallic surfaces shall be free of cracks, foreign material, and sharp or rough edges. Voids, blisters, pinholes, or mold marks shall not exceed 10 percent of the total nonmetallic surface area.

3.16.2.1 Molded or potted surfaces. Molded or potted surfaces shall be continuous, homogeneous, and fully cured, and shall cover all components, leads, and circuitry. Pits, pinholes, or voids not exceeding 0.030 inch (0.76 mm) diameter and 0.010 inch (0.25 mm) deep or scratches not exceeding 0.020 inch (0.51 mm) wide by 0.5 inch (12.70 mm) long and 0.010 inch (0.25 mm) deep are permissible, provided no components or circuitry are exposed. Maximum concentration of defects shall not exceed 5 percent of the total molded or potted surface area.

3.16.3 Soldering and printed wiring assemblies. Soldering and printed board assemblies workmanship shall be in accordance with MIL-STD-2000. For power supplies, any item requiring approval or concurrence by the contracting officer shall also require approval by the SPSP-QAA.

3.17 Documentation. Documentation shall be provided as specified herein and in the statement of work and the contract. The power supplies shall conform to the released drawings.

3.18 Request for deviation or waiver. Requests for deviation or waiver from the materials, processes, and requirements specified herein, and from any applicable drawings, specifications, publications, and materials or processes referenced herein shall be submitted to the SPSP-QAA and the command or agency concerned in accordance with MIL-STD-481 for approval.

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4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier (see 6.5.3) is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 SPSP-QAA verification. All quality assurance operations performed by the supplier will be subject to SPSP-QAA verification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.4).

(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Inspections shall be performed in accordance with the test conditions specified herein.

4.3.1 Standard test conditions. Unless otherwise specified herein, tests shall be performed in an area having a relative humidity of up to 95 percent and a barometric pressure

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of between 24 and 32 inches (610 and 813 mm) of mercury. Temperature of the test area shall be maintained at 25 ± 5°C.

4.3.2 Thermal test conditions.

4.3.2.1 Family S1. Tests requiring specific power supply mounting surface temperatures shall be performed with the power supply mounted to a cooling plate in accordance with 3.11.6.1. The cooling plate shall have the same finish requirements as the power supply mounting surface (see 3.11.7.1.1) and shall maintain a uniform temperature with a maximum deviation of plus or minus 2°C from nominal. Cooling plate surface temperatures may be monitored in lieu of the power supply mounting surface temperatures. If so monitored, the average cooling plate temperature shall be assumed to be 7.5°C lower than the power supply mounting surface.

4.3.2.2 Family S2. Tests requiring specific power supply temperatures shall have the fin or rib temperature (as applicable, see appendices A through E of MIL-STD-1389) controlled. Power supplies with multiple fins or ribs shall have the temperature measured on the cooling fin or rib most clearly approaching the mean temperature of all the cooling fins or ribs.

4.3.3 Test equipment. The ratio of specification tolerances to test equipment accuracy shall at least be 10:1 for tolerances of 3 percent or greater and at least 4:1 for tolerances less than 3 percent. Exceptions, due to accuracy limitation of available test equipment, shall be specified in the associated detail specification. Deviations to either of the ratios as specified herein or the specified exceptions shall be recorded and approved in accordance with appendix B. Test equipment shall be accurately calibrated. Frequency of recalibration shall be in accordance with MIL-STD-45662. The test equipment may be laboratory instruments connected in a breadboard arrangement, a console type of equipment, or an automatic tester. Test setups are provided for laboratory instruments in the associated detail specification. The test equipment specified in the associated detail specification is typical of equipment adequate to perform the tests and shall be deemed as satisfying all accuracy requirements. Test equipment shall be in accordance with MIL-STD-1665 and the associated detail specification. Test equipment not specified in MIL-STD-1665 may be utilized for testing power supplies, provided accuracy tables for the equipment specified in the test procedure are submitted to the SPSP-QAA for approval.

4.3.4 Gauges. The supplier shall provide gauges as necessary to ensure that each power supply meets all dimensional

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requirements. The Government representative shall be permitted to use any of the supplier's manufacturing gauges at no cost to the Government and shall be permitted to check such gauges when necessary. However, the fact that manufacturing gauges may have been so checked does not relieve the supplier of the responsibility of meeting all dimensional requirements.

4.3.5 Calibration standards. Standards used for calibration of inspection and test equipment shall be of an accuracy at least four times greater than the accuracy of the equipment being calibrated.

4.3.6 Electrostatic discharge. The supplier shall protect electrical and electronic parts, assemblies, and equipment from ESD in accordance with MIL-STD-1686 and DOD-HDBK-263.

4.4 Qualification inspection. Qualification inspection will be performed by the SPSP-QAA or its designated representative. Qualification inspection shall be conducted in accordance with the procedures described herein.

4.4.1 Initial qualification tests. Tests shall be performed to determine the capability of the power supply to meet the requirements specified herein and in the associated detail specification. The tests specified in table X (for family S1) and table XI (for family S2) and in the associated detail specification shall be performed on the quantity of samples listed. These samples shall be electrically and mechanically interchangeable with production power supplies but need not be necessarily manufactured under production conditions. Materials and processes shall be equivalent (equivalency shall be approved by the SPSP-QAA prior to submittal) to those used in production. The supplier is not required to perform the tests marked "Q" in the associated detail specification, however, the samples shall meet the "Q" test requirements and will be tested for compliance by the SPSP-QAA during qualification testing (including periodic check). Unless otherwise specified, it remains as the suppliers option to perform any test in addition to the required quality conformance testing.

4.4.1.1 Test data. The initial qualification samples shall be accompanied by test data showing compliance with the performance characteristics of the associated detail specification (25°C, 100 percent, and sample tests). Any deviation shall require prior written approval from the SPSP-QAA. The supplier shall retain test data for a minimum of 1 year after completion of all applicable tests and identified by type and serial number of the power supplies.

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TABLE X. Qualification inspection (family S1).

Inspection or test 1/	Requirement	Test	Qual samples 4/ Initial	Qual samples 4/ Periodic
Visual	3.4, 3.11, 3.14, 3.16	4.7.1.1	3	2
Initial electrical	3.3	4.7.2	3	2
Operating temperature	3.5.2.1	4.7.3.2	3	2
Long-term stability	3.5.2.4	4.7.3.5	1	-
CCT/TCCT	3.12.1	4.7.8	1	-
Dripproof	3.5.2.6	4.7.3.7	1	-
Vibration 2/	3.5.2.2.1	4.7.3.3.1	1	1
Mechanical shock 2/	3.5.2.3	4.7.3.4	1	1
Air/structure noise	3.5.2.7	4.7.3.8	1	-
Radiation hardness 3/	3.5.2.5 3.5.3.7	4.7.3.6 4.7.3.15	1	-
EMI 5/	3.6	4.7.4	1	1
Thermal shock	3.5.3.2	4.7.3.10.1	1	1
Humidity	3.5.3.1	4.7.3.9.1	1	1
Storage temperature	3.5.3.6	4.7.3.14	1	-
Salt fog	3.5.3.4	4.7.3.12	1	-

1/ Individual qualification samples shall be subjected to the inspections or tests in the sequence indicated in the table.

2/ These tests may have their testing order reversed.

3/ These tests may be performed at any point during qualification inspection after completion of all tests within a subgroup, provided that they are immediately preceded by tests that satisfy the initial 100 percent and sample electrical requirements specified in the detail specification.

4/ Three power supplies of each type shall be submitted for initial qualification and 2 samples of each power supply type shall be submitted for periodic check.

5/ For periodic testing, these tests will be performed on the sample which undergoes vibration and mechanical shock and will be first in the sequence.

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TABLE XI. Qualification inspection (family S2).

Inspection or test 1/	Requirement	Test	Qual samples 4/ Initial	Qual samples 4/ Periodic
Visual	3.4, 3.11, 3.14, 3.16	4.7.1.1	3	2
Initial electrical	3.3	4.7.2	3	2
Operating temperature	3.5.2.1	4.7.3.2	3	2
Long-term stability	3.5.2.4	4.7.3.5	1	-
CCT/TCCT	3.12.1	4.7.8	1	-
Vibration 2/	3.5.2.2.2	4.7.3.3.2	1	1
Mechanical shock 2/	3.5.3.3	4.7.3.11	1	1
Mechanical	3.11.10	4.7.6	1	1
Durability	3.5.3.5	4.7.3.13	1	1
Radiation hardness 3/	3.5.2.5 3.5.3.7	4.7.3.6 4.7.3.15	1	-
EMI 5/	3.6	4.7.4	1	1
Thermal shock	3.5.3.2	4.7.3.10.2	1	1
Humidity	3.5.3.1	4.7.3.9.2	1	1
Storage temperature	3.5.3.6	4.7.3.14	1	-
Salt fog	3.5.3.4	4.7.3.12	1	-

1/ Individual qualification samples shall be subjected to the inspections or tests in the sequence indicated in the table.

2/ These tests may have their testing order reversed.

3/ These tests may be performed at any point during qualification inspection after completion of all tests within a subgroup, provided that they are immediately preceded by tests that satisfy the initial 100 percent and sample electrical requirements specified in the detail specification.

4/ Three power supplies of each type shall be submitted for initial qualification and 2 samples of each power supply type shall be submitted for periodic check.

5/ For periodic testing, these tests will be performed on the sample which undergoes vibration and mechanical shock and will be first in the sequence.

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4.4.1.2 Inspection and test procedures. The supplier shall prepare inspection and test procedures for all inspections and tests performed, including inspections and tests of incoming materials, in-process inspections and tests, and final inspections and tests. Final inspection and test procedures shall be submitted to the SPSP-QAA for approval. These procedures shall include as a minimum the following:

(a) Quantity to be inspected or tested.

(b) Parameters of materials to be inspected and tested.

(c) Limits of acceptance and rejection.

(d) Equipment to be utilized in making inspections and tests.

(e) Detailed instructions on how to perform inspection and tests.

(f) Data to be recorded.

4.4.2 Failure rate prediction and parts derating. The supplier shall submit information to permit evaluation of the failure rate prediction and of parts derating (see 4.7.5.1). This failure rate data shall be supplied with the initial qualification samples to the SPSP-QAA.

4.4.3 Thermal analysis. Each power supply type shall have a thermal analysis (see 4.7.8) performed by the supplier prior to submission of initial qualification samples. This thermal analysis is to predict or verify that the designs meet the thermal requirements herein (see 3.12). The detailed thermal analysis shall be submitted to the SPSP-QAA with the initial qualification samples.

4.4.4 Design conformance. The supplier shall prepare and maintain power supply assembly drawings. The drawings shall consist of a pictorial view of the power supply printed wiring board assemblies specifying component location and identification, printed wiring boards, power supply structure, electrical schematic, parts list, and any other features necessary for power supply assembly. A note or detail view shall denote orientation of polar diodes, capacitors, and symmetrical multi-lead components. The SPSP-QAA will review power supply designs prior to or during initial qualification tests for conformance to all applicable specifications. This review shall be performed from the power supply assembly drawings, the parts lists, the schematics or logic diagrams, and all other

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documentation used by the supplier to build the power supplies. The supplier shall submit the design package prior to or with the initial qualification samples. Complete design disclosures shall be made for SPSP-QAA use only.

4.4.4.1 Change control. After successful initial qualification, the supplier shall forward to the SPSP-QAA and the command or agency concerned, all changes to documents that impact design, qualification or correlation status. These changes shall be submitted to SPSP-QAA for approval. All changes to test procedures and manufacturing flow charts shall be approved by the SPSP-QAA.

4.4.5 Periodic check. Periodic checks shall be performed to verify that the supplier is manufacturing power supplies that meet all the requirements specified herein and in the associated detail specification. Periodic checks shall be performed on sample power supplies, selected at random by the Government quality assurance representative from power supplies produced during the 12 month period since the last submission, and shall be performed by the SPSP-QAA or its designated representative. Each sample of two power supplies submitted for periodic check shall be subjected to those functional and environmental tests listed in the periodic column of table X or XI. Additional tests as listed in the initial column may be performed as deemed necessary by the SPSP-QAA. Upon request, the supplier shall make available to the SPSP-QAA a cross-reference matrix of date code versus serial number versus contract, by power supply part number, on all power supplies delivered. The matrix shall be updated with each submittal.

4.4.5.1 Submission schedule. The supplier shall submit periodic check samples at 12 month intervals following successful completion of initial qualification inspection. After the first two periodic check intervals, the periodic check interval may (at the discretion of the SPSP-QAA) be changed to have the interval increased to 18 months. Criteria for this decision shall be past performance, design changes made, and engineering judgement. If production is interrupted and the 12 or 18 month interval is exceeded, the supplier shall submit the first two power supplies produced when production resumes.

4.4.5.2 Noncontinuous production. When production has been discontinued during a qualification period and periodic check samples are not submitted on a timely basis, qualification status shall only be maintained upon receipt of a letter of intent that power supply production will be resumed within an 18 month period.

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4.4.6 Acceptance and rejection criteria. Successful completion of initial qualification inspection or periodic check shall be based on no defects which may affect function, reliability, or interchangeability (as determined by the SPSP-QAA). The supplier and command or agency concerned shall be notified in writing of the test results within 10 days following completion of testing. In case a qualification sample does not pass the tests, the supplier shall be notified at the time of failure.

4.4.7 Corrective action. Upon notification of failure of qualification inspection, the supplier shall submit a letter indicating the corrective action. The corrective action may or may not require a resubmission. After approval by the SPSP-QAA and completion of corrective action, the qualification status will be reinstated as appropriate.

4.4.8 Correlation. Test equipment shall be correlated by the SPSP-QAA in accordance with appendix B. Power supplies submitted to the SPSP-QAA for qualification may be used for correlation samples.

4.4.9 Quality assurance requirements. The contractor shall develop and maintain a quality program in accordance with MIL-Q-9858 (see 6.3). The program should include, as a minimum, information on the following topics:

(a) Manufacturing flow chart.

(b) Inspection and test procedures.

(c) Manufacturing instructions.

(d) Process control procedures.

(e) Process certifications.

(f) Workmanship samples.

(g) Accumulation and analysis of defect data.

(h) Power supply assembly drawings.

(i) Electrostatic discharge control program.

4.4.9.1 Manufacturing flow chart. A manufacturing flow chart shall be prepared and submitted to the SPSP-QAA for review and approval. This flow chart shall start with the receipt of materials and show in graphic form the sequence of manufacturing

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operations, inspections, and tests performed in producing the power supplies, and shall end with the shipment of the completed power supplies. The flow chart shall include references to the procedures and instructions to be followed at each operation and shall include the stations at which workmanship samples are located and defect data is recorded. This chart shall have a title, a document number, revision status, and approval blocks.

4.4.9.2 Manufacturing instructions. There shall be instructions provided for each manufacturing operation. These instructions shall be in the form of written procedures and visual aids. These instructions are not required to be submitted to the command or agency concerned, but shall be submitted to the SPSP-QAA for review. These instructions shall be retained at the manufacturing facilities of the supplier until final acceptance by the Government of all power supplies manufactured in conformity therewith. These instructions shall be used in support of inspection by personnel of the command or agency concerned and Government inspectors to ensure compliance with all quality assurance provisions herein. Use of the information in this manner shall be on a proprietary basis.

4.4.9.3 Processes requiring certification. The following processes (if used in the manufacturing of power supplies) shall be certified to their applicable military specifications:

(a) Nickel plating: QQ-N-290.

(b) Gold plating: MIL-G-45204.

(c) Anodizing: MIL-A-8625.

(d) Resistance welding of electronic circuit modules: MIL-W-8939.

(e) Special technological processes.

Process certifications will be approved by the SPSP-QAA.

4.4.10 Workmanship samples. The supplier shall prepare workmanship samples for the evaluation of in-process workmanship. These samples may be in the form of drawings, photographs, or hardware and shall be available for review and approval by the command or agency concerned. Samples shall be maintained as standards of quality.

4.4.11 Maintenance data. Maintenance data for test, fault isolations, and repair of the power supplies shall be prepared by the supplier and submitted to SPSP-QAA for review.

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4.4.12 Retention of qualification. Upon successfully completing initial qualification inspection, a supplier retains qualification status until one of the following occurs:

(a) Change of design (including components, materials, and processes).

(b) Failure of periodic check.

(c) Failure to submit samples and associated data on a timely basis (maximum interval under no production conditions will be 18 months) (see 4.4.5).

4.5 Quality conformance inspection. The following quality conformance inspection requirements apply.

4.5.1 Quality conformance criteria. Quality conformance acceptance of a power supply shall be based on no defects. The inspection and test plan in table XII (for family S1) or table XIII (for family S2) shall be used. Any characteristics failing to conform to the requirements specified herein and in the associated detail specification shall be considered a defect. Power supplies subjected to quality conformance inspection shall be completed samples, requiring no other manufacturing steps.

4.5.2 Sampled quality conformance inspection. For sampled inspections (S tests), acceptance or rejection of a lot shall be based on the sampling procedures specified in table XII (for family S1) or table XIII (for family S2), as applicable. If a failure occurs in a sampled electrical quality conformance inspection, the entire inspection lot shall be screened for that parameter before shipment of the lot. Power supplies failing during the screen shall be rejected.

4.5.3 One hundred percent inspection. For 100 percent quality conformance inspection parameters (X tests), any power supply not meeting the requirements specified herein and in the associated detail specification shall be rejected.

4.5.4 Failure analysis. Unless otherwise specified by SPSP-QAA, failure analysis shall be conducted by the supplier on all power supplies that fail tests required by the quality conformance test plan specified in 4.5.1. Component failure analysis is not required on failed parts resulting from recurring failure modes provided that component analysis has been performed on prior failures and the necessary and acceptable corrective action has been determined.

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TABLE XII. Quality conformance inspection and test plan (family S1).

Inspection 1/	Requirement	Test	Sampling
Electrical requirements at 25°C mounting surface temperature	3.7.2	4.7.5.2.1	100 percent for "X" tests in accordance with the associated detail specification.
Temperature cycling 2/	3.7.2	4.7.5.2.1	100 percent
Random vibration	3.7.2	4.7.5.2.2	100 percent
Electrical requirements at 25°C mounting surface temperature	3.3	4.7.2	100 percent for "X" tests in accordance with the associated detail specification. Sample testing per 3/.
Visual:			
Profile dimensions	3.4.1.1	4.7.1.1	Two power supplies or 20 percent of the inspection lot, whichever is greater.
Keying position	3.11.9	4.7.1.1	
Marking	3.14	4.7.1.1	
Workmanship	3.16	4.7 1.1	

1/ Order of performance shall be as shown. Within the subgroup containing visual inspection, the order of testing is optional.

2/ Temperature cycling and random vibration shall be completed prior to the final 25°C electrical test.

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TABLE XII. Quality conformance inspection and test plan (family S1) - Continued.

3/ For those parameters which are to be sample inspected, two power supplies or 1 percent of the inspection lot, whichever is greater, shall be randomly selected from each inspection lot. The electrical parameters, marked "S" in the associated detail specification, shall be tested at 25°C mounting surface.

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TABLE XIII. Quality conformance inspection and test plan (family S2).

Inspection 1/	Requirement	Test	Sampling
Electrical requirements at 25°C fin or rib temperature	3.7.2	4.7.5.2.1	100 percent for "X" tests in accordance with the associated detail specification.
Temperature cycling 2/	3.7.2	4.7.5.2.1	100 percent
Electrical requirements at 25°C fin or rib temperature	3.3	4.7.2	100 percent for "X" tests in accordance with the associated detail specification
Visual:			Sample testing per 3/.
Profile dimensions	3.4.1.2	4.7.1.1	Two power supplies or 20 percent of the inspection lot, whichever is greater.
Keying position	3.11.10.6	4.7.1.1	
Marking	3.14	4.7.1.1	
Workmanship	3.16	4.7 1.1	

1/ Order of performance shall be as shown. Within the subgroup containing visual inspection, the order of testing is optional.

2/ Temperature cycling shall be completed prior to the 25°C electrical test.

3/ For those parameters which are to be sample inspected, two power supplies or 1 percent of the inspection lot, whichever is greater, shall be randomly selected from each inspection lot. The electrical parameters, marked "S" in the associated detail specification, shall be tested at 25°C ambient.

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4.5.5 Inspection lot. An inspection lot shall consist of all power supplies produced for quality conformance inspection at any one time. An inspection lot may be defined in a contract or purchase order.

4.6 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with MIL-P-116 groups A and B quality conformance inspection requirements. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Test procedures.

4.7.1 Visual examination. Power supplies shall be examined with no more than 10 power magnification lens to ensure compliance with this specification and the associated detail specification.

4.7.1.1 Initial visual examination. The following characteristics shall be examined to ensure initial compliance:

(a) Power supply profile. Dimensions and weight shall be checked to ensure that they are within specified tolerances.

(b) Keying. Keying positions shall be checked to ensure correct positioning.

(c) Marking. Markings shall be checked to ensure completeness, correct positioning, and legibility.

(d) Workmanship. Power supplies shall be checked for compliance with the requirements of 3.16.

4.7.1.2 Visual degradation examination. Power supplies shall be visually examined upon completion of each operating and nonoperating environmental test (see 4.7.3). If a test chamber or fixture is used in a test, a visual examination of the power supply while in the chamber or on the fixture and at test conditions shall be performed, when possible. Upon completion of a test, a visual examination of the power supply shall be performed again, after the power supply has been returned to standard ambient conditions.

4.7.2 Electrical performance tests. The initial electrical requirements of each power supply shall be tested to the extent specified herein and in the associated detail

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specification. Following each test specified in table X or XI, the electrical quality conformance tests shall again be performed as specified herein or in the associated detail specification. When the electrical tests to be performed are not specified, the 100 percent quality conformance inspection tests shall be performed.

4.7.2.1 Spike voltage source impedance. For design, test, and evaluation purposes, the voltage spike shall be 1000 volts peak and the source impedance of the voltage spike shall be assumed to be 5 ohms (see 3.3.1.1.2). The voltage spike shall be superimposed on the power line waveform of both the AC and auxiliary DC power input leads (but not simultaneously). Both polarities of the spike shall be injected (but not simultaneously).

4.7.2.2 Step input test. For the purposes of this test (see 3.3.1.9.1), neither the power "off" period nor the power "on" period shall be less than 1 second. The bandwidth of the measurement shall be 100 kHz $\pm$ 10 percent.

4.7.2.3 Input harmonic current test. For the purposes of this test (see 3.3.1.10), the input power source shall not have a harmonic voltage content in percent RMS at any frequency which is greater than 25 percent of the allowable RMS harmonic current percentage at any frequency. If the input power source does not meet this harmonic voltage requirement, then the voltage distortion must be factored into the result of the harmonic current measurement. The accuracy of measurement of harmonic currents shall be plus or minus 5 percent of the harmonic being measured.

4.7.2.4 Power factor test. The power factor requirement (see 3.3.1.11) shall apply when the power supply is operated at rated load. Power factor shall be computed by real power divided by apparent power, as indicated in the following formula:

$$\text{Power factor} = \frac{W}{(V_1^2 I_1^2 + V_2^2 I_2^2 + V_3^2 I_3^2)^{1/2}}$$

where: W = watts

V_1, V_2, V_3 = true RMS line-to-line voltages

I_1, I_2, I_3 = true RMS line currents

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4.7.2.5 Total ripple and noise measurement. Total ripple and noise (see 3.3.2.3) shall be measured differentially over all applicable wideband frequency ranges. At least one measurement on each output shall be made using a device whose 3 dB bandwidth shall extend from a minimum frequency, no higher than 12 Hz, to at least 20 MHz.

4.7.2.5.1 Low frequency ripple and noise measurement. The low frequency ripple and noise (see 3.3.2.3.1) shall be measured differentially using a device whose 3 dB bandwidth shall extend from a minimum frequency, no higher than 12 Hz, to 10 kHz ± 20 percent.

4.7.2.5.2 High frequency ripple and noise measurement. The high frequency ripple and noise (see 3.3.2.3.2) shall be measured differentially using a device whose 3 dB bandwidth shall extend from 10 kHz ± 20 percent to 1 MHz ± 10 percent.

4.7.2.6 Common mode output current measurement. The common mode output current (see 3.3.2.4) measurement shall be made using a current measuring device whose bandwidth is at least 20 MHz. A 10 microfarad ± 10 percent plastic film (or equivalent) capacitor shall be connected between the power supply output and its return. The return shall be connected to the power supply chassis. The current measuring device shall be connected around the wire going from the output return to chassis.

4.7.2.7 Static line and load regulation test. Line and load regulation (see 3.3.2.5) shall be measured for all combinations of minimum and maximum line voltage and no load (or minimum load) and full load. For multiple output power supplies, the load regulation shall be measured as above as well as measuring that a load change on one output has a minimal effect on another output (cross regulation).

4.7.2.8 Dynamic load regulation test. The step change (see 3.3.2.7) in load current shall be equal to 10 percent of the power supply full rated load current. Unless otherwise specified in the associated detail specification, the step change shall occur in 60 microseconds or less. The positive step change shall be applied at any nominal load current between 10 and 90 percent of full rated load. The negative step change shall be applied at any nominal load between 20 and 100 percent. When required, dynamic load regulation for load step changes greater than 10 percent shall be specified in the associated detail specification.

4.7.2.9 Dynamic line regulation test. The step change (see 3.3.2.8) in input voltage shall be equal to 21 percent of the nominal input voltage and applied to the nominal input

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voltage with a rise time between 50 and 500 us.

4.7.2.10 Efficiency measurement. The efficiency (see 3.3.5) shall be measured at nominal input voltage and at maximum rated output load current.

4.7.3 Environmental test procedures. The test procedures for environmental exposures to determine conformance to the requirements specified in 3.5, shall be in accordance with 4.7.1.2, 4.7.2, the following subparagraphs and the associated detail specification.

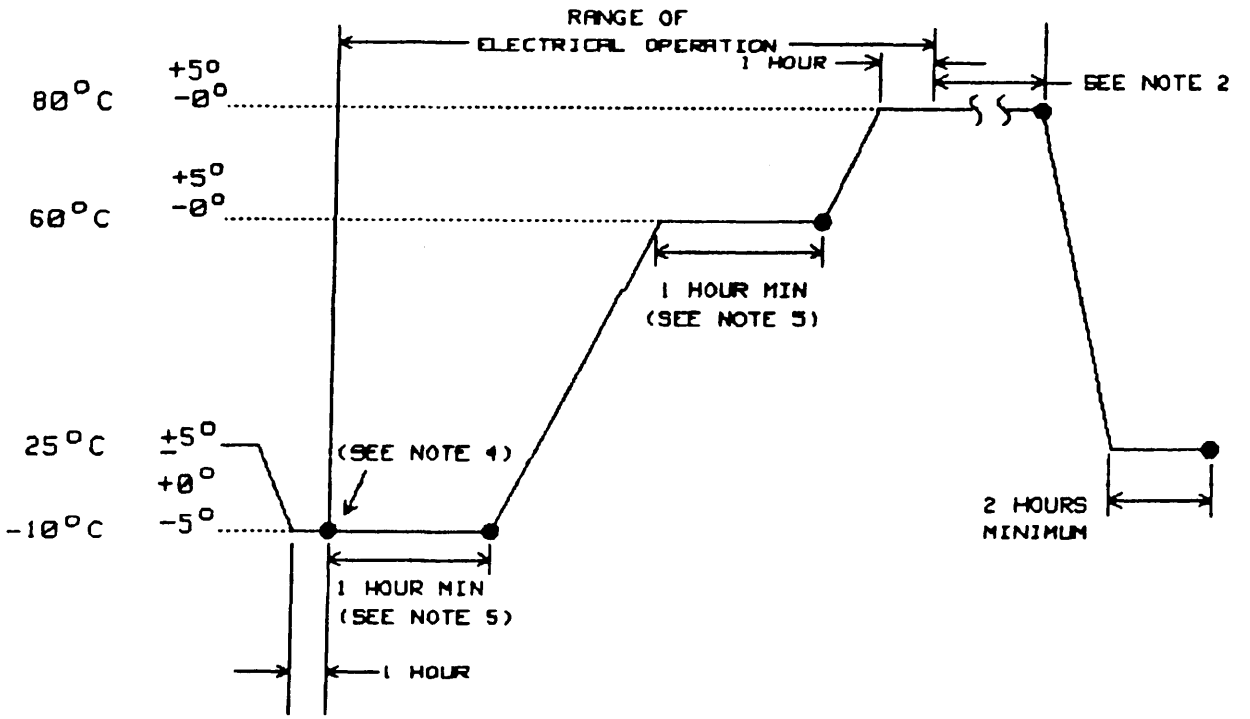
4.7.3.1 Mounting requirements. Power supplies shall not be required to be mounted on a cooling plate (family S1) or in a card cage (family S2) during environmental exposure for those environments which do not require electrical operation. The power supplies shall be mounted and restrained by normal mounting methods during vibration and mechanical shock.

4.7.3.2 Operating temperature tests. Power supplies shall be subjected to the operating temperature tests (see 3.5.2.1). For family S1, the thermal interface shall be the power supply mounting surface (see figure 1). For family S2, the thermal interface shall be the ribs or fins in accordance with appendices A through E of MIL-STD-1389. Power supplies with multiple fins or ribs shall have the temperature measured on the cooling fin or rib most closely approaching the mean temperature of all the fins or ribs. The power supply shall be set up and temperature tested according to the associated detail specification, figure 9 and the following;

(a) Low temperature test. The power supply's average thermal interface temperature shall be lowered, with the power supply unenergized, to minus 10, plus 0, minus 5°C. Following a 1 hour stabilization period, the power supply shall be exercised with a cold start under rated load conditions and shall meet the specified performance (see figure 9). The power supply shall meet all specified performance requirements following an operating stabilization period of 1 to 8 hours, as required, at the low temperature.

(b) High temperature test. With the power supply operating continuously at rated load, the average temperature of the thermal interface shall be increased to 60, plus 5, minus 0°C. The power supply shall meet all specified performance requirements following an operating stabilization period of 1 to 8 hours, as required, at 60°C.

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NOTES:

1. The maximum temperature gradient shall be 1°C per minute.
2. The total time allowed for the transient thermal test shall be 1 hour of power supply operation plus optional 0 to 8 hours of nonoperating storage plus 1 hour for electrical tests.
3. Electrical tests shall be performed at the indicated nodes (●).
4. Following a 1 hour temperature stabilization period, the power supply shall be energized. Testing shall determine conformance to 3.3.2.3, 3.3.2.6, and 3.3.2.11 (turn-on).
5. If there are indications the power supply has not stabilized after 1 hour, up to 8 hours will be allowed prior to performing electrical tests.

FIGURE 9. Operating temperature cycle.

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(c) Transient thermal operation. With the power supply operating at rated load, the average temperature of the thermal interface shall be increased to 80, plus 5, minus 0°C and maintained for 1 hour. Electrical testing may immediately be started or may be delayed for up to 8 hours if the power supply is turned off. Electrical test time shall not exceed 1 hour. The power supply shall meet the high temperature end-of-life (EOL) limits as specified in 3.5.4.2 and the associated detail specification.

4.7.3.3 Vibration

4.7.3.3.1 Family S1. The power supplies shall be subjected to the 10 gravity peak sinusoidal vibration test (see 3.5.2.2.1) in accordance with MIL-STD-202, method 204, test condition A. The frequency range shall be 10 to 500 Hz with no resonances below 50 Hz. The power supplies shall be energized with a load current greater than or equal to 10 percent of the rated output current. The output voltages shall be monitored during the vibration test and the details of any failure recorded. Following the vibration test, the power supply covers may be removed for visual examination (see 4.7.1.2).

4.7.3.3.2 Family S2. The power supplies shall be subjected to separate simple harmonic motion and random vibration (see 3.5.2.2.2) in accordance with 4.7.3.3.2.1 and 4.7.3.3.2.2. Power supply connectors shall be inserted into a backplane utilizing appropriate mating connectors (tuning fork type connectors shall be in accordance with MIL-C-28754 or MIL-C-28859). The power supplies shall be securely clamped along both ribs and shall be energized with a load current greater than or equal to 10 percent of the output current. The output voltages shall be monitored during the vibration test and the details of any failure recorded. Following the vibration test, the power supply covers may be removed for visual examination.

4.7.3.3.2.1 Simple harmonic motion vibration. The power supplies shall be subjected to vibration in accordance with MIL-STD-202, method 201, for 3 minutes in each axis.

4.7.3.3.2.2 Random vibration. Power supplies shall be subjected to random vibration in accordance with MIL-STD-202, method 214, test condition I, letter D, for 3 minutes in each axis.

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4.7.3.4 Mechanical shock (family S1). The power supplies shall be shock tested (see 3.5.2.3) in accordance with grade A, type A, class I, of MIL-S-901. The power supplies shall be energized with a load current greater than or equal to 10 percent of the rated output current. The output voltages shall be monitored during the shock test and the details of any failures recorded. Following the shock test, the power supply covers may be removed for visual examination.

4.7.3.5 Long term stability. The output voltages of each power supply shall not change more than 0.2 percent (see 3.5.2.4) when operated for a minimum of 1,000 hours, under isothermal ambient conditions, at maximum mounting surface or thermal interface temperature, at full rated load current, and at nominal input voltage. The baseline output voltages shall be measured 20 plus or minus 2 hours after initial power supply turn-on.

4.7.3.6 Radiation hardness. The test sample shall consist of power supplies which have passed the electrical requirements specified in the associated detail specification for the 100 percent and sample acceptance test initial limits at 25°C. Post irradiation electrical tests shall be these same electrical requirements, using the end-of-life (see 3.5.4.2) limits at 25°C. The test sample shall be subjected to the following levels of radiation exposure (see 3.5.2.5):

(a) Ionizing dose rate: 1×10^{9} rad (Si)/second (20 to 100 nS pulse width)

(b) Total ionizing dose: 3×10^{3} rad (Si).

The power supplies shall pass the end-of-life requirements after each radiation exposure test. The combination of these radiation exposure environments are considered destructive and, therefore, test sample power supplies subjected to them shall not be used in any operational system. For the purpose of this specification, all terms used in the referenced test methods of 4.7.3.6.1, 4.7.3.6.2, and 4.7.3.15 such as semiconductor device, linear microcircuit, integrated circuit or part, and so forth, meaning the part under test or irradiation, shall be interpreted or construed as meaning power supply.

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4.7.3.6.1 Ionizing dose rate. The power supplies shall be tested in accordance with MIL-STD-883, method 1023. The radiation pulse width shall be 20 to 100 nS with 100 rad (Si) maximum absorbed total dose per pulse. The power supplies shall be operated at full rated load and at maximum input voltage during irradiation. They shall be irradiated once on each side, except for the mounting surface/connector location side, for a total of 5 pulses. The output voltage shall be monitored immediately after each irradiation, with an out of end-of-life regulation limit reading counted as a failure (the test shall only be stopped before conclusion, however, if there is a catastrophic failure). The power supplies shall pass the post-irradiation electrical tests which shall be started within 2 hours of the last exposure.

4.7.3.6.2 Total ionizing dose. The power supplies shall be tested in accordance with MIL-STD-883, method 1019. The power supplies shall be operated at full rated load and at maximum input voltage during irradiation. Post-irradiation electrical tests shall be started within 2 hours of the termination of exposure.

4.7.3.7 Dripproof test (family S1). The power supplies shall be subjected to the dripproof test (see 3.5.2.6) in accordance with MIL-STD-108. The power supplies shall be mounted to a cold plate with finish requirements in accordance with 3.11.7.1.1 and the mounting fasteners torqued in accordance with the associated detail specification. The power supplies shall be operated in accordance with MIL-STD-108 and with 10 percent minimum of rated load. The power supplies shall be orientated in 10 different positions including normal (10.8 inch dimension vertical, connector on top), inverted normal, and at 45 degree angles from normal and inverted normal. Accumulation of moisture (greater than the area of a 2 inch diameter circle on a horizontal surface/gasketed portion of the cooling plate or more than a couple of drops in a recessed area) or failure of the power supply to operate satisfactorily shall be cause for rejection. Following the dripproof test, the power supply covers may be removed for visual examination.

4.7.3.8 Airborne sound and structureborne vibratory acceleration (family S1). The airborne sound test (see 3.5.2.7) shall be conducted in accordance with MIL-STD-740-1 and the structureborne vibratory acceleration test shall be conducted in accordance with MIL-STD-740-2. The power supply shall be operated at maximum rated load current and nominal AC input voltage.

4.7.3.9 Humidity.

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4.7.3.9.1 Family S1. Humidity tests (see 3.5.3.1) shall be in accordance with MIL-STD-810, method 507, procedure III, except that the power supplies shall not be energized while in the humidity chamber. The power supplies shall be placed in the chamber on either of the sides measuring 2.48 by 10.8 inches. At the conclusion of the exposure to the humid environment, excess water shall be shaken and wiped from the external surfaces of the power supplies. Electrical testing shall be started within one hour and be completed within five hours after removal of the power supplies from the chamber. The power supplies shall be tested to the EOL limits of the 100 percent electrical acceptance tests specified in the associated detail specification.

4.7.3.9.2 Family S2. The power supplies shall be subjected to the humidity test as specified for family S1, except the power supplies shall be placed in the chamber with the module span at a 45° angle to the horizontal plane. Also, the power supplies shall have their mechanical dimensions checked and they shall be energized within 1 hour after removal from the test chamber.

4.7.3.10 Thermal shock.

4.7.3.10.1 Family S1. The power supplies shall be subjected to the thermal shock test (see 3.5.3.2) in accordance with MIL-STD-202, method 107, condition A-3. The temperature extremes shall be minus 55, plus 0, minus 3°C and 85, plus 3, minus 0°C. The test shall consist of 100 cycles (less than or equal to 100 cycles for periodic check at the discretion of the SPSP-QAA), each one requiring stabilization at each temperature for 1 hour.

4.7.3.10.2 Family S2. The power supplies shall be subjected to thermal shock as specified for family S1, except the stabilization at each temperature may be reduced to 0.5 hour if the weight of the power supply is 0.3 pound or less.

4.7.3.11 Mechanical shock (family S2). The power supplies shall be shock tested (see 3.5.3.3) in accordance with MIL-STD-202, method 213, test condition I. Following the shock test, the power supply covers may be removed for visual examination.

4.7.3.12 Salt fog. The power supplies shall be exposed for 48 hours to a salt-sea atmosphere (see 3.5.3.4) in accordance with MIL-STD-810, method 509, procedure I, and as modified herein. For family S1, the power supplies shall be placed in the chamber on either of the sides measuring 2.48 by 10.80 inches.

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The connector area shall not be directly exposed to the salt fog spray nozzle. The power supplies shall be examined with the aid of a 10 power magnification following a gentle wash in warm (20 to 38°C) water upon removal from the test chamber (to remove visible salt deposits) and storage for 48 hours at room ambient conditions to allow for evaporation of excess moisture. Surface corrosion products shall not be evidence of failure. The power supply covers shall be removed for visual examination. Electrical tests are not required following the salt fog test except as a method to detect the corrosion failure mechanisms.

4.7.3.13 Durability (family S2). Power supplies shall be subjected to 500 cycles of mating in accordance with 3.5.3.5 at the rate of not more than 10 cycles per minute. During the test, power supplies shall be mounted so the insertion and extraction forces are transmitted directly from the fin or header to the connector. Formats C, D, and E power supplies shall be inserted into a backplane meeting the requirements of MIL-A-28870. For power supplies which exercise the lateral displacement option (floating), the card cage test system shall provide a clearance of 0.006 plus or minus 0.001 inches (0.15 plus or minus 0.03 mm) between the surface of the power supply guide rib and the interfacing rail surface on the card cage. For power supplies which do not exercise the lateral displacement option (nonfloat), the clearance above should only be 0.0030 plus 0.0005 or minus 0.0000 inches (0.076 plus 0.013 or minus 0.000 mm) between the surface of the guide rib and the interfacing rail surface on the card cage. The clamping pressure in both cases shall be 75 pounds per square inch to each guide rib. Power supplies shall be forced against the rail surface on each cycle after they are seated into the backplane. This clamping force shall be removed prior to extraction of the power supplies. This clamping force shall be uniformly applied to the power supply guide ribs on the surface opposite the one in contact with the rail surface. The clamping testing may be included in the insertion and extraction sequences or it may be performed as a separate test. To enhance the visibility of any exposed underplating, the contacts may be dipped in a 5 percent sodium sulfide solution for 2 minutes, plus or minus 15 seconds, and then rinsed prior to examination. The power supply connector contacts shall be examined under 3 power magnification (10 power for referee inspection) for underplating exposure at the completion of the durability test.

4.7.3.14 Storage temperature. The power supply shall be subjected to the nonoperating temperature extremes of minus 55, plus 0, minus 5°C and 85, plus 5, minus 0°C (see 3.5.3.6). The

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power supply shall be maintained at each temperature extreme for 24 hours with verification of electrical performance at a stabilized temperature of 25°C following each temperature soak. Temperature excursions shall not exceed 3°C per minute.

4.7.3.15 Non-operating radiation hardness. The power supplies shall be subjected to a neutron fluence of 2×10^{12} neutrons per (cm^2) (see 3.5.3.7). They shall be tested in accordance with 4.7.3.6 and MIL-STD-883, method 1017. The power supplies shall not be operated during irradiation.

4.7.3.16 Solvents test. Power supplies which use adhesive backed labels shall meet the requirements of 3.5.3.8 when tested in accordance with MIL-STD-202, method 215.

4.7.4 Electromagnetic interference measurements. Each power supply type shall be tested using its primary output polarity and all applicable types of input power. The testing requirements and procedures of MIL-STD-462 shall be used to determine compliance with applicable emission and susceptibility requirements (see 3.6) of parts 5 and 6 and section 5 of part 1 of MIL-STD-461. Data gathered as a result of performing tests in one electromagnetic discipline may be sufficient to satisfy requirements in another. Therefore, to avoid unnecessary duplication, a single test program should be established with similar tests conducted concurrently whenever possible. The following details shall apply:

(a) Conducted EMI. CE01 and CE03 do not apply to the power supply output leads.

(b) Susceptibility to EMI. The susceptibility criteria for the power supply shall be the output DC voltage going out of regulation, or the output ripple and noise exceeding the specified limits when using an oscilloscope having a 20 MHz plus or minus 20 percent bandwidth. When testing, the thresholds of susceptibility shall be determined and recorded in the EMI test results, whether below the contractual EMI requirements or above, within the maximum capability of the test equipment used. Testing to verify compliance to requirement CS11 shall apply only to the input power leads.

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4.7.5 Reliability.

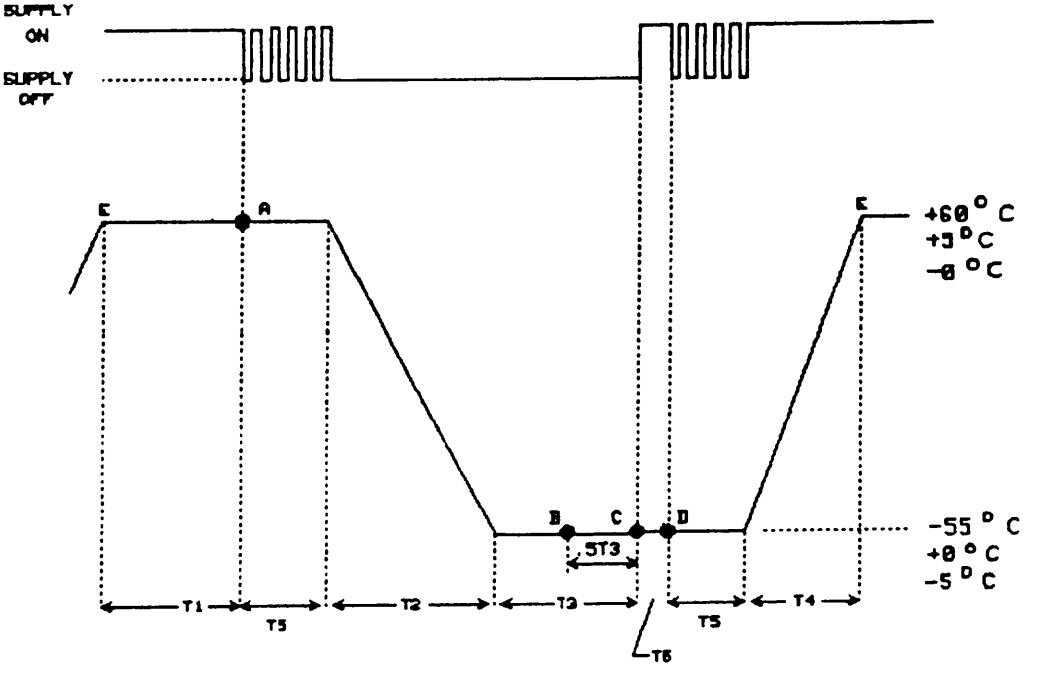
4.7.5.1 Reliability prediction. A reliability prediction (see 3.7.1) shall be performed in accordance with MIL-STD-756, task 201, method 2005. MIL-HDBK-217 shall be used as the basic data source. The prediction shall be based on a detailed thermal/electrical/mechanical stress analysis of each part. It shall show that the part derating requirements (see 3.10.2) and the component junction and case temperature requirements (see 3.12) are being met. Unless otherwise specified in the associated detail specification, the reliability prediction shall be made at nominal input voltage, 60°C thermal interface temperature, Navy sheltered environment, and maximum rated output power.

4.7.5.2 Environmental stress screening. Environmental stress screening shall be performed on all power supplies (see 3.7.2). Random vibration is not required to be performed on family S2 power supplies. Electrical testing prior to and following environmental stress screening shall be in accordance with 4.5.1.

4.7.5.2.1 Temperature cycling. Each power supply shall be subjected to the temperature cycling test as shown on figure 10. The power supplies shall be temperature cycled a minimum of 12 times with the last two cycles being failure free. The thermal stabilization period at each temperature shall be as shown on figure 10. A thermal survey shall be performed to verify that the temperatures achieved meet the specified profile.

4.7.5.2.2 Random vibration (family S1). Each power supply shall be subjected to a random vibration screen as specified herein and shall meet all performance requirements after the random vibration screening. The power supply under test shall be hard mounted to a shake table capable of reproducing random vibration having the power spectral density (PSD) characteristics shown on figure 11. A pseudorandom or synthetic random shaker capable of reproducing this power spectral density function will be an acceptable substitute for a true random vibration fixture. Each power supply shall be shaken in a

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NOTES:

1. T1 = T3 = 2 hours. Power supply thermal stability shall be considered to have been reached only when the part with the greatest thermal inertia (excluding magnetics) has a rate of change of less than 5°C per hour or is within 10°C of the power supply average thermal interface temperature. Power supply thermal stability shall be attained a minimum of 20 minutes prior to points A or C. Points E and B apply to the average temperature of the power supply thermal interface.

2. For T2 and T4, the average rate of change of the chamber ambient temperature shall be a minimum of 5°C per minute.

3. T5 = 10 minutes maximum. T6 = 2 minutes minimum.

4. The thermal cycle shall begin at point E with the power supply energized at the maximum rated load and nominal input voltage.

5. At point A, turn the power supply off and on 5 times for a maximum time of 1 minute in each state. Monitor the output status signal (see 3.3.4.4). Alternately, verify output voltage for compliance with 3.3.2.6.

FIGURE 10. Temperature cycling screen.

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NOTES: - Continued.

6. Turn the power supply off and reduce the temperature to point B.

7. At point C, the power supply shall be turned on. Monitor the output status signal (see 3.3.4.4).

8. At point D, turn power supply off and on 5 times for a maximum time of 1 minute in each state. Monitor the output status signal (see 3.3.4.4). However, the power supply is only required to be in accordance with 3.3.2.6 when the mounting surface temperature is greater than or equal to minus 15°C (see 3.5.2.1 (a)).

9. Increase the temperature to point E and repeat the cycle.

FIGURE 10. Temperature cycling screen - Continued.

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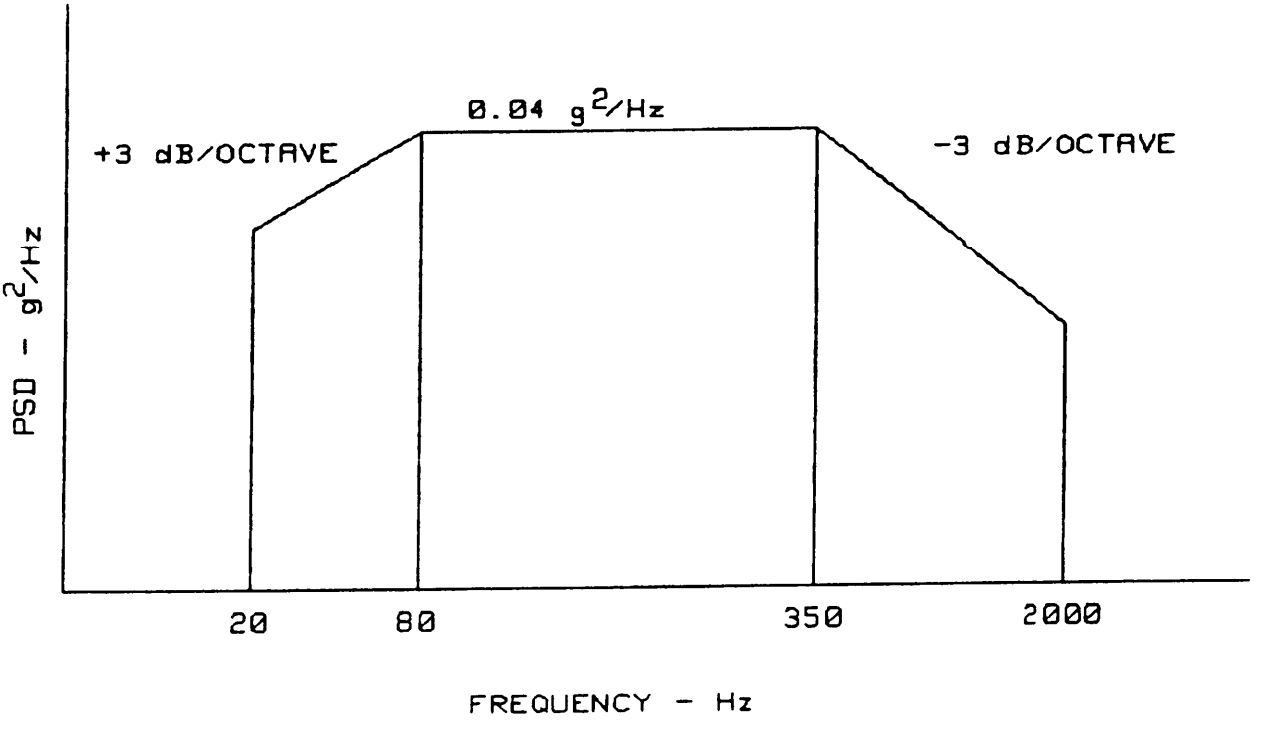


FIGURE 11. Random vibration screen.

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minimum of 2 axes. At least 1 axis shall be perpendicular to the component mounting surfaces of the internal modules. The duration of random vibration shall be at least 10 minutes in each axis. The power supply shall be energized during this test. The load current shall be a minimum of 10 percent of full rated load.

4.7.5.2.3 Environmental stress screen failures. Any power supplies which do not successfully complete the temperature cycling and random vibration (family S1) shall be considered as nonconforming material (see 6.9).

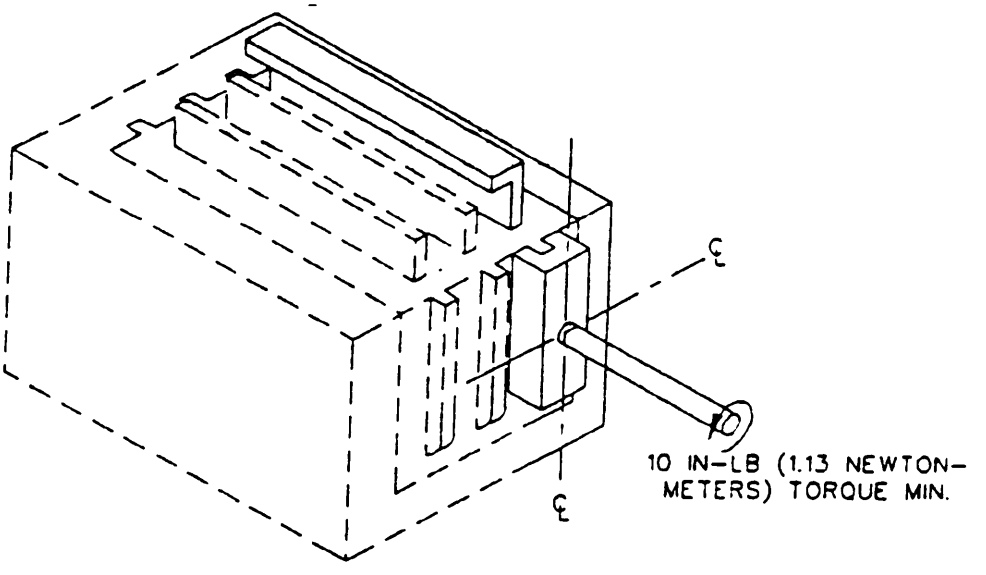
4.7.6 Maintenance and repair. Test procedures, fault isolation, and subassembly replacement for maintenance (see 3.8.1) and repair should be simple and capable of being performed by semi-skilled, user-trained personnel employing a minimal amount of test equipment and tools. No special tools (in accordance with MIL-STD-454, requirement 63) shall be required.

4.7.7 Mechanical integrity tests (family S2). Following the tests specified in 4.7.7.1 through 4.7.7.11, the power supply shall meet all the configuration requirements specified herein, in the associated detail specification, and the electrical requirements of 4.7.2.

4.7.7.1 Rib strength. The individual power supply ribs shall withstand the torque specified in 3.11.10.2.1 applied in a direction perpendicular to the plane of the ribs and along the full length of the rib using a method similar to that shown on figure 12. The torque shall be applied at a rate such that the specified torque is applied in 2 to 10 seconds from the time the torque is first applied to the rib and maintained for 10 to 15 seconds.

4.7.7.2 Pin shield retention. Each pin shield shall withstand the force specified in 3.11.10.3.1. The force shall be applied to each end of each pin shield in a direction normal to a plane passing through the centerline of both keying pins in each of two directions 180° apart. The point of application of force shall be midway between the first and second electrical contacts on each end up 0.06, plus 0.00 or minus 0.03 inch (1.5, plus 0.0 or minus 0.8 mm) from the bottom edge of the pin shield (see figure 13). The rate of application of force shall be such that the specified force is applied in 2 to 5 seconds and maintained for 10 to 15 seconds at 25, plus or minus 5°C. For multiple span power supplies employing connectors having rows of 50 contacts, the foregoing tests are also required at points opposite the midpoint between contacts 25 and 26 and between 75 and 76.

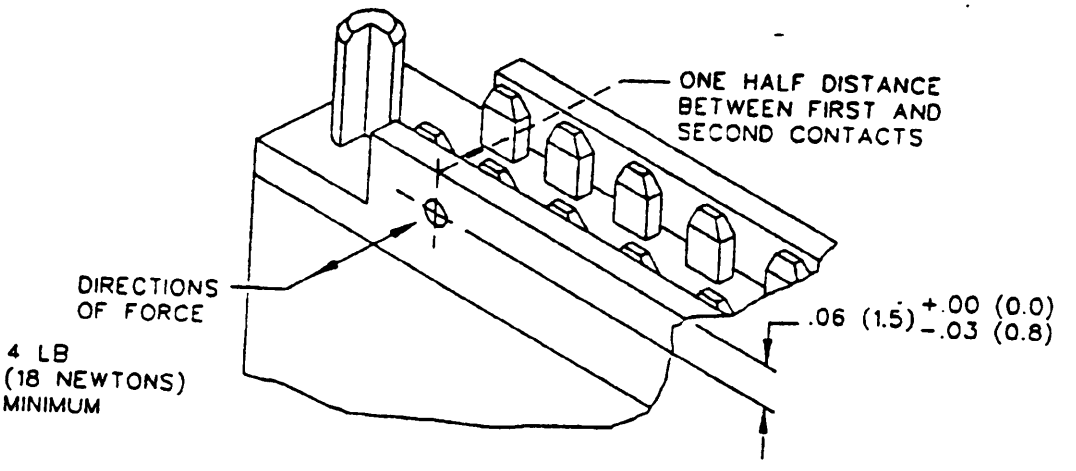
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NOTES:
1. Required for all formats.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parenthesis.

FIGURE 12. Rib strength.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parenthesis.
4. On modules with 20 contact row connectors, the force is applied at four different points for a total of eight measurements. On modules with 50 contact row connectors, the force is applied at six different points for a total of 12 measurements.
5. Required for all formats.

FIGURE 13. Pin shield retention.

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4.7.7.3 Power supply torque. The power supply shall be mounted in a fixture which restrains the movement of the base of the power supply (see figure 14) and shall have a torque as specifed in 3.11.10.4.1 applied to the power supply fin or header in each of two opposing directions. The torque shall be applied at a rate so that the specified torque is applied in 2 to 10 seconds from the time the torque is first applied to the module fin or header and maintained for 10 to 15 seconds. During the period of time in which torque is being applied, the power supply shall be rigidly supported within a zone between the interface plane and 0.50 inch (12.7 mm) above the interface plane for formats A and B, or 1.0 inch (25.4 mm) above the interface plane for formats C, D, and E.

4.7.7.4 Header torque. The power supply shall be supported along the entire length of the power supply ribs (see figure 15). The torque specified in 3.11.10.4.2 shall be applied in both directions along the header on formats C, D and E power supplies. The torque shall be applied in 2 to 10 seconds and maintained for 10 to 15 seconds.

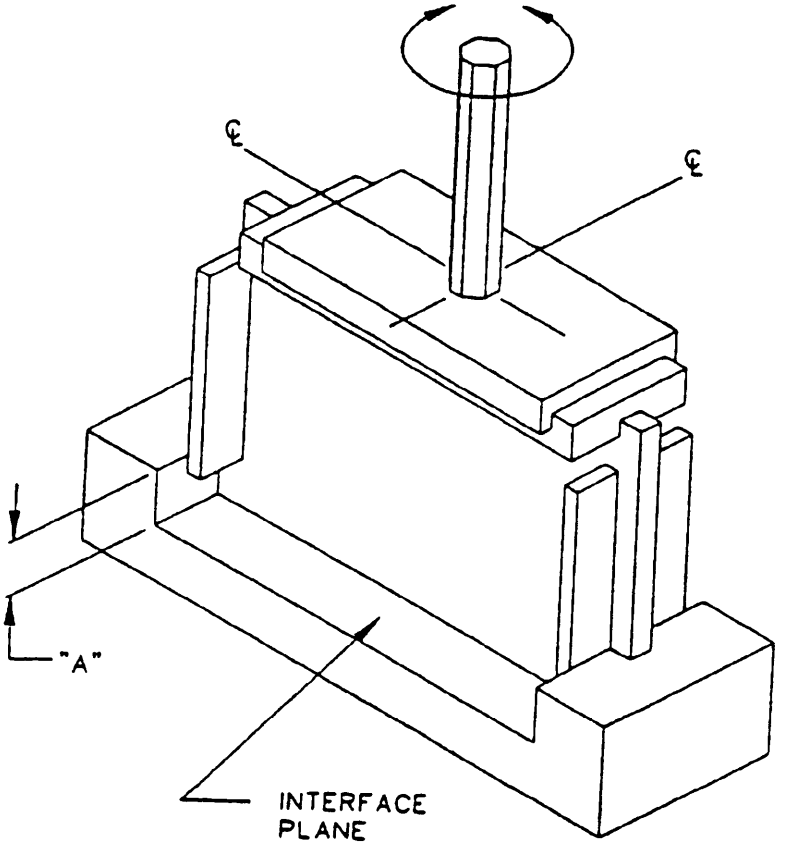
4.7.7.5 Fin/header cantilever load. The force specified in 3.11.10.4.3 shall be applied at the center of the fin or header along the centerline between the two extractor holes and perpendicular to the plane of the fin or header in two directions (see figure 16). The power supply shall be rigidly supported along the power supply ribs within a zone between the interface plane and 0.50 inch (12.7 mm) above the interface plane. The specified force shall be applied in 2 to 10 seconds from the time the force is first applied and maintained for 10 to 15 seconds.

4.7.7.6 Connector integrity. With the power supply ribs supported to prevent movement of the unit, the force specifed in 3.11.10.5.1 or 3.11.10.5.2 shall gradually be applied to the connector in both directions of the longitudinal axis of the contact. The force shall be applied at a rate such that the specified force is applied in 2 to 10 seconds from the time the force is applied to the connector and maintained for 10 to 15 seconds.

4.7.7.7 Connector contact integrity. With the body of the connector supported to prevent movement of the connector, the force specified in 3.11.10.5.4 shall gradually be applied to each contact individually in the connector in both directions of the longitudinal axis of the contact. The force shall be applied at a rate such that the specified force is applied in 2 to 10 seconds from the time the force is first applied to the contacts and maintained for 10 to 15 seconds.

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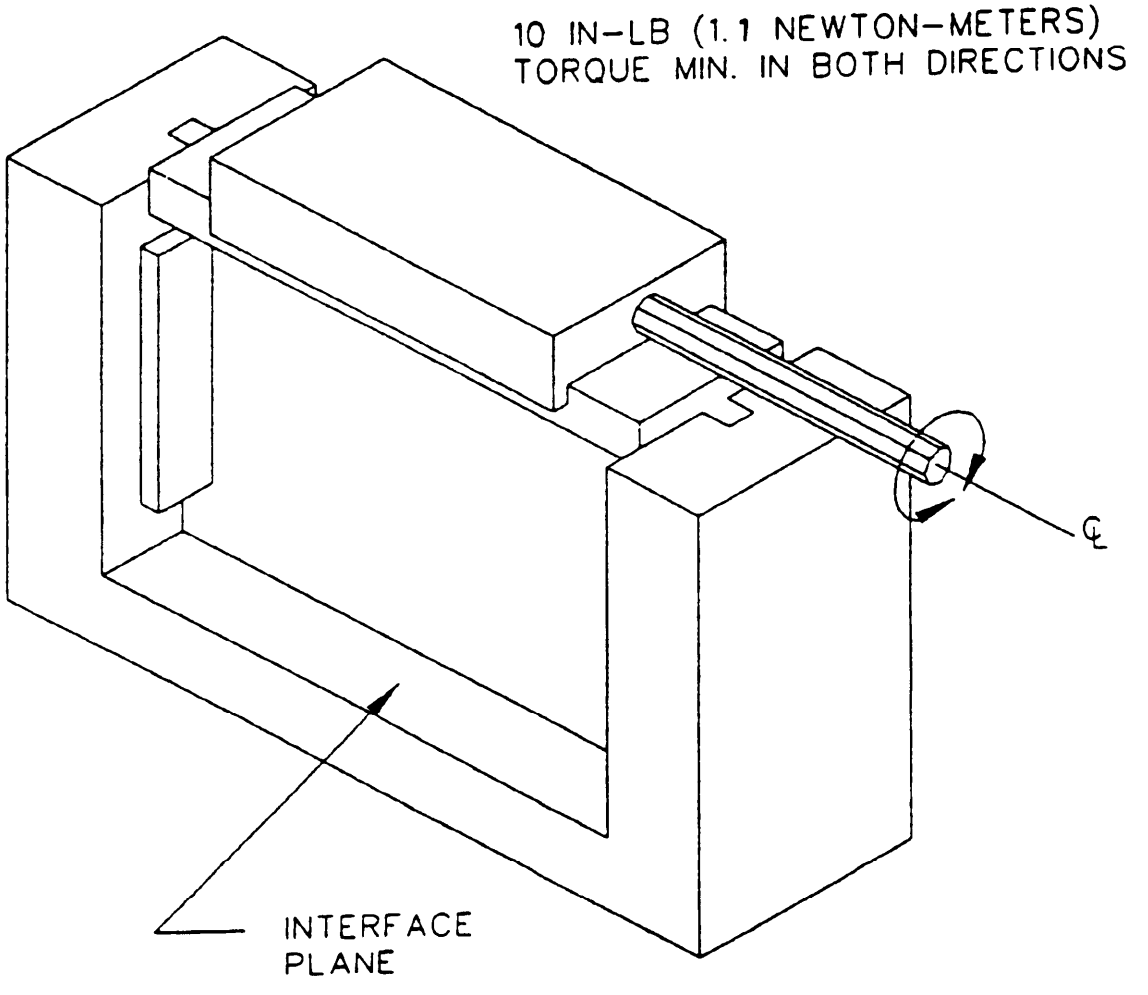
FORMATS A,B: 6 IN-LB (Ø.68 NEWTON-METERS) TORQUE MIN. IN BOTH DIRECTIONS
FORMATS C,D,E: 25 IN-LB (2.82 NEWTON-METERS) TORQUE MIN. IN BOTH DIRECTIONS



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parenthesis.
4. "A" is 0.5 inch (12.7 mm) for formats A and B, and 1.0 inch (25.4 mm) for formats C, D, and E.

FIGURE 14. Power supply torque.

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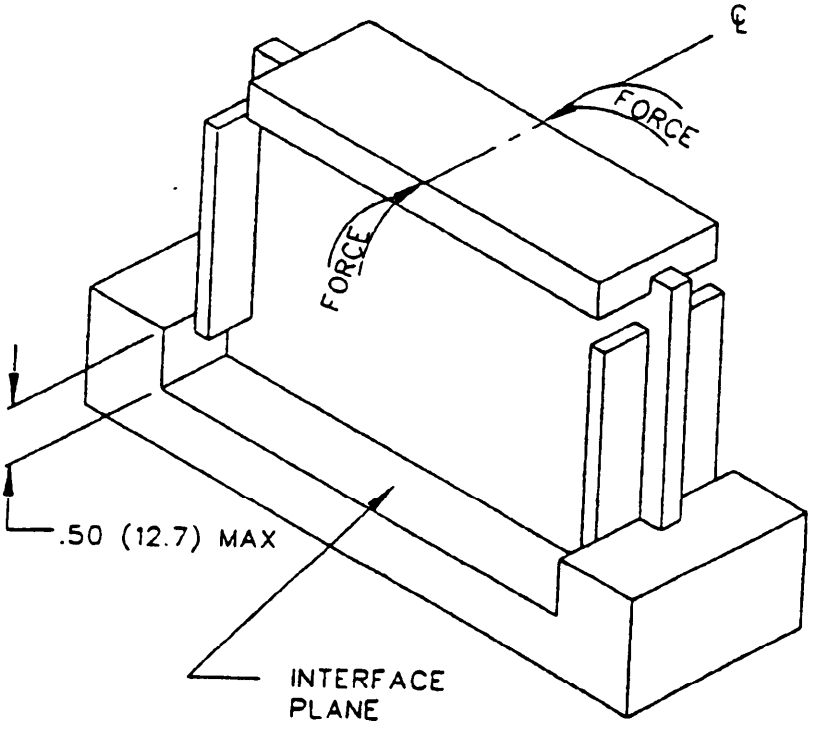


NOTES:
1. Metric equivalents are given for general information only.
2. Metric equivalents are in parenthesis.

FIGURE 15. Header torque.

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FORMATS A,B: 2 LB (9 NEWTONS)
FORMATS C,D,E: 5 LB (22.3 NEWTONS)
MINIMUM IN BOTH DIRECTIONS



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parenthesis.

FIGURE 16. Fin/header cantilever load.

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4.7.7.8 Keying pin torque. With the power supply body supported to prevent rotation of the unit about the longitudinal axis of the keying pin to be tested, the torque specified in 3.11.10.6.5.1 shall be applied in a plane perpendicular to the longitudinal axis of the keying pin. The torque shall be applied so that the specified torque is applied in 2 to 10 seconds from the time torque is first applied to the pin and maintained for 10 to 15 seconds. The longitudinal axis of the keying pin is considered the center of the keying pin diameter extending the length of the keying pin.

4.7.7.9 Keying pin pullout. With the power supply body supported to prevent movement of the unit, the force specified in 3.11.10.6.5.2 shall be applied to the keying pin in the longitudinal axis of the keying pin. The force shall be applied at a rate such that the specified force is applied in 2 to 10 seconds from the time the force is first applied to the keying pin and maintained for 10 to 15 seconds.

4.7.7.10 Keying pin cantilever load. With the power supply body supported to prevent movement of the unit, the force specified in 3.11.10.6.5.3 shall gradually be applied perpendicular to the longitudinal axis of the keying pin at a minimum distance of 0.200 inch (5.08 mm) from the interface surface. The force shall be applied at a rate such that the specified force is applied in 2 to 10 seconds from the time the force is first applied to the keying pin and maintained for 10 to 15 seconds. The keying pin under test shall have no external support along the keying pin length extending from the interface surface during the test.

4.7.7.11 Keying pin pushout. With the power supply body supported to prevent movement of the unit, the force specified in 3.11.10.6.5.4 shall be applied to the keying pin in the longitudinal axis of the keying pin. The force shall be applied at a rate such that the specified force is applied in 2 to 10 seconds from the time the force is first applied to the keying pin and maintained for 10 to 15 seconds. The force shall be applied in the opposite direction to the force in 4.7.7.9. This test shall be applied to format D modules only.

4.7.8 Thermal measurements. Component temperature (see 3.12.1) verification shall be performed by SPSP-QAA as specified in 4.7.8.1 through 4.7.8.4.1.

4.7.8.1 Critical component temperature. With the power supply operating in accordance with the associated detail specification (if no circuit is specified, the long term stability test

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circuits shall be used), the temperatures of the components shall be monitored while maintaining the average mounting surface temperature at 60, plus 5, minus 0°C. At 60°C, the CCT of any component shall not exceed 105°C (100°C for magnetic components) in accordance with 3.12.1.1. At 65°C maximum, the CCT shall not exceed 110°C (105°C for magnetic components) consistent with 3.12.1.1.

4.7.8.2 Transient critical component temperature. With the power supply operating in accordance with 4.7.8.1, all input voltages shall be maintained constant and the average mounting surface temperature shall be increased to 80, plus 5, minus 0°C at a maximum rate of 1°C per minute. Increased mounting surface temperature shall be maintained for a minimum of 60 minutes while monitoring component temperature. The TCCT of any component shall not exceed 125°C or the maximum rating, whichever is less, in accordance with 3.12.1.2.

4.7.8.3 Heat removal (family S1). For the purposes of testing, analysis, and prediction, the total heat generated by the power supply shall be removed through the power supply mounting surface.

4.7.8.4 Heat removal (family S2). Unless otherwise specified in the associated detail specification, component temperature tests shall be performed for each applicable thermal interface (including direct air impingement for formats B, D, and E) (see appendixes A through E of MIL-STD-1389). For the purposes of testing, analysis, and prediction, the total heat generated by the power supply shall be removed from the thermal interface being utilized (rib or fin) without any heat loss from the other interface. During the thermal measurements, the fin or rib temperature shall be monitored as near as possible to the thermal interface without disrupting the integrity of the thermal interface.

4.7.8.4.1 Direct air impingement tests (family S2). Inlet air temperature shall be in accordance with the 4.7.3.2 interface temperature. The inlet air shall be monitored in such a manner that it does not significantly alter the flow of air to the power supply under test. The direction of air flow across the power supply shall be from the pin 1 (alpha) end of the power supply. All power supplies shall be tested in an air flow duct sized in accordance with the particular design requirements of MIL-STD-1389, appendix B, D or E, as applicable. Provisions must be incorporated in the ducts to measure component temperature and air flow pressure drop. The ducts shall be provided with adequate transition regions up stream and down stream of the power supply test location.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1. Process method. The method of preservation shall be in accordance with method IIb of MIL-P-116 and the requirements specified herein. Preservation and packing shall be in accordance with level A of MIL-STD-2073-1.

5.2 Preservation.

5.2.1 Cleaning. Power supplies shall be cleaned in accordance with MIL-P-116, process C-1.

5.2.2 Drying. Power supplies shall be dried in accordance with MIL-P-116.

5.2.3 Preservation application. Contact preservatives shall not be used.

5.2.4 Desiccants and humidity indicators. Desiccant shall be provided in accordance with MIL-P-116. Humidity indicators are not required.

5.2.5 Interior package. A power supply shall be contained in a snug-fitting carton conforming to PPP-B-00636, class weather resistant, type, variety, grade, and style optional, normal requirements. Box closure shall be in accordance with method IV of the appendix to PPP-B-00636.

5.2.6 Barrier. The interior package shall be enclosed in a sealed bag conforming to MIL-B-117, type I, class E, style 1. Power supplies requiring protection from electrostatic field forces shall be enclosed in sealed bags conforming to MIL-B-117, type I, class F, style 1.

5.3 Packing.

5.3.1 Cushioning. Cushioning shall be provided by shock and vibration-absorbing materials or devices that adequately protect the power supplies from physical damage during handling, shipment, and storage.

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5.3.2 Exterior package. An exterior package shall be provided which conforms to PPP-B-00636, class weather resistant, type, variety, grade, and style optional, special requirements. Closure of the exterior package shall be in accordance with method V of the appendix to PPP-B-00636 and shall not damage the barrier bag. Exterior containers shall be of a minimum tare and cube consistent with the protection required.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Power supplies specified herein are intended for use in military systems and subsystems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.

(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

(c) Title, number and date of the associated detail specification and the PIN.

(d) Levels of preservation, packaging, and packing required (see section 5).

(e) QUAL certification, as required (see 3.14.1.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.1.1	DI-NDTI-80809	Test/inspection reports	
4.4.2	DI-R-7082	Reliability predictions report	
4.4.4	DI-DRPR-81000	Product drawings and associated lists	
4.4.9	UDI-R-23743	Quality program plan	
4.4.11	DI-T-25837	Document, maintenence requirement	10.4.3f Applies without exceptions

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3.1 Data requirements for qualification. Data items identified in this specification are required to be delivered to the SPSP-QAA for the purpose of facilitating qualification testing. The data will not be disclosed outside the Government nor duplicated, used, or disclosed, in whole or in part, for any purpose other than to evaluate the power supply submitted for qualification testing (see 6.3.1.1). Generally, these data requirement deliveries coincide with qualification submission schedules and are not a function of power supply acquisitions. Deliverable data required (but not restricted to those listed) by this specification for qualification are cited in the following paragraphs.

Paragraph no.	Data requirement title
3.10.4	Printed wiring and printed wiring assemblies (equivalent materials, processes and requirements)
4.4.1	Initial qualification tests (equivalent materials and processes)
4.4.1.1	Test data
4.4.1.2	Inspection and test procedures

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4.4.2	Failure rate prediction and parts derating
4.4.3	Thermal analysis
4.4.4	Design conformance (design package)
4.4.4.1	Change control
4.4.5	Periodic check (cross-reference matrix)
4.4.5.2	Noncontinuous production
4.4.7	Corrective action
4.4.9	Quality assurance requirements
4.4.9.1	Manufacturing flow chart
4.4.9.2	Manufacturing instructions
4.4.9.3	Processes requiring certification
4.4.11	Maintenance data
10.5, appendix B	Notification
50.3, appendix B	Test procedure documentation
60.4, appendix B	Correlation requirements
70.2, appendix B	Recorrelation criteria

6.3.1.1 Rights in technical data. All data and information provided for use or specified to be delivered for qualification purposes will be furnished for use by the SPSP-QAA only. This data must be delivered to the requirements of SD-6, "Provisions Governing Qualification" and therefore no DID's are required.

6.4 Qualification. With respect to products requiring qualification, awards will be made only to power supply contractors (1) whose products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL No. 24764 or (2) who have at least one power supply listed on QPL-24764, at the time set for opening of bids and deliver with their bid a statement of certification that the procured items will be qualified prior to the required delivery date. The attention of the power supply contractor is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Weapons Support Center, Code 603, Crane, Indiana 47522-5060. Application for qualification tests shall be made in accordance with SD-6, "Provisions Governing Qualification".

6.5 Definitions. The following definitions are applicable to terms used herein.

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6.5.1 Command or agency concerned. The command or agency concerned is the organizational element of the Government which contracts for articles, supplies, or services, or it may be a contractor or subcontractor when the organizational element of the Government has given specific written authorization to such a contractor or subcontractor to serve as agent of the command or agency concerned. A contractor or subcontractor serving as agent of the command or agency concerned does not have the authority to grant waivers, deviations, or exceptions to this specification unless specific written authorization to do so has also been given by the Government.

6.5.2 SPSP Quality Assurance Activity (SPSP-QAA). The SPSP-QAA is the activity that is responsible for QPL maintenance. This function is being performed by the Naval Weapons Support Center, Crane, Indiana 47522-5060 (Code 603, 812-854-1695).

6.5.3 Power supply supplier. A power supply supplier is any person, partnership, company, corporation, or association who owns, operates, or maintains a factory or establishment that produces, on the premises, the power supply required under a contract with the command or agency concerned. If a power supply is manufactured by an organization within the command or agency concerned, that organization is considered a power supply supplier. Supplier and manufacturer are used interchangeably throughout this specification.

6.5.4 Material Review Board (MRB). The MRB will be comprised of at least three members as follows:

(a) A Government member or his designated representative will be the chairman and should approve the qualifications of all other members and alternates.

(b) A power supply supplier's representative whose primary responsibility is the quality of the product.

(c) A power supply supplier's representative whose primary responsibility is the design of the product.

Any concurrence of a disposition by the board must be unanimous. A member may have alternates approved by the chairman. An SPSP-QAA representative may constitute the fourth MRB member at the discretion of the SPSP-QAA.

6.5.5 Specified performance. The terms "specified performance", "specified requirements", and "specified limits"

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are used interchangeably and refer to all applicable power supply requirements specified herein and in the associated detail specification.

6.5.6 MIL-STD-1389 and MIL-M-28787. References herein to MIL-STD-1389 and MIL-M-28787 are made for requirements as applicable to the power supplies. When these documents are used for family S1, the term "Standard Electronic Module (SEM)" is to be interpreted as Shipboard Power Supply Products (SPSP), and the term "module" is to be interpreted as power supply.

6.5.7 Power supply thermal interface. For family S1, the thermal interface is the mounting surface. For family S2, the thermal interface is the fin or rib cooling surface or the component surface of the power supply for direct air impingement cooling.

6.5.8 SEM Design Review Activity (SEM-DRA). The SEM activity which is currently responsible for the review and classification of module designs and assigns manufacturer's designations is the Naval Avionics Center, 6000 East 21st Street, Indianapolis, Indiana 46219-2189 (Code 814, 317-353-3807).

6.6 Part or identifying number (PIN). The PIN to be used for power supplies acquired to this specification is created as follows:

M	24764	/123	A	(1)
Prefix to indicate military specification	Specification number	Detail specification	Detail specification revision letter	Detail specification amendment number

For example, a power supply may be marked M24764/123A(1). The example PIN is not intended to designate a length of field requirement. The length of the PIN will vary according to the applicable associated detail specification but in no case will exceed 15 characters. The associated detail specification revision letter and amendment number in the PIN are to be left blank if none exists.

6.7 Sub-contracted materials and parts. The packaging requirements of reference documents listed in section 2 do not apply when materials and parts are acquired by the manufacturer for incorporation into the power supplies and lose their separate identity when the equipment is shipped.

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.10 Efficiency measurement. The efficiency (see 3.7.2

6.8. Examples of plans and additional requirements. Test plans, documentation, and additional requirements that may be required by the statement of work or contract may include, but not be limited to, the examples specified in 6.8.1 through 6.8.7.

6.8.1 Reliability program plan. A reliability program plan should be prepared that identifies and ties together all program management tasks required to achieve the reliability requirement of 3.7.1. The task description section of MIL-STD-785, task 101, Reliability Program Plan, should be considered in developing the plan. The plan should be submitted to the command or agency concerned for approval.

6.8.2 Failure reporting, analysis, and corrective action. A closed loop system should be established for reporting deficiencies found in accordance with MIL-Q-9858, for analysis of failures to determine cause (see 4.5.4), and for recording corrective action taken. Classification of failures should be in accordance with MIL-STD-2074.

6.8.3 Electronic parts/circuits tolerance analyses. Parts/circuits tolerance analyses should be conducted on each power supply type. These analyses should verify that, given reasonable combinations of specification characteristics and parts tolerance buildup, the power supply being analyzed will perform within specification requirements. Component characteristics (life-drift and temperature) should be factored into the analyses. Results of these analyses and actions taken should be made available to the command or agency concerned.

6.8.4 Environmental stress screening plan. A detailed environmental stress screening test plan should be prepared as part of the reliability program plan and should consider the requirements of task 301 of MIL-STD-785 and NAVMAT-P-9492. The test plan should include the 100 percent screening requirements defined herein as temperature cycling (4.7.5.2.1) and random vibration (4.7.5.2.2).

6.8.5 Reliability development/growth (RDGT) test program. An RDGT program should be implemented in accordance with task 302 of MIL-STD-785.

6.8.6 EMI test plan. An EMI test plan should be prepared to implement the testing requirements and procedures of MIL-STD-462 as defined in 3.6. Formal testing is not to commence without approval of the test plan by the SPSP-QAA.

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6.8.7 Government source inspection. The command or agency concerned should notify the Defense Contract Management Administration Office (DCMAO) representative servicing the command or agency prior to placement of an order invoking this specification. The recommended DCMAO actions should include review of the supplier's quality program and the means employed to control quality and to comply with contract requirements, initiation of required corrective action, and the inspection of the power supplies.

6.9 Nonconforming materials. Any variance of material from this specification and from the associated detail specification should be disposed of in accordance with the decision reached by the MRB (see 6.5.4). The supplier may use in-house established procedures documented in his control plan for such disposition. Records of all such variances and disposition should be retained on file and available for Government review. SPSP-QAA should be notified of MRB decisions which result in using nonconforming materials in the manufacture of power supplies which cause the power supplies to not meet the requirements specified herein or in the associated detail specification.

6.9.1 Disposition of nonconforming material. The disposition of nonconforming material should be in accordance with the decisions reached by the MRB. The dispositions will generally fall into the following categories:

(a) Use as is. Material that, in its present condition, can be used without compromising form, fit, function, performance, reliability, safety, and so forth. The disposition will be by the MRB.

(b) Repair. Material, after repair, which meets the definition of (a) but does not fully meet drawing and specification requirements. This disposition, including approval of the repair procedure, will be by the MRB.

(c) Rework. Material which can economically be restored to a conforming condition. This disposition does not require MRB approval.

(d) Scrap. Material not useable in its present condition and which cannot be economically reworked or repaired. This disposition does not require MRB approval.

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6.9.2 Shipment of nonconforming material. Power supplies containing nonconforming material will not be shipped until the applicable waiver or deviation has been approved by the command or agency involved or the final MRB disposition has been received. Modules requiring a waiver or deviation will not be marked with the qualification certification mark "QUAL".

6.10 QUAL marking. The SPSP-QAA will notify in writing each power supply supplier on a PIN basis when "QUAL" may be marked on a power supply. Conversely, when a power supply supplier fails to meet all of the requirements herein, and in the associated detail specification, the vendor will be notified in writing to stop marking "QUAL". Such notification will come from the SPSP-QAA.

6.11 Conflicting requirements. Conflicting requirements arising between the requirements specified herein and those in any specification, standard, handbook or publication listed herein should be referred in writing to the command or agency concerned and the SPSP-QAA for interpretation, clarification, resolution or correction.

6.12 Supersession. This document and associated detail specifications supersede Naval Sea Systems Command (NAVSEA) drawing SE 010-AA-SPN-010 Standard Power Supply Program, General Specification for, Revision A, dated September 1985 and associated detail drawings.

6.13 Subject term (key word) listing.

Converter, AC to DC
Converter, DC to DC
SHARP
Shipboard Power Supply Products
SPS
SPSP
Standard Hardware Acquisition and Reliability Program
Standard Power Supplies

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APPENDIX A

FAMILY S1 POWER SUPPLY DIMENSIONS AND KEYING

10. SCOPE

10.1 Scope. This appendix establishes the case dimensional requirements, keying locations, and key coding for power supplies in the family S1, Shipboard Power Supply Products (SPSP). This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specification. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

MIL-C-28748	-	Connectors, Electrical, Rectangular, Rack and Panel, Solder Type and Crimp Type Contacts, General Specification for.

(Unless otherwise indicated, copies of military specifications are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

20.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5M	-	Dimensioning and Tolerancing. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3308.)

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APPENDIX A

30. GENERAL REQUIREMENTS

30.1 Dimensions. Each power supply case style shall conform to the dimensions and notes shown on figure 17 and in table XIV.

30.1.1 Connectors. The input and output connectors shall be in accordance with MIL-C-28748. The position, style, and number of contacts shall be in accordance with table XV. The location of connector guide pins and guide sockets shall be as shown on figure 17. The connector contact height dimensions and engagement depth requirements shall be as shown on figure 17.

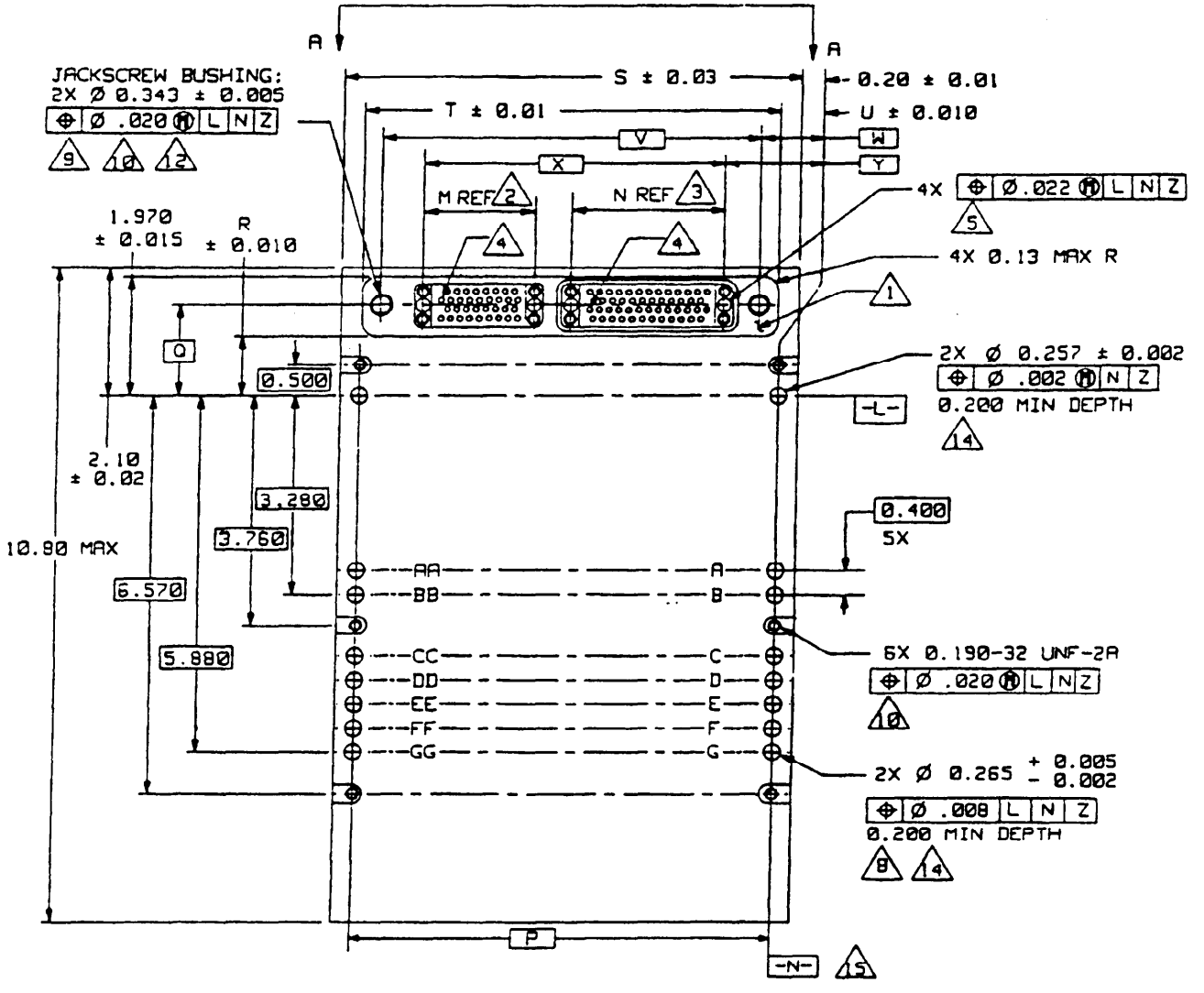
30.1.2 Hold-down screws. The location and number of hold-down screws shall be as specified. Case style 1 does not require jackscrews in the V dimension of figure 17. Jackscrew bushings shall be as shown on figure 17. Hold-down screws and jackscrews shall be socket head cap screws as shown on figure 17.

30.2 Keying. Each power supply shall have four keying hole locations.

30.2.1 Stationary keying. Each power supply shall have two keying holes located on the connector end of the supply and positioned as shown on figure 17 and in table XIV.

30.2.2 Personality keying. Each power supply shall have two keying holes positioned according to the coding assigned to the power supply type and as shown on figure 17 and in table XIV. Figure 17 shows the alternate locations for the personality keying holes.

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BOTTOM VIEW

FIGURE 17. Dimensions and keying.

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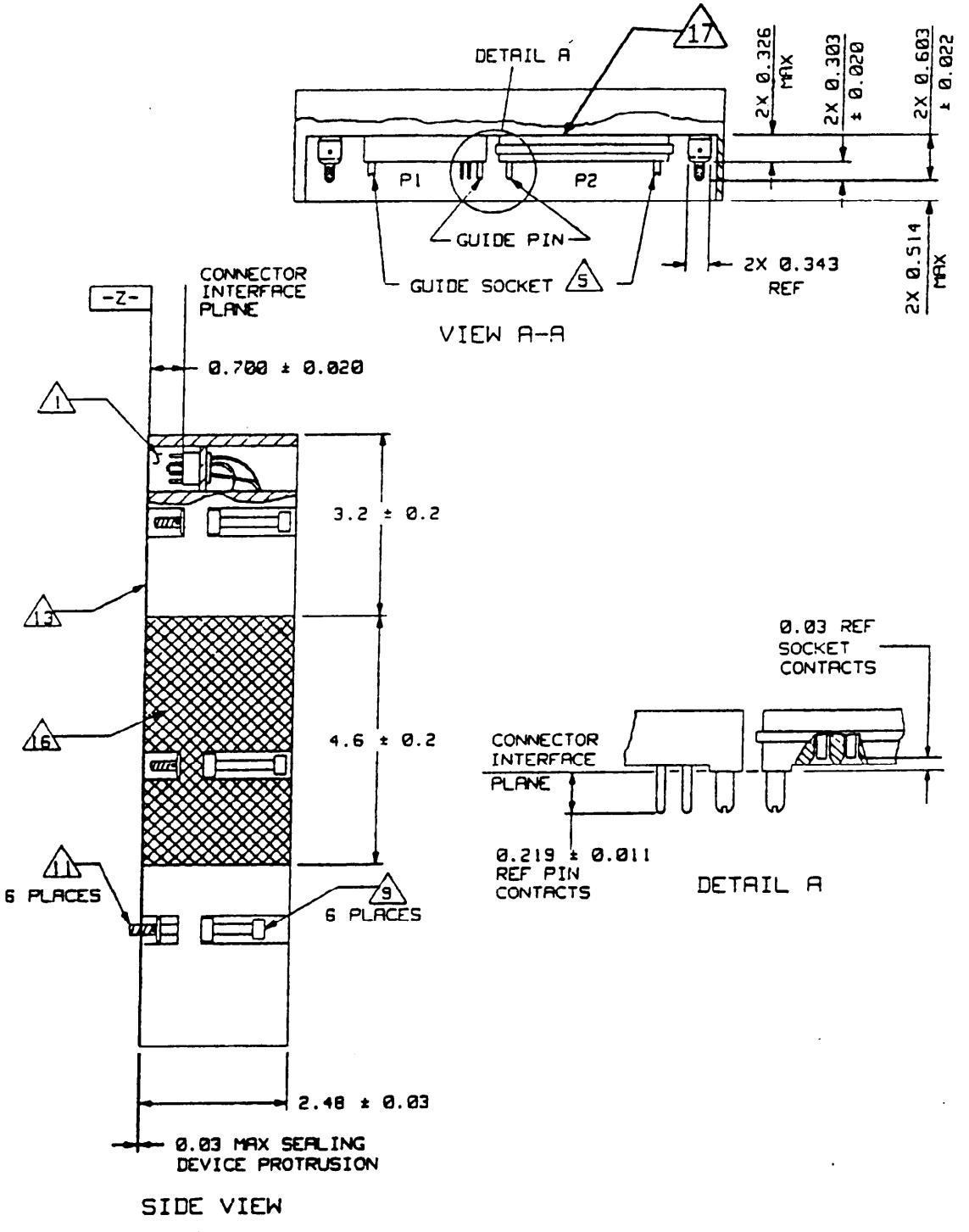


FIGURE 17. Dimensions and keying - Continued.

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NOTES:

1. Connector interface recess. Connector recess dimensions shall be held to a depth of 1.020 minimum to allow clearance for the mating connector.

2. Input connector. See table XV for part number. Dimension M for reference only. Actual dimensions to be obtained from MIL-C-28748.

3. Output connector. See table XV for part number. Dimension N for reference only. Actual dimensions to be obtained from MIL-C-28748.

4. Indicates pin "A" on connector.

5. Install guide socket in this location.

6. 0.062 minimum contact wiping engagement is required between the power supply connectors and the mating connectors for full current carrying capacity of the contacts.

7. The entire case shall provide environmental protection (including corrosion prevention) and electric and magnetic EMC shielding as prescribed in the associated detail specification.

8. View illustrates all possible personality keying hole locations. Refer to associated detail specification for actual key code and corresponding keying hole locations.

9. 0.190-32 UNF-3A socket head cap screw. Hexagonal keyway in screw head to measure 0.1562 + 0.0025 - 0.0000 across flats.

10. Dimensions and tolerances control location of mounting hole for captive screw. Captive fasteners must provide a minimum of 0.015 and a maximum of 0.04 radial float from nominal position.

11. 0.340 $\pm$ 0.030 penetration of captive fasteners in the secured position. In the retracted position, the ends of the fasteners shall recess 0.010 minimum into the mounting surface.

12. Jackscrew fasteners at the connectors are deleted on style 1 power supply.

FIGURE 17. Dimensions and keying - Continued.

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NOTES: - Continued.

13. 0.0005 inch/inch and 0.005 overall flatness and 32 microinches finish tolerances to be maintained on mounting surface.

14. 1.00 min pin clearance past |-Z-| shall be provided. The initial 0.200 of hole depth shall be as specified. The diameter of the remainder of the pin clearance hole may be enlarged, but shall not be less than the minimum diameter of the initial 0.200 of hole depth and shall be positioned to guarantee clearance around the initial hole perimeter.

15. |-N-| is perpendicular to |-L-| at the center of the 0.257 dia. hole.

16. A gripping surface shall be located in the designated area on both sides of the power supply (see 30.4).

17. This surface, as defined by the dimensions on the connector interface plane and connector body, is not required to be (but is not prohibited from being) the same plane as defined by the dimensions on the jackscrew and bushing.

18. Dimensions are in inches.

19. Dimensioning is in accordance with ANSI Y14.5M.

FIGURE 17. Dimensions and keying - Continued.

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APPENDIX A

TABLE XIV. Dimensions for figure 17.

Style	Dimensional variables												
	P	Q	R	S	T	U	V	W	X	Y	M	N
1	4.600	1.480	0.980	5.00	4.40	0.100	N/A	N/A	3.450	0.575	1.255	1.694
2	5.400	1.480	0.980	5.80	5.35	0.025	4.750	0.325	3.880	0.760	1.694	1.694
3	6.400	1.480	0.980	6.80	6.38	0.012	5.900	0.250	4.437	0.982	1.694	2.290
4	8.400	1.163	0.360	8.80	6.80	0.800	6.300	1.050	4.850	1.775	1.976	2.389
5	11.400	1.163	0.360	11.80	6.80	2.300	6.300	2.550	4.850	3.275	1.976	2.389

All dimensions are in inches.

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TABLE XV. Connector type.

Power supply style	MIL-C-28748 connector type			
	Input		Output	
	Type	Contacts	Type	Contacts
1	/1-C00G1A	20	/2-D00G1A	34
2	/1-D00G1A	34	/2-D00G1A	34
3	/1-D00G1A	34	/2-F00G1A	50
4	/1-G00G1A	66	/2-J00G1A	104
5	/1-G00G1A	66	/2-J00G1A	104

30.2.3 Engaging pins. The mating thermal and mechanical interface shall provide appropriately placed pins to engage the keying holes in the power supply. The length of the engaging pins is optional except that they must be long enough to prevent installation and energizing of the wrong power supply, yet not exceed the minimum pin clearance depth referenced in note 14 on figure 17.

30.3 Key coding.

30.3.1 Key code. A key code for a particular power supply type shall include the identifier for the keying hole on the input connector side of the supply (bottom view), the case style, and the identifier for the keying hole on the output connector side of the supply (for example, AA3F).

30.3.2 Key code assignment. A keycode shall be assigned in accordance with MIL-STD-2038 for each new power supply type.

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greatest thermal inertia (excluding magnetics),

30.4 Gripping area. A textured gripping surface or indentation shall be provided within the area shown on figure 17 and located on both sides of the power supply. A textured surface finish in the designated areas shall be 500 microinches minimum. An indented area shall be a maximum of 0.40 inches deep. The indentation shall be located in the area between 0.90 and 2.10 inches from the Z datum plane, but shall be a minimum of 0.50 inches wide. Any method used to provide the gripping area shall not degrade the performance of the power supply. The finish shall not be degraded or abraded by handling or testing.

30.5 Sealing device. The sealing device shall be placed around the entire perimeter of the power supply within the allowable area shown on figure 18. The sealing device shall be installed on each power supply during manufacturing prior to environmental stress screening tests.

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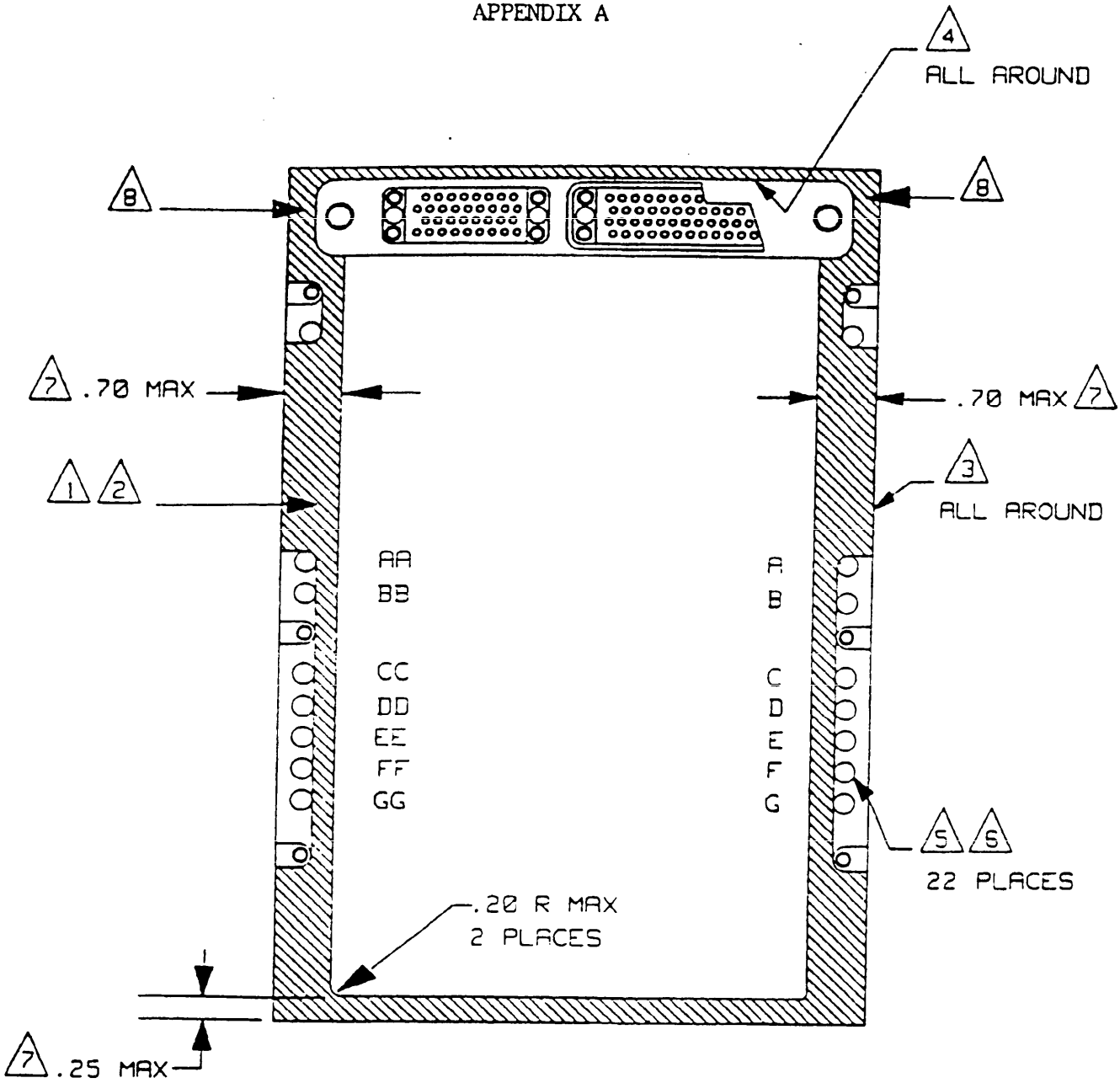


FIGURE 18. Location for mounting surface environmental seal.

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NOTES:

1. Shaded area indicates allowable locations for sealing devices (gasket, O-ring, and so forth). Sealing devices shall not degrade the performance of the power supply.

2. The sealing device shall be captivated to the power supply and must withstand a 90° peel test with an applied load of 15 pound-force per inch of material width or 0.75 pound-force, whichever is greater. Adhesion of the sealing device shall not be degraded by environmental exposure.

3. Sealing device edges shall not extend beyond outside edges of power supply.

4. Sealing device edges shall not extend past inside edges of connector mounting recess.

5. Sealing device shall not enclose any mounting screw or key pin hole unique to the power supply key code.

6. Sealing device location shall not preclude the use of any mounting screw or key pin hole unique to the power supply key code.

7. Dimensions apply from the outside edge of the power supply to the inside edge of the sealing device.

8. Allowable area width adjacent to connector recess not to exceed 0.60 for case styles 4 and 5.

FIGURE 18. Location for mounting surface environmental seal - Continued.

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APPENDIX B

SHIPBOARD POWER SUPPLY PRODUCTS TESTER CORRELATION

10. SCOPE

10.1 Scope. This appendix describes and states criteria for correlation, the method of proving, for a specific power supply PIN, that all production testers shall yield test data which matches the correlation data within close tolerance and determines if the power supply meets the requirements of the power supply associated detail specification. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

10.2 Performance. Correlation shall be performed on all production testers used to accept, qualify, or recertify SPSP. Successful correlation shall be required to certify the use of a tester and its peripheral equipment (adapter, switching cards, and so forth).

10.3 Method. Correlation is accomplished by using a common medium between the production tester and the quality assurance provision of the power supply associated detail specification. This medium provides a means of comparing the results of a specific power supply having known recorded parameters with the results of the production testers. This common medium is a power supply hereafter termed correlation power supply.

10.4 Responsibility. The SPSP-QAA shall be responsible for correlation and shall determine the need for correlation of production testers associated with a given power supply PIN. The SPSP-QAA shall also maintain control over the correlation power supplies and only take them to the vendor's facility during correlation attempts.

10.5 Notification. In the event of proposed changes to test fixturing, test procedures, tester software, equipment, and so forth, the testing facility should notify the SPSP-QAA in writing as soon as the changes are proposed. The SPSP-QAA will determine the requirement for re-correlation.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX B

30. CORRELATION CRITERIA

30.1 Correlation criteria. The following are criteria to be used in determining the need for correlation of a production tester for a specific power supply PIN.

(a) Qualification certification. Qualification certification for a SPSP power supply requires that all test systems be correlated.

(b) Accuracy requirements. The associated detail specification has critical parameters with close tolerances requiring accurate measurements.

(c) Test method deviations. A tester with a test method which deviates from procedures specified in the power supply associated detail specification.

(d) Factory or field failure. Power supply types which have experienced excessive factory or field failures.

40. CORRELATION AND PROOF POWER SUPPLIES

40.1 Acquisition. The initial developer or contractor shall provide the SPSP-QAA with two production power supplies of the newly developed PIN for use as correlation and proof power supplies. Or, when agreed to by SPSP-QAA, power supplies submitted for qualification may be used for correlation samples. Prior to the availability of production power supplies, prototype power supplies meeting the above requirements may be submitted. Each power supply shall meet the product design configuration of the associated detail specification.

40.2 Development. The SPSP-QAA shall test the two power supplies to requirements of the associated detail specification. The data shall be recorded and a data file shall be maintained and updated. When the power supplies are tested by a bench setup, all test equipment and associated information shall be recorded. When power supplies are tested by automatic test equipment (ATE), a copy of the program and a drawing of all required interface circuitry shall be maintained. All data shall be referenced to the applicable test conditions of the associated detail specification. The determination to use bench or ATE shall be made by the correlation activity.

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40.2.1 Correlation power supply. One of the power supplies for which correlation sheets are made shall be designated as the correlation power supply. The purpose of the correlation power supply shall be to obtain comparison data for use in correlating an existing tester. The correlation power supply shall display a conspicuous identification sticker marked "CORRELATION POWER SUPPLY".

40.2.2 Proof power supply. The remaining power supply which was tested on a bench test setup or a correlated tester shall be designated a proof power supply. The proof power supply shall be used to check out new or modified testers in preparation for correlation testing. The proof power supply protects the correlation power supply from damage and is used for laboratory investigation of requested tester deviations. The proof power supply shall display a conspicuous identification sticker marked "PROOF POWER SUPPLY".

40.3 Repair and retesting of correlation and proof power supplies. A Government designated facility will be responsible for repairs or modifications to correlation and proof power supplies. Correlation and proof power supplies which have been repaired, modified or otherwise adjusted will require retesting in a correlation activity laboratory in order to obtain new correlation power supply data which will supersede previous data.

40.4 Storage and control. Correlation and proof power supplies and their associated data sheets shall be stored by and under control of the SPSP-QAA.

50. TESTER FACILITY REQUIREMENTS

50.1 Facility requirements. A tester facility shall be considered ready to test a correlation power supply and record correlation data when the facility has complied with the following and received approval from the SPSP-QAA:

(a) The tester test document has been written to the required revision of the power supply associated detail specification.

(b) The tester is compatible with its own documentation which includes a top assembly drawing and has been made available to the SPSP-QAA upon request.

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(c) The top assembly drawing number and revision status of the production tester and its peripheral equipment is displayed on the applicable equipment.

(d) The production tester and its associated equipment have been serialized and the serial number displayed on each item along with the current calibration tag.

50.2 Tester documentation. The tester documentation shall contain the following:

(a) Manufacturer and model number of each piece of test equipment used.

(b) Serial number of tester to be correlated.

(c) Tester measurement offsets and associated parasitic impedance data along with an explanation of the derivations.

(d) Test system configuration which includes equipment used, equipment specifications, location of equipment in the test system, and any optional equipment used.

50.3 Test procedure documentation. The tester facility shall prepare a test procedure for each power supply PIN to be correlated. The test procedure shall be submitted to the SPSP-QAA for review at least 2 weeks prior to the correlation attempt. The test procedure shall contain the following:

(a) Power supply specification number and revision to which tester is testing power supplies.

(b) Top assembly drawing number and revision of tester and power supply related peripheral equipment (adapters, switch cards, test tapes, load boards). Drawings of peripheral equipment will be made available to the SPSP-QAA on request.

(c) ATE test program documentation which includes:

(1) System software revision.

(2) List of all files and sub-files.

(3) Test measurement routines.

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(4) Order of testing.

(5) Flow charts for particular test sequences, that is, algorithms.

(6) Programming constraints.

(7) Data logging and operator routines.

(d) A definition of each deviation from the associated detail specification test method which is made in the tester test procedure.

(e) Measurement limits to which tester is testing power supplies.

(f) Determination of test as sample, 100 percent, or qualification only.

(g) Proper order of testing per the associated detail specification.

(h) Comparison of the tester obtained accuracies and required accuracies for each unique test condition of the associated detail specification (this is not required if the method and test equipment used are the same as in the associated detail specification).

(i) Schematic drawing of the test fixture(s) for bench test systems.

60. CORRELATION TESTING

60.1 Correlation testing. The SPSP-QAA shall require the following items for correlation testing at the location of the applicable tester facility:

(a) The correlation data sheet for each correlation power supply with the information specified in 40. and already recorded (supplied by SPSP-QAA).

(b) A copy of the PIN associated detail specification for which the tester is to be correlated.

(c) Tester documentation as specified in 50.

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(d) One correlation power supply and one proof power supply of the PIN for which the tester is to be correlated (supplied by SPSP-QAA).

60.2 Using the proof power supply. Prior to inserting a correlation power supply into the tester, a proof power supply of the same PIN shall be inserted into the tester to protect the correlation power supply. When automatic testers are used, a proof power supply shall be tested twice to verify the power supply has not been damaged. If this requirement is not met, then the responsible tester facility personnel shall take the necessary corrective action. If the corrective action requires that changes be made in tester documentation, then these changes shall be sent in writing to the SPSP-QAA. When the tester test results on the proof power supply have met the requirements of this appendix, the tester is ready to have a correlation power supply inserted. Correlation testing shall be done by tester facility personnel at the tester facility and witnessed by the SPSP-QAA.

60.3 Recording information on the correlation data sheet. The correlation personnel shall properly complete the data and information required on the correlation data sheet corresponding to the correlation power supply of the PIN for which the tester is being correlated.

60.4 Correlation requirements. The tester test results shall be considered to have met the correlation requirements when they agree with the requirements of 90.3 through 90.4.

60.4.1 Discrepancies. Tester facility personnel shall be responsible for taking corrective action on each discrepancy found. Before correlation data is retaken on the tester, tester facility personnel shall explain to the SPSP-QAA in writing the reason for and action taken on each discrepancy. All changes and revisions to tester documentation shall be submitted in writing to the SPSP-QAA. Upon approval of these changes by the SPSP-QAA, correlation data shall be taken on those test conditions deemed necessary by the SPSP-QAA.

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60.4.2 Failure with correlation power supply. If a tester measurement fails the correlation criterion, the proof power supply may be used to confirm or deny the failure to correlate. If, when using the proof power supply, the tester measurement passes the correlation criterion, the measurement will not be considered to have failed correlation until the data of the correlation power supply has been verified on a SPSP-QAA laboratory bench test setup.

60.5 Correlation not successful. When correlation is not successful, the reason and planned corrective action shall be entered in the tester correlation report (see figure 19) and forwarded to the acquisition activity.

60.6 Correlation successful. When correlation is successfully completed, the SPSP-QAA shall issue a correlation report (see figure 19) which is the tester's official certification for use in testing the specific power supply PIN.

60.7 Report. The SPSP-QAA shall send copies of the completed correlation report to the cognizant Government activity at the tester facility and to the acquisition activity.

70. RECORRELATION

70.1 Responsibility. The SPSP-QAA shall be responsible for recorrelation of a tester or any part thereof.

70.2 Recorrelation criteria. The following criteria shall be used to determine the necessity for recorrelation:

(a) The tester or any part thereof has undergone modification which directly influences the electrical testing of power supplies.

(b) Repeated failures of a specific power supply PIN are indicated by tester measurements.

(c) Repeated failures of a specific power supply PIN are indicated by system usage.

(d) A power supply of a specific PIN has undergone modification which results in a change in the test procedure, the test results, or the tester, or any part thereof.

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APPENDIX B

TESTER CORRELATION REPORT
SHIPBOARD POWER SUPPLY PRODUCTS

POWER SUPPLY KEY CODE	MIL-P-24764				
	MIL-P-24764	/ SHEET NO.	REV. LTR.	AMENDMENT NO.	☐ YES X & S TESTS (& Q TEST FOR NAVSCC) ☐ NO ☐ OTHER
	EQUIVALENT OR SPECIFICATION CONTROL				
	DWG. NO.		REV. LTR.		

VENDOR AND LOCATION	CORRELATION
	☐ INITAL ☐ RECORRELATION (DATA RETAKEN) ☐ RECORRELATION (PAPER UPDATE)

TEST PROCEDURE		DATE (DATA TAKEN)
NUMBER	REVISION	

TEST EQUIPMENT CORRELATED

VENDOR TESTER PERSONNEL

CORRELATION STANDARDS AND SERIAL NOS.

COMMENTS

AUTHORIZED DEVIATIONS

SPSP APPROVAL	DATE
CORRELATION ENGR.	
SPSP APPROVAL	

EXAMPLE

FIG. 19 TESTER CORRELATION REPORT FORMAT .

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(e) Elapse of qualification status criteria as specified in this specification.

The SPSP-QAA shall be notified when recorrelation is required.

80. CORRELATION CONTROL DOCUMENTATION

80.1 Information required. For every SPSP power supply PIN, the following information shall be documented and maintained by the SPSP-QAA:

(a) Tester correlation data.

(b) Tester test equipment data.

(c) Test procedure documentation (see 50.3).

(d) Tester documentation (see 50.2)

(e) Tester correlation report (see figure 19).

90. CALCULATION AND APPLICATION OF TESTER OFFSETS AND CORRELATION TOLERANCES

90.1 General. The tester test results shall be compared to the correlation power supply bench test data plus known tester offsets. The tester results will be considered correlated if they agree with the correlation power supply bench test data plus known tester offsets within the correlation tolerances.

90.2 Tester offsets. A tester measurement offset is a constant quantity with a known physical explanation associated with that tester's measurement of a particular electrical parameter. An offset may be extra tester wiring in a resistance measurement, a fixed tester delay in a time measurement, a deviation in test method or other such tester related examples.

(a) Example 1.

A power supply transition time is found to be equal to 125 nanosecond (ns) as measured in the laboratory with an oscilloscope. On a production tester, the transition time measurement on the same power supply is found to be equal to 145 ns due to 95 picofarad of parasitic capacitance associated with the tester adapter output pins into which the power supply is plugged. The resultant offset is therefore 145 minus 125 equals 20 ns.

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(b) Example 2.

A power supply test specification states that a voltage is to be measured at a test point. At the test point, this voltage is found to be 9.8 volts as measured in the laboratory. The production tester cannot make measurements at the test point. It must make the voltage measurement at a power supply contact on the other side of a conducting diode. The diode voltage drop is found to be equal to 0.6 volt. The output measured by the tester is therefore equal to 9.8 volts minus 0.6, equaling 9.2 volts. The tester offset is then minus 0.6 volt.

(c) Offset.

The offset is used for two purposes. First, it serves in the correlation between the bench test data and the tester data on the correlation power supply. Second, the final programmed tester pass or fail limits are calculated using the offset. This is shown as follows:

Final minimum limit equals minimum test specification limit plus offset.

Final maximum limit equals maximum test specification limit plus offset.

90.3 Correlation tolerance. The tester results from the correlation power supply are compared to the results obtained by adding the appropriate tester offsets to the bench test data (taken to power supply test specification) on this same power supply. These results should be identical within a tolerance called the correlation tolerance. One of the following two methods is used to calculate the correlation tolerance depending upon the associated detail specification parameter requirement.

90.3.1 Parameter tolerance equal to or greater than 3 percent. In cases where the parameter tolerance is greater than or equal to 3 percent of the parameter, 22 percent of the parameter tolerance shall determine the correlation tolerance. This can be expressed as follows:

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APPENDIX B

Correlation tolerance = ± 0.22

Correlation tolerance = ± 0.11 $\left[\left|\frac{\text{Max test specification limit} - \text{min test specification limit}}{2}\right|\right]$.

[Maximum test specification limit -minimum test specification limit].

(a) Example.

A test specification states that the requirements for a parameter are tr equals 125 ns, plus or minus 15 percent. The maximum limit would be 125 plus 0.15 (125) equaling 143.8 ns. The minimum limit would be 125 minus 0.15 (125) equaling 106.2 ns. The correlation tolerance is 0.11 times the quantity, 143.8 minus 106.2, equaling plus or minus 4.14 ns. This means that the tester measured tr should agree with the bench test measured tr plus tester offset (if applicable) within 4.14 ns. If the upper or lower limit is infinity, or minus infinity, then the parameter will not be correlated.

90.3.2 Parameter tolerance less than 3 percent. In cases where the parameter tolerance is less than 3 percent of the parameter, 25 percent of the parameter tolerance plus the bench test equipment accuracy will determine the correlation tolerance. This can be expressed as follows:

Correlation tolerance = $\pm\{ 0.25 \left|\frac{\text{Max test spec limit} - \text{min test spec limit}}{2}\right| + \left|\text{bench test eqpt accuracy}\right| \}$.

(a) Example.

A test specification states that the requirements for a module parameter are Vp equals 10 volts, plus or minus 1 percent. The maximum limit is 10 plus the quantity 0.01 times 10, equaling 10.1 volts. The minimum limit is 10 minus the quantity, 0.01 times 10, equaling 9.9 volts.

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In addition, the voltmeter used to measure Vp in the laboratory during the recording of correlation data has an accuracy of plus or minus 0.1 percent of full scale as specified by the manufacturer. The accuracy of a 10 volt reading is, therefore, plus or minus 0.1 percent times 10 volts, equals plus or minus 0.01 volt.

$$\text{Correlation tolerance} = \pm \left(0.25 \left|\frac{(10.1 - 9.9)}{2}\right| + [.01]\right).$$

Correlation tolerance = ± 0.035 volt.

90.4 Parameter tolerance undefined in terms of percent. In cases where the parameter tolerance cannot be defined in terms of percent due to the parameter requirement being centered about zero, 22 percent of the parameter tolerance shall determine the correlation tolerance. In cases where the parameter tolerance cannot be defined in terms of percent due to the reading being taken in decibel (dB), 1.1 times the sum of the bench test equipment accuracy and the tester test equipment accuracy shall determine the correlation tolerance.

(a) Example.

A test specification states that the requirement for a module parameter is voltage offset equals 0, plus or minus 10 mV. The maximum limit is 0 plus 10, equals 10 mV. The minimum limit is 0 minus 10, equals minus 10 mV.

$$\text{Correlation tolerance} = \pm 0.22 \left|\frac{10 - (-10)}{2}\right| .$$

Correlation tolerance = ± 2.2 mV.

(b) Example.

A test specification states that the requirements for a power supply parameter are Gain equals 20 dB, plus or minus 2 dB (in the electrical requirements table and in the test table). The test equipment used in the laboratory and the tested test equipment has an accuracy of plus or minus 0.2 dB.

Correlation tolerance = ± 1.1 (0.2 + 0.2).

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Correlation tolerance = ± 0.44 dB.

For those test conditions where the electrical requirements table limits are specified in dB and in the test table limits specified in volts, the correlation limits shall be determined in volts.

90.5 Test equipment accuracy deviation. In the case where a deviation from the test equipment accuracy requirements specified herein and MIL-STD-1665 has been granted either through the power supply specification or through waiver acceptance, 1.1 times the sum of the bench test equipment accuracy and the tester test equipment accuracy shall determine the correlation tolerance. This can be expressed as follows:

Correlation tolerance = ±1.1 (bench test equipment accuracy + tester equipment accuracy)

(a) Example.

A test specification states that the requirements for a power supply parameter are 8 nanoamperes (nA), plus or minus 25 percent. The required test accuracy is plus or minus 25 divided by 10, equals plus or minus 2.5 percent. The power supply specification states that the test accuracy may be as great as plus or minus 5 percent. The test equipment used in the laboratory has an accuracy of plus or minus 3 percent on the 10 nA scale. The accuracy of this reading is therefore plus or minus 0.03 times 10, equaling 0.3 nA. The test equipment used at the tester facility has an accuracy of plus or minus 4 percent on the 10 nA scale. The accuracy of this reading is therefore plus or minus 0.04 times 10, equaling plus or minus 0.4 nA.

Correlation tolerance = ±1.1 (0.3 + 0.4).

Correlation tolerance = ±0.77 nA.

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